SOFTWARE REQUIREMENTS SPECIFICATION

Pixora

Online Image Converter

MVP release: AVIF to JPG conversion

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How to use this document This SRS is intended as the single source of truth for building Pixora. Requirements carry unique IDs (FR, NFR, SR, TC, R, A, D). Where a statement uses SHALL it is mandatory for the MVP; SHOULD and MAY indicate lower priority. Items labelled "Future" are explicitly out of scope for the MVP.

Document Control and Version History

Document control

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Approvals

Role	Name	Signature	Date
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Table of Contents

Document Control and Version History **2**

4. Introduction **6**

4.1 Document conventions 6

4.2 Overview of the document 6

5. Purpose **6**

5.1 Business objectives 6

6. Scope **7**

6.1 In scope (MVP) 7

6.2 Out of scope (MVP) 7

6.3 Assumptions and constraints 7

7. Definitions and Abbreviations **9**

8. Overall Description **10**

8.1 Design principles 10

8.2 Key design decisions 10

9. Product Perspective **11**

9.1 System context 11

9.2 External dependencies 11

10. Product Functions **12**

11. User Classes **13**

12. Operating Environment **14**

12.1 Client environment 14

12.2 Server environment 14

12.3 Development environment 14

13. Functional Requirements **15**

13.1 Homepage and static content 15

13.2 Image upload 15

13.3 Image conversion 16

13.4 Image preview 17

13.5 Download system 18

13.6 User interface 18

13.7 Extensibility (architecture only) 19

13.8 User workflow 19

13.9 Edge cases and alternative workflows 20

14. External Interface Requirements **22**

14.1 User interfaces 22

14.2 Hardware interfaces 22

14.3 Software interfaces 22

14.4 Communications interfaces 22

14.5 Browser platform interfaces 22

15. API Specifications **23**

15.1 Conventions 23

15.2 Error response structure 23

15.3 Endpoint summary 24

15.4 Batch and file lifecycle (conversion status handling) 24

15.5 Endpoint specifications 24

15.6 Rate limit tiers 32

15.7 Error code catalogue 32

16. Data and File Handling Requirements **34**

16.1 Data inventory and classification 34

16.2 Batch and file data model 34

16.3 File naming rules 34

16.4 Image processing rules (conversion pipeline)....34
16.5 Default limits and configuration values (single source of truth)....35
17. Non-Functional Requirements....36
17.1 Measurement basis....36
17.2 Requirements....36
18. UI/UX Specifications....39
18.1 Design direction....39
18.2 Colour palette....39
18.3 Typography....40
18.4 Spacing, radius, elevation and motion....41
18.5 Components....41
18.6 Responsive breakpoints and layout....42
18.7 Light and dark mode....43
18.8 Wireframe descriptions....43
18.9 Homepage content structure and microcopy....45
18.10 SEO and business-readiness requirements....46
19. Security and Privacy Requirements....48
19.1 Threat overview....48
19.2 Security and privacy requirements....48
19.3 Rate limiting tiers....50
19.4 Temporary storage lifecycle and cleanup....50
19.5 ZIP file handling and cleanup....51
19.6 Upload processing pipeline....51
19.7 Privacy by design....51
19.8 Logging policy....52
20. System Architecture....53
20.1 Architectural overview....53
20.2 Technology stack and rationale....53
20.3 Module responsibilities....54
20.4 Conversion sequence....55
20.5 Frontend architecture....55
20.6 Project structure....55
20.7 Purpose of key files and folders....57
21. Database and Storage Strategy....59
21.1 No database in the MVP....59
21.2 In-memory state....59
21.3 File storage layout....59
21.4 Retention policy summary....59
21.5 Capacity planning....59
21.6 Scale-out path (post-MVP)....60
22. Error Handling....61
22.1 Principles....61
22.2 Scenario catalogue....61
22.3 Error levels in the user interface....62
22.4 Server-side behaviour....62
23. Testing Strategy....63
23.1 Objectives and principles....63
23.2 Test levels....63
23.3 Test environments and CI gating....63
23.4 Test data (fixtures)....64
23.5 Defect severity and exit criteria....64
24. Test Cases....66
24.1 Unit tests....66
24.2 API tests....67
24.3 Integration tests....69
24.4 Frontend component tests....71

24.5 End-to-end tests....72
24.6 Security tests....73
24.7 Performance tests....75
24.8 Responsive tests....75
24.9 Accessibility tests....76
24.10 SEO tests....76
24.11 Usability test....77
24.12 Reviews....77
25. Deployment Requirements....78
25.1 Deployment topologies....78
25.2 Frontend hosting requirements....78
25.3 Backend hosting requirements....78
25.4 Environment configuration....79
25.5 Production build....79
25.6 Reverse proxy and platform settings....79
25.7 CORS, HTTPS and domain configuration....80
25.8 File storage....80
25.9 CI/CD pipeline....80
25.10 Logging, monitoring and alerting....81
25.11 Go-live checklist....81
26. Development Roadmap....82
26.1 Definition of done (all phases)....83
27. Acceptance Criteria....84
28. Risks and Mitigation....85
29. Future Enhancements....86
29.1 Additional conversion tools....86
29.2 Additional image tools....86
29.3 Monetisation roadmap....86
29.4 Architectural seams reserved in the MVP....87
30. Glossary....88
31. References....89
Appendix A: Consistency Review and Open Decisions....90
A.1 Automated consistency checks (run when this document was generated)....90
A.2 Issues found and resolved during the review....90
A.3 Open decisions and unresolved assumptions....91
A.4 Recommended first actions....91
Appendix B: Requirement-to-Test Traceability Matrix....92
B.1 Functional requirements....92
B.2 Security and privacy requirements....95
B.3 Non-functional requirements....95

4. Introduction

Pixora is a web-based image conversion platform. The MVP converts images from the AVIF format (AV1 Image File Format) to JPG/JPEG through a fast, private and accessible web interface that requires no account. The product is designed so that further formats and image tools can be added later and so that it can grow into a commercial image-processing SaaS.

4.1 Document conventions

- **Requirement language.** SHALL = mandatory, SHOULD = recommended, MAY = optional. Priority labels Must, Should, Could follow MoSCoW.
- **Identifiers.** `FR-nnn` functional requirement, `NFR-nnn` non-functional requirement, `SR-nnn` security and privacy requirement, `TC` test case (prefix shows type), `R-nn` risk, `A-nn` assumption, `D-nn` decision.
- **MVP versus Future.** Anything marked Future is documentation of intent only and MUST NOT be built in the MVP.
- **Defaults.** Numeric limits (file size, retention and so on) are defaults of configurable settings; Section 25.4 lists every setting. All values in this document come from one table (Section 16.5) to keep them consistent.
- **Proposed targets versus guarantees.** Measurable targets in Section 17 are engineering targets validated by tests on a stated reference environment. They are not service-level guarantees.

4.2 Overview of the document

Sections 5 to 12 define purpose, scope, context and users. Sections 13 to 16 specify what the system does (functions, interfaces, API, data handling). Sections 17 to 22 specify quality attributes, user interface, security, architecture, storage and error handling. Sections 23 to 28 cover testing, deployment, roadmap, acceptance and risks. Sections 29 to 31 cover future enhancements, glossary and references. Appendix A records the consistency review and open decisions; Appendix B is the requirement-to-test traceability matrix.

5. Purpose

This SRS defines the complete functional, non-functional, interface, security and quality requirements of the Pixora MVP. Its purposes are to:

- give developers and AI coding tools enough precision to implement the product without further clarification;
- give testers a basis for deriving and tracing tests;
- give the business a clear scope boundary, a phased plan and a monetisation path;
- document the architecture decisions and their rationale for technical review.

5.1 Business objectives

ID	Objective	How the MVP addresses it
BO-1	Provide a simple and reliable AVIF-to-JPG service	Server-side Sharp conversion with fixture-based reliability target (NFR-rel.success)
BO-2	Offer a professional user experience	Design system, premium SaaS look, light and dark themes (Section 18)
BO-3	Be fast and easy without registration	No accounts; three-step flow; performance budgets (Section 17)
BO-4	Support multiple image conversions	Batches of up to 20 files, ZIP download

ID	Objective	How the MVP addresses it
BO-5	Be optimised for Google search	Pre-rendered pages, metadata, structured data, sitemap (Section 18.10)
BO-6	Be suitable for future monetisation	Reserved layout slots, configurable limits, API-first design (Section 29)
BO-7	Create a foundation for a larger image SaaS	Converter registry and tool-driven configuration (FR ext.*)

6. Scope

6.1 In scope (MVP)

- Public website with homepage (including the converter), About, Contact, Privacy Policy and Terms pages.
- Upload of one or many AVIF files by drag-and-drop or file picker, with client and server validation.
- Server-side conversion of AVIF to JPEG with quality and background-colour options, orientation and colour handling.
- Per-file status, converted preview, individual downloads and ZIP download.
- Temporary storage with automatic deletion; no permanent storage of user images.
- REST API (`/api/v1`) supporting batch processing.
- Rate limiting, input validation, security headers, structured logging.
- SEO foundation: pre-rendered pages, metadata, sitemap, robots, structured data.
- Deployment-ready configuration (environment variables, container image, CI pipeline).

6.2 Out of scope (MVP)

- User accounts, authentication, authorisation, and any database (see decision D-01).
- Payments, subscriptions and advertising (documented as a roadmap only).
- Other conversion formats, image compression, resizing, cropping and format comparison (Future).
- Permanent storage, galleries, sharing links, or history of past conversions.
- Native mobile applications, browser extensions, public developer API keys.
- Languages other than English (strings are structured for later localisation).
- Client-side (in-browser) conversion; conversion is server-side by design (D-02).

6.3 Assumptions and constraints

ID	Assumption or constraint	Impact if wrong
A-01	The service is free to use at launch; monetisation starts after the MVP is validated.	Limits and architecture may need earlier rate/quotas.
A-02	A production domain will be registered; `https://pixora.example` is used as a placeholder in this document.	Update canonical URLs, CORS, sitemap, Open Graph.
A-03	Sharp prebuilt binaries available for the chosen OS/CPU decode AVIF (libheif/libaom) and encode JPEG (libjpeg-turbo or mozjpeg).	A different base image or a custom libvips build is required (R-07).
A-04	The MVP runs as a single backend instance with local disk and in-memory batch state.	Horizontal scaling requires the migration in Section 21.6.
A-05	Typical AVIF input is below 10 MB; the 25 MB per-file limit covers rare high-resolution files.	Adjust limits through configuration only.

ID	Assumption or constraint	Impact if wrong
A-06	Expected MVP traffic is up to a few thousand conversions per day.	Scale-out and CDN offload may be needed sooner.
A-07	Contact is provided through a published e-mail address; no contact form or mail backend.	A form would require storing personal data and spam control.
A-08	Web analytics, if used, is privacy-friendly and cookieless, and is disabled unless configured.	A consent banner may become necessary under local law.
A-09	Legal documents (Privacy Policy, Terms) will be reviewed by qualified counsel before launch; this SRS is not legal advice.	Launch blocker.
A-10	Node.js (current Active LTS), React, Vite, Tailwind and Sharp versions are pinned at Phase 1 start after checking current releases.	Version-specific instructions here may need small adjustments.

7. Definitions and Abbreviations

Domain terms are explained in the Glossary (Section 30). Abbreviations used in this document:

Abbreviation	Meaning	Abbreviation	Meaning
API	Application Programming Interface	LCP	Largest Contentful Paint
AVIF	AV1 Image File Format	MIME	Multipurpose Internet Mail Extensions
CDN	Content Delivery Network	MoSCoW	Must / Should / Could / Won't prioritisation
CI/CD	Continuous Integration / Delivery	MVP	Minimum Viable Product
CLS	Cumulative Layout Shift	NFR	Non-Functional Requirement
CORS	Cross-Origin Resource Sharing	OG	Open Graph
CSP	Content Security Policy	OWASP	Open Worldwide Application Security Project
CSPRNG	Cryptographically Secure Pseudo-Random Number Generator	REST	Representational State Transfer
E2E	End-to-end (test)	SaaS	Software as a Service
EXIF	Exchangeable Image File Format	SEO	Search Engine Optimisation
FR	Functional Requirement	SPA	Single-Page Application
HDR	High Dynamic Range	SR	Security (and privacy) Requirement
HEX	Hexadecimal colour notation	SRS	Software Requirements Specification
ICC	International Color Consortium (profile)	TC	Test Case
INP	Interaction to Next Paint	TTL	Time To Live
JPEG/JPG	Joint Photographic Experts Group format	WCAG	Web Content Accessibility Guidelines

8. Overall Description

Pixora is a stateless-feeling web tool: a visitor arrives, converts images, downloads the results and leaves. There are no accounts, no history and no permanent storage. All persistent state is limited to short-lived files and in-memory metadata that disappear after the retention period (default 30 minutes).

8.1 Design principles

Principle	Meaning for the implementation
Private by default	Originals deleted right after conversion; outputs deleted at expiry; no image data or file names in logs.
Simple first	No database, authentication, queue broker or object storage in the MVP; add only when a stated requirement demands it.
Safe with untrusted input	Content validation, resource limits, bounded queue, non-root container.
Extensible by design	Converter registry, tool identifiers in the API, config-driven UI.
Accessible and fast	WCAG 2.2 AA, Core Web Vitals budgets, pre-rendered public pages.
AI-implementable	Explicit contracts (API, error codes, limits, directory layout), unique IDs, testable criteria.

8.2 Key design decisions

ID	Decision	Rationale
D-01	No database, authentication or payments in the MVP.	Brief requires it; stateless lifecycle with TTL files is enough. Revisit when premium features start (Section 29).
D-02	Conversion happens on the server using Sharp.	Consistent output across devices, access to a robust decoder, measurable performance; avoids shipping an AV1 decoder to browsers.
D-03	The converter lives on the homepage `/` in the MVP.	Fastest path to value and strongest SEO focus on one keyword set. When a second tool launches, `/` becomes a hub and the AVIF tool moves to `/avif-to-jpg` with a 301 redirect.
D-04	Public pages are pre-rendered at build time (static HTML), then hydrated by React.	Satisfies SEO needs for a Vite SPA without moving to a different framework. Tooling is confirmed in a Phase 9 spike (R-03).
D-05	Frontend polls batch status every 1 s instead of using WebSockets or SSE.	Simplest reliable approach through proxies and CDNs; payload is tiny. SSE can replace it later.
D-06	Two-step upload then convert (`POST /batches`, `POST /batches/{id}/conversions`).	Separates validation feedback from processing, keeps requests short, and supports options in JSON.
D-07	ZIP is prepared as a temporary file, then served with Content-Length.	Detects ZIP errors before the download starts and avoids truncated downloads (Section 19.5).
D-08	Originals are deleted immediately after their conversion attempt; changing quality means re-uploading from the browser (files are still in memory there).	Minimises retention of sensitive originals; simplifies state.
D-09	Metadata (EXIF, GPS) is stripped from outputs by default.	Privacy-protective; matches what most users expect from a web converter.
D-10	npm workspaces monorepo with a shared types package.	One repository, shared API types and error codes keep frontend and backend consistent.

9. Product Perspective

Pixora is a new, self-contained product with no dependency on other Anthropic or third-party business systems. It consists of a browser-based frontend and a small HTTP API backend that share nothing except the REST contract.

9.1 System context

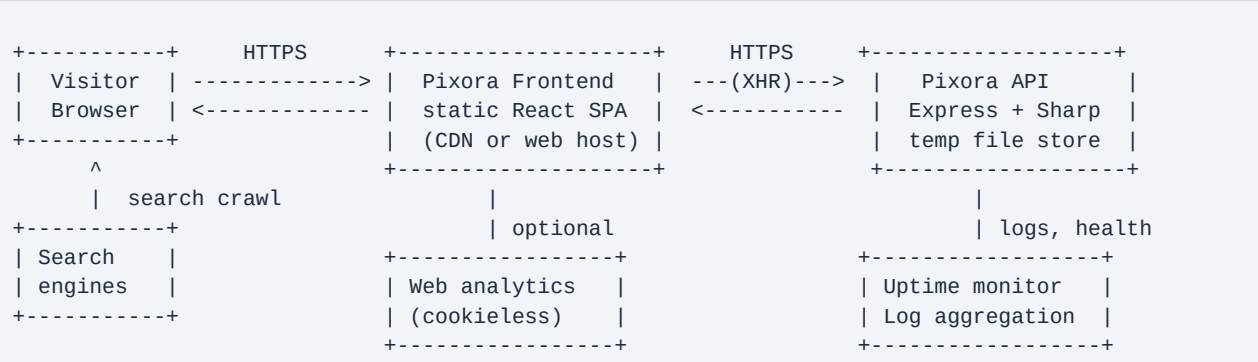


Figure 1: System context. Dashed optional components are disabled unless configured.

9.2 External dependencies

Dependency	Purpose	Required for MVP?
Static hosting / CDN	Serve built frontend and pre-rendered pages	Yes
Container or Node host with disk	Run the API with local temporary storage	Yes
DNS and TLS certificates	Domain and HTTPS	Yes
Uptime monitor	Availability checks on `/api/v1/health`	Recommended
Log aggregation	Searchable structured logs, 14-day retention	Recommended
Privacy-friendly analytics	Traffic insights without cookies	Optional (A-08)

10. Product Functions

The table summarises the major functions. Detailed requirements are in Section 13.

No.	Function	Summary	Requirement group
F1	Landing and information	Explain the service, show limits, FAQ and trust information	Homepage and static content
F2	Upload and validation	Select, review, validate and upload AVIF files	Image upload
F3	Conversion	Convert to JPEG with quality, background, orientation and colour handling	Image conversion
F4	Status and preview	Show progress, metadata and previews of results	Image preview
F5	Download	Download single files or a ZIP; delete files on demand	Download system
F6	Experience	Responsive, themed, accessible interface with clear states	User interface
F7	Housekeeping	Automatic and manual deletion of temporary data	Section 19.4
F8	Extensibility	Plug-in architecture for future converters	Extensibility

11. User Classes

Class	Description	Typical needs	Priority
UC-1 General visitor	Anyone who has an AVIF image that an app or site does not accept	Convert one file fast; no sign-up; understandable messages	Primary
UC-2 Student / professional	Needs to submit images in widely supported formats	Reliable results; batch; ZIP	Primary
UC-3 Designer / photographer	Cares about quality, colour, transparency	Quality control, sRGB, background choice, high resolution	Primary
UC-4 Web developer / marketer	Handles many assets	Batches, predictable file names, ZIP	Primary
UC-5 Small business owner	Occasional converter for catalogues and social posts	Simplicity, privacy assurance	Secondary
UC-6 Operator (deployer)	Runs and monitors the service; no admin UI in MVP	Environment configuration, logs, health, alerts	Secondary
UC-7 Search crawler	Indexes public pages	Pre-rendered content, sitemap, structured data	Secondary
UC-8 Developer / AI agent	Implements and extends the system	Unambiguous specification, modular code	Secondary

All human users are anonymous; no role-based access exists in the MVP.

12. Operating Environment

12.1 Client environment

- Browsers: latest two major versions of Chrome, Edge, Firefox, Safari (macOS, iOS 16.4 or later) and Samsung Internet.
- Devices: phones, tablets, laptops and desktops from 320 px to 1920 px width and above.
- JavaScript must be enabled for the converter; public content pages are readable without JavaScript.
- The browser is not required to decode AVIF; optional local thumbnails of originals appear only where supported.

12.2 Server environment

Item	Specification
Runtime	Node.js current Active LTS (pin exact version in `.nvmrc` and Dockerfile)
Operating system	Linux container (Debian-based `node:*-slim` recommended for Sharp prebuilt binaries); x86-64 or arm64
Reference hardware	2 vCPU, 4 GB RAM, 20 GB SSD for temporary storage (see NFR reference environment)
Process model	One Node.js process per container; supervisor or orchestrator restarts on failure
Network	HTTPS terminated by reverse proxy or platform; API reachable by frontend origin (same origin recommended)
Temporary storage	Dedicated writable directory (`STORAGE_DIR`), excluded from backups

12.3 Development environment

- Node.js and npm (workspaces), Git, GitHub, Docker.
- Test tooling: Vitest, Supertest, React Testing Library, Playwright with axe-core, k6.
- Image tooling for fixtures: an AVIF encoder (for example `avifenc`) and an EXIF utility.

13. Functional Requirements

This section contains 59 functional requirements. Every requirement is testable; the "Verified by" column lists test cases from Section 24 (Appendix B gives the full matrix). All defaults come from the configuration table in Section 16.5.

13.1 Homepage and static content

ID	Requirement	Priority	Verified by
FR-001	The homepage SHALL show a hero section with one `h1` containing the phrase "AVIF to JPG" and a sub-headline of at most 25 words that states that no registration is needed and that files are deleted automatically.	Must	E2E-15, SEO-01
FR-002	The drag-and-drop upload area SHALL be visible without scrolling on a 1280 x 720 desktop viewport, and its top edge SHALL be within the first 640 px of the page on a 360 x 800 mobile viewport.	Must	E2E-15, RSP-02
FR-003	The upload area SHALL contain a "Browse files" button that opens the operating-system file picker with `multiple` enabled and `accept=".avif,image/avif"`.	Must	FE-01
FR-004	The homepage SHALL display supported-format information: input AVIF (.avif), output JPG, maximum size per file, maximum files per batch and maximum batch size. These limit values SHALL be read from `GET /api/v1/config` and SHALL NOT be hard-coded in the frontend.	Must	API-02, E2E-15
FR-005	The homepage SHALL include a "How it works" section with exactly three steps: Upload, Convert, Download.	Must	E2E-15
FR-006	The homepage SHALL include a "Key features" section of at least six feature cards (for example: batch conversion, no sign-up, automatic deletion, quality control, transparency handling, ZIP download).	Must	E2E-15
FR-007	The homepage SHALL include an FAQ of at least eight questions using accessible disclosure widgets (native `details`/`summary`), with matching `FAQPage` structured data in the pre-rendered HTML.	Must	E2E-15, A11Y-01, SEO-01, SEO-03
FR-008	Every page SHALL include a footer with links to About, Contact, Privacy Policy and Terms, plus a copyright line.	Must	E2E-15
FR-009	The MVP SHALL provide static content pages at `/about`, `/contact`, `/privacy-policy` and `/terms`. The Contact page SHALL expose an e-mail address (mailto link) and SHALL NOT collect personal data through a form in the MVP.	Must	E2E-16
FR-010	Every page SHALL include a header with the Pixora logo (link to `/`), anchor links to How it works, Features and FAQ (on the homepage), and the theme toggle. On viewports below 768 px the links SHALL collapse into an accessible menu button.	Should	E2E-15

13.2 Image upload

ID	Requirement	Priority	Verified by
FR-011	The user SHALL be able to add AVIF files by drag-and-drop onto the upload area or through the file picker; multiple files SHALL be selectable in one action, and additional selections SHALL be appended to the current list.	Must	API-03, API-04, FE-01, E2E-01, E2E-02
FR-012	The upload area SHALL provide a distinct drag-over visual state, SHALL be operable by keyboard (Enter and Space open the picker), and SHALL ignore dropped content that is not a file (for example selected text).	Must	A11Y-02

ID	Requirement	Priority	Verified by
FR-013	Each selected file SHALL be listed with its file name (truncated with an ellipsis when too long, full name available to assistive technology), its size in human-readable units with one decimal place (KB or MB), and its client-side validation state.	Must	UT-07, FE-01
FR-014	The frontend SHALL pre-validate selections for UX purposes only: extension `.avif` (case-insensitive) or MIME `image/avif`, per-file size, file count, total batch size, and duplicates (same name, size and last-modified time). Rejected files SHALL NOT be added and SHALL produce an inline message naming the file and reason.	Must	UT-08, FE-02, E2E-12
FR-015	The backend SHALL validate the actual content of every uploaded file (ISO-BMFF `ftyp` signature with an AVIF brand and a successful decoder metadata read reporting an AV1-coded image) independently of file extension and declared MIME type.	Must	UT-03, UT-04, API-03, API-05, SEC-01
FR-016	The backend SHALL reject unsupported or corrupted files individually with a machine-readable error code and a user-safe message; valid files in the same request SHALL continue to be processed. If no file in a request is valid, the request SHALL fail with `NO_VALID_FILES`.	Must	UT-04, API-05, API-07, INT-10, E2E-03
FR-017	The user SHALL be able to remove any single file, or all files ("Clear all"), from the selection before conversion starts. Removal SHALL be immediate and SHALL NOT require a server call.	Must	FE-03
FR-018	Upload limits SHALL be configurable through environment variables with these defaults: 25 MB per file, 20 files per batch, 150 MB per batch, 50 megapixels (50,000,000 pixels) per image.	Must	UT-08, UT-12, API-02, API-08, API-09, E2E-12, SEC-05
FR-019	While files are uploading, the UI SHALL show overall upload progress (percentage and bytes sent) and SHALL allow the user to cancel the upload; cancellation SHALL abort the request and leave the selection intact.	Should	E2E-05
FR-020	The "Convert" button SHALL be disabled while the selection is empty, and an empty submission attempted by any other means SHALL produce the `NO_FILES_PROVIDED` error.	Must	API-06, FE-04

13.3 Image conversion

ID	Requirement	Priority	Verified by
FR-021	The system SHALL convert each accepted AVIF image to a JPEG file (`image/jpeg`, extension `.jpg`) using server-side processing with the Sharp library.	Must	API-12, INT-01, E2E-01
FR-022	By default the output pixel dimensions SHALL equal the oriented input dimensions (no resampling). No resize option exists in the MVP.	Must	INT-01, E2E-10
FR-023	The JPEG quality SHALL be an integer from 1 to 100, defaulting to 90 (environment variable `DEFAULT_JPEG_QUALITY`). Chroma subsampling SHALL be 4:4:4 when quality is 90 or higher and 4:2:0 otherwise.	Must	UT-05, API-12, API-13, INT-04
FR-024	Images that contain an alpha channel SHALL be flattened onto a solid background colour supplied as a six-digit HEX string (pattern `^#[0-9A-Fa-f]{6}$`), defaulting to #FFFFFF. Images without alpha SHALL be unaffected by this option.	Must	UT-05, API-13, INT-02, E2E-04
FR-025	Image orientation SHALL be applied to the pixel data, honouring both AVIF container transforms (`irot`, `imir`) and EXIF Orientation values 1 to 8, so the output visually matches the intended display orientation. The output SHALL NOT carry a non-default orientation tag.	Must	INT-03

ID	Requirement	Priority	Verified by
FR-026	Output SHALL be 8-bit sRGB. Embedded ICC profiles in the input SHALL be applied when converting to sRGB.	Must	INT-01, INT-16
FR-027	By default, EXIF, XMP and embedded thumbnails (including GPS location data) SHALL be removed from the output. There is no user option to retain metadata in the MVP.	Must	INT-13
FR-028	Batches SHALL be processed with bounded concurrency (default 2 simultaneous conversions, environment variable `CONVERSION_CONCURRENCY`) in first-in-first-out order. The failure of one file SHALL NOT stop or alter the processing of other files.	Must	UT-11, API-04, INT-10, INT-17, E2E-02
FR-029	Each file SHALL expose one of the statuses `uploaded`, `queued`, `processing`, `completed`, `failed` or `rejected`, and the UI SHALL reflect status changes within 2000 ms of the server state changing under normal network conditions.	Must	API-14, API-15, FE-05, E2E-06, E2E-17
FR-030	The output file SHALL be written to a temporary name and atomically renamed only after the encoder finishes successfully. Preview and download SHALL be available only for files in status `completed`, so that incomplete files are never served.	Must	API-17, INT-05
FR-031	Every conversion failure SHALL be caught, mapped to an error code from the error catalogue, recorded in the file result, and its partial output removed. Stack traces, file-system paths and library messages SHALL NOT be exposed to the client.	Must	UT-09, API-24, INT-05, INT-10, SEC-11
FR-032	For animated or multi-image AVIF (image sequences) the system SHALL output a single JPEG of the primary (first) image and SHALL attach the warning code `ANIMATION_FIRST_FRAME_ONLY` to the file result. The implementation SHALL be confirmed by a spike in Phase 4 (see risk R-09).	Should	INT-12
FR-033	AVIF files with 10-bit or 12-bit depth, including wide-gamut or HDR signalling, SHALL convert without error to 8-bit sRGB. Tone-mapping fidelity for HDR content is not guaranteed in the MVP (see risk R-08).	Must	INT-11
FR-034	A conversion that exceeds 60 seconds (environment variable `CONVERSION_TIMEOUT_SECONDS`) SHALL be marked `failed` with code `CONVERSION_TIMEOUT`, and any result produced later SHALL be discarded.	Must	INT-14
FR-035	The UI SHALL provide an "Advanced settings" panel with a quality control (presets High 90, Balanced 80, Small 70, and a 1 to 100 slider) and a background-colour control (White, Black, custom HEX). Settings apply to the whole batch and are sent when conversion starts.	Must	API-12, FE-12, E2E-04

13.4 Image preview

ID	Requirement	Priority	Verified by
FR-036	For each file the results view SHALL show: file name, original size, original dimensions, output size, output dimensions, size change in percent, and a status badge.	Must	API-15, FE-06, E2E-01
FR-037	For each completed file the results view SHALL show a thumbnail of the converted JPEG loaded from the preview endpoint, lazily loaded, with a skeleton placeholder while loading and descriptive `alt` text containing the file name.	Must	API-17, FE-06
FR-038	Activating a thumbnail SHALL open an accessible modal preview (native `dialog` element) showing the converted image at "fit to screen" with a toggle for 100% size; the dialog SHALL trap focus, close on Escape, and return focus to the trigger.	Must	FE-08

ID	Requirement	Priority	Verified by
FR-039	When the user's browser can decode AVIF, the file list MAY show a small client-side thumbnail of the original; otherwise a neutral placeholder SHALL be shown. Original thumbnails SHALL be generated locally (object URLs) and never uploaded separately.	Could	FE-14
FR-040	For a failed or rejected file the card SHALL show the user-facing error message for its error code and SHALL NOT show a download button.	Must	FE-06, E2E-03

13.5 Download system

ID	Requirement	Priority	Verified by
FR-041	Each completed file SHALL have a "Download JPG" action that downloads that single file with `Content-Disposition: attachment`.	Must	API-18, E2E-01
FR-042	When at least one file is completed, a "Download all (ZIP)" action SHALL be available. It SHALL first request ZIP preparation (`POST .../archive`), and on success start the download of the prepared archive containing only completed files.	Must	API-19, API-20, E2E-02, E2E-03, E2E-13
FR-043	Output file names SHALL be `<sanitised original base name>.jpg`. Name collisions within a batch (case-insensitive) SHALL be resolved deterministically in upload order by appending `-1`, `-2`, ... The same names SHALL be used for individual downloads and ZIP entries. The ZIP file SHALL be named `pixora-converted-YYYYMMDD-HHmmss.zip`.	Must	UT-01, UT-02, API-19, E2E-02, E2E-09
FR-044	Before starting any download the frontend SHALL verify availability with a `HEAD` request. On success it SHALL start the download and show a "Download started" notification; on failure it SHALL show the error message for the returned error code (for example expired files).	Must	API-18, FE-13, E2E-01, E2E-07
FR-045	The results view SHALL show a countdown to the batch expiry (computed from `expiresAt`), SHALL warn when fewer than 5 minutes remain, and SHALL replace the results with an "expired" state when time runs out.	Must	UT-10, API-16, INT-06, FE-10, E2E-07
FR-046	The results view SHALL provide a "Delete my files now" action that calls `DELETE /api/v1/batches/{batchId}` and, when successful, confirms deletion to the user and returns to the empty upload state.	Must	API-21, E2E-08
FR-047	The frontend MAY store the opaque `batchId` in `sessionStorage` so that a page refresh during processing can resume the results view until the batch expires. No file names or image data SHALL be stored in the browser.	Could	E2E-11

13.6 User interface

ID	Requirement	Priority	Verified by
FR-048	The interface SHALL be fully usable from 320 px to 1920 px viewport width with no horizontal page scrolling, using the breakpoints defined in Section 18.	Must	RSP-01, RSP-02, RSP-03, RSP-04
FR-049	The interface SHALL support light and dark themes, default to the operating-system preference, allow manual override through a toggle, persist the choice in `localStorage` (key `pixora-theme`), and apply the theme before first paint to avoid a flash of the wrong theme.	Must	FE-07, E2E-14, A11Y-04
FR-050	Animations SHALL last no longer than 300 ms, SHALL animate only `opacity` and `transform` where possible, and SHALL be disabled or reduced when the user prefers reduced motion.	Must	E2E-14, A11Y-05

ID	Requirement	Priority	Verified by
FR-051	The UI SHALL show loading indicators for: uploading (progress bar), processing (per-file spinner and an overall progress bar), ZIP preparation (button spinner), and thumbnails (skeleton).	Must	FE-05, E2E-17
FR-052	The UI SHALL provide a designed empty state for the upload area (no files selected) and for the results area (all files deleted or expired).	Must	FE-03, E2E-08
FR-053	The UI SHALL provide distinct error states at three levels: per file (inline), per batch (banner with retry action), and application (full-page fallback with a reload action).	Must	FE-09, E2E-03, E2E-06, E2E-13
FR-054	Transient feedback SHALL be shown as toasts: success and info toasts auto-dismiss after 5 s, error toasts after 8 s or on dismissal; toasts SHALL be announced through an ARIA live region (`polite` for success and info, `assertive` for errors).	Must	FE-11, A11Y-03
FR-055	The UI SHALL conform to WCAG 2.2 Level AA: every control keyboard-operable, visible focus indicators, programmatic labels, status changes announced to screen readers, and the contrast ratios defined in Section 18.	Must	FE-08, A11Y-01, A11Y-02, A11Y-03, A11Y-04
FR-056	All user-facing strings SHALL be defined in a central module (`content/strings.ts`) so that localisation can be added later. The MVP language is English.	Should	UT-14

13.7 Extensibility (architecture only)

ID	Requirement	Priority	Verified by
FR-057	The backend SHALL implement conversions behind an `ImageConverter` interface and a `ConverterRegistry`, so that a new tool (for example `png-to-jpg`) can be added by registering a converter, its option schema and its input validator, without modifying the upload, storage, queue, archive or cleanup modules.	Must	UT-13, REV-01
FR-058	Batches SHALL carry a `tool` identifier (MVP value: `avif-to-jpg`). The API SHALL reject unknown tool values with `UNSUPPORTED_TOOL`, and the frontend SHALL read tool metadata (labels, accepted extensions, output extension) from configuration rather than literals.	Must	UT-13, API-11
FR-059	The architecture SHALL NOT preclude the future features listed in Section 29 (additional formats, compression, resizing, cropping, comparison). These features are explicitly NOT part of the MVP and SHALL NOT be implemented in it.	Should	REV-01

13.8 User workflow

The complete journey from first visit to automatic deletion:

Step	User action	System behaviour	Notes and requirement links
1	Opens Pixora	Static pre-rendered page loads; the app fetches `GET /api/v1/config`; limits and defaults appear in the format panel	Hero, upload area, limits from config
2	Selects AVIF files or drags them in	Browser adds files to the selection; drag-over state is shown	Multiple files, appended selections

Step	User action	System behaviour	Notes and requirement links
3	Application validates selected files	Client-side checks of extension, size, count, total size, duplicates; invalid files are skipped with a message	UX checks only; server re-validates
4	Reviews the list	List shows name, size, remove buttons; Advanced settings are available; Convert becomes enabled	Remove one or all
5	Clicks Convert	Frontend uploads files (`POST /batches`) with progress, then starts the conversion (`POST /batches/{id}/conversions`)	Server validates content per file; invalid files rejected individually
6	Waits	Backend queues and converts with bounded concurrency; each original is deleted after its attempt	Frontend polls `GET /batches/{id}` every 1 s
7	Sees progress and results	Per-file statuses and overall progress bar; cards update to Completed or Failed	Metadata, sizes, dimensions shown
8	Previews converted files	Thumbnails load from the preview endpoint; click opens lightbox	Original info is shown beside converted info
9	Downloads	Single download (HEAD check then download) or ZIP (prepare then download)	Names are sanitised and unique
10	Leaves, or clicks Delete my files now	Outputs are removed at expiry (default 30 minutes), or immediately on request; sweepers guarantee cleanup after crashes	Countdown shown; expired state after deadline

13.9 Edge cases and alternative workflows

ID	Situation	Expected behaviour
EC-01	User selects the same file twice	Second copy is skipped with a notice ("already added").
EC-02	User selects 25 files when the limit is 20	First 20 are added; the rest are refused with one summary message naming the limit.
EC-03	One file in a batch is corrupt	Other files convert; the corrupt file shows its error; ZIP contains only successful files; batch is `partially_completed`.
EC-04	All files are invalid after upload	API returns `NO_VALID_FILES`; UI shows a batch-level error and keeps the selection so the user can fix it.
EC-05	User closes the tab during processing	Server completes processing and deletes everything at expiry; nothing is stored beyond retention.
EC-06	User refreshes during processing	Without the optional resume feature the results are lost and expire server-side; with it (FR dld.resume) the view resumes.
EC-07	User wants a different quality after seeing results	User opens settings and clicks Convert again; the browser still holds the original files and re-uploads them as a new batch (D-08).
EC-08	Connection drops during upload	Upload fails with a network error; selection is kept; user can retry; the server deletes the partial upload.
EC-09	Connection drops during processing or polling	UI shows "Reconnecting"; polling retries with backoff (1, 2, 4, 8 s) for up to 60 s; then offers manual retry; batch remains valid until expiry.
EC-10	Download clicked after expiry	HEAD returns 410; UI shows the expired message and the Start over button.
EC-11	Two files have the same name	Output names get `-1`, `-2` suffixes in upload order.
EC-12	File name contains emoji, RTL text or path characters	Display name is sanitised and shown safely; stored under a random ID; download name uses the sanitised name.

ID	Situation	Expected behaviour
EC-13	Image with transparency	Flattened on the chosen background (default white); an informational note is shown on the card ("Transparency replaced by background").
EC-14	Very large image (above pixel limit)	Rejected with `IMAGE_DIMENSIONS_EXCEEDED` before decoding; message states the maximum.
EC-15	Server is busy or storage is low	API returns 503 with `Retry-After`; UI shows a friendly retry message; selection kept.
EC-16	User opens two tabs	Each tab has its own batch; rate limits are per IP; both are independent.
EC-17	JavaScript disabled	Public content remains readable; a `noscript` message explains that JavaScript is required for conversion.

14. External Interface Requirements

14.1 User interfaces

The user interface follows Section 18. Pages: Home (`/`), About (`/about`), Contact (`/contact`), Privacy Policy (`/privacy-policy`), Terms (`/terms`) and a 404 page. The converter is a single-page experience on `/` with these views: empty, files selected, uploading, processing, results, expired and error.

14.2 Hardware interfaces

None. The system uses no special hardware. Server file storage is a local disk or mounted volume.

14.3 Software interfaces

Component	Version policy	Interface use
Sharp (libvips, libheif, libaom)	Pinned; updated for security fixes	AVIF decode, orientation, flatten, colour conversion, JPEG encode, metadata read
Multer 2.x	Pinned; maintained major only	Multipart parsing with disk storage in per-batch temporary directories
Archiver (or equivalent)	Pinned	ZIP creation with store-only (level 0) compression
Express 5	Pinned	HTTP routing and middleware
Zod	Pinned	Request validation and shared schemas
Pino	Pinned	Structured JSON logging with redaction
express-rate-limit, helmet, cors	Pinned	Rate limiting, security headers, CORS

> **Version note** Library majors named here reflect current practice at the time of writing; confirm the latest stable releases and their security advisories in Phase 1 (assumption A-10).

14.4 Communications interfaces

- HTTPS only in production; HTTP redirected to HTTPS. HTTP/1.1 and HTTP/2 supported by the proxy.
- REST over JSON for control messages; `multipart/form-data` for uploads; binary responses for previews, downloads and ZIP.
- Status updates by polling (D-05). Default interval 1 s, advertised by the server as `pollIntervalMs`.
- Request IDs: clients MAY send `X-Request-Id` (pattern `^[A-Za-z0-9-]{8,64}$`); the server always returns one.
- Time values use ISO 8601 UTC. Sizes use bytes. Identifiers are opaque strings.

14.5 Browser platform interfaces

- File API and Drag-and-Drop API for selecting files; `XMLHttpRequest` for upload progress; `fetch` for other calls.
- `dialog` element for the lightbox; `details` element for FAQ.
- `localStorage` for the theme and `sessionStorage` for the optional resume feature.
- `URL.createObjectURL` for optional local thumbnails of originals.

15. API Specifications

The API is a JSON/REST interface versioned under `/api/v1`. It is designed for batch processing: one batch groups up to 20 files, which are uploaded together, converted together and downloaded individually or as one ZIP.

15.1 Conventions

Topic	Rule
Base URL	`/api/v1` on the API origin (same origin as the site when deployed behind one proxy, otherwise `https://api.pixora.example`)
Identifiers	Batch ID `bch_` + 32 lowercase hex characters (128-bit random). File ID `fil_` + 16 lowercase hex characters. IDs are opaque; clients MUST NOT parse them.
Content types	Requests: `multipart/form-data` (upload) or `application/json` (others). Responses: `application/json; charset=utf-8`, `image/jpeg`, `application/zip`.
Caching	All API responses carry `Cache-Control: no-store`.
Request ID	Every response has `X-Request-Id`. Errors repeat it in the body.
Rate limiting	Standard `RateLimit-*` headers on every response; `429` includes `Retry-After` (seconds).
Idempotency	`GET`, `HEAD`, `DELETE` and `POST .../archive` are idempotent. `POST /batches` and `POST .../conversions` are not.
Unknown properties	JSON bodies with unknown properties are rejected with `VALIDATION_ERROR`.
Authentication	None in the MVP. Access is by possession of the unguessable batch ID (capability URL), see SR ids.
Documentation	An OpenAPI 3.1 document (`apps/api/openapi.yaml`) mirrors this section and is checked by a contract test.

15.2 Error response structure

All errors, including framework-level ones, use this structure:

```
{
  "error": {
    "code": "FILE_TOO_LARGE",
    "message": "\"scan.avif\" is 31.2 MB, which is over the 25 MB limit per file.",
    "details": [ { "field": "files", "fileName": "scan.avif", "limitBytes": 26214400 } ],
    "requestId": "9d3c6a0e-7e11-4b38-a3a0-5c2d9f1b6f52"
  }
}
```

Field	Type	Description
`error.code`	string	Stable machine-readable code from the catalogue in Section 15.7
`error.message`	string	User-safe English message; MAY include the sanitised file name; never includes paths or internals
`error.details`	array (optional)	Structured hints such as invalid fields or limits
`error.requestId`	string	Correlation ID, equal to the `X-Request-Id` header

15.3 Endpoint summary

No.	Method and URL	Purpose	Success
1	GET /api/v1/health	Liveness and storage health	200
2	GET /api/v1/config	Public limits and defaults	200
3	POST /api/v1/batches	Upload files; validate content; create batch	201
4	POST /api/v1/batches/{batchId}/conversions	Start conversion with options	202
5	GET /api/v1/batches/{batchId}	Batch and per-file status	200
6	GET /api/v1/batches/{batchId}/files/{fileId}/preview	Inline preview of converted JPEG	200
7	GET /api/v1/batches/{batchId}/files/{fileId}/download	Download one JPEG	200
8	POST /api/v1/batches/{batchId}/archive	Prepare ZIP of completed files	200
9	GET /api/v1/batches/{batchId}/archive	Download prepared ZIP	200
10	DELETE /api/v1/batches/{batchId}	Delete batch and all files now	204

15.4 Batch and file lifecycle (conversion status handling)

Object	Status	Meaning	Next
File	`rejected`	Failed content validation at upload; not stored	Terminal
File	`uploaded`	Stored and validated; waiting for conversion to be requested	`queued`
File	`queued`	Conversion requested; waiting for a free worker slot	`processing`
File	`processing`	Being converted	`completed` or `failed`
File	`completed`	Output exists, is finalised, and can be previewed and downloaded	Terminal
File	`failed`	Conversion failed; `error` explains why	Terminal
Batch	`uploaded`	Created; conversion not started	`processing`
Batch	`processing`	At least one accepted file is `queued` or `processing`	Terminal state below
Batch	`completed`	All accepted files completed	Terminal
Batch	`partially_completed`	At least one completed and at least one failed	Terminal
Batch	`failed`	All accepted files failed	Terminal

Rejected files appear in the batch `files` list for display but do not count towards the batch status. After `expiresAt` (or `DELETE`) every endpoint except `config` and `health` returns `410 BATCH_EXPIRED` for 24 hours (tombstone), then `404 BATCH_NOT_FOUND`.

15.5 Endpoint specifications

15.5.1 GET /api/v1/health

Used by monitors and orchestrators. Performs a cheap writable-storage check; does not touch image libraries.

Request parameters	None

Request body	None
Headers	`Accept: application/json` (optional)
Response	`status` ("ok" or "unavailable"), `version` (application version), `uptimeSeconds`, `storage` ("writable" or "unavailable"), `time`
Status codes	200 healthy; 503 when storage is unavailable
Validation	None

Example response

```
HTTP/1.1 200 OK
{ "status": "ok", "version": "1.0.0", "uptimeSeconds": 86412, "storage": "writable", "time":
"2026-11-01T10:30:00.000Z" }
```

15.5.2 GET /api/v1/config

Returns public limits and defaults so the UI never hard-codes them.

Request parameters	None
Request body	None
Headers	None required
Response	`tools[]` (id, label, inputExtensions, inputMimeTypes, outputExtension), `limits`, `defaults`, `quality` range, `retentionMinutes`
Status codes	200
Validation	None

Example response

```
{
  "tools": [ { "id": "avif-to-jpg", "label": "AVIF to JPG", "inputExtensions": [".avif"],
               "inputMimeTypes": ["image/avif"], "outputExtension": ".jpg" } ],
  "limits": { "maxFileSizeBytes": 26214400, "maxFilesPerBatch": 20,
              "maxBatchSizeBytes": 157286400, "maxPixels": 50000000 },
  "defaults": { "quality": 90, "background": "#FFFFFF" },
  "quality": { "min": 1, "max": 100 },
  "retentionMinutes": 30
}
```

15.5.3 POST /api/v1/batches

Uploads one or more files, validates the content of each, stores valid files in a new batch and returns the batch. Conversion does not start yet.

Request parameters	None (query string ignored)
Request body	`multipart/form-data`. Repeated file field `files` (1 to max files). Optional text field `tool` (default `avif-to-jpg`, max 64 bytes).
Headers	`Content-Type: multipart/form-data; boundary=...` (required), `Content-Length` (required; checked against the batch limit before streaming)
Response	Batch object (see the example under endpoint 5) with status `uploaded`, `counts`, and `files[]` in upload order including `rejected` entries

Status codes	201 at least one valid file; 400 `NO_FILES_PROVIDED` / `UNSUPPORTED_TOOL` / `VALIDATION_ERROR`; 413 `FILE_TOO_LARGE` / `BATCH_TOO_LARGE`; 415; 422 `TOO_MANY_FILES` / `NO_VALID_FILES`; 429; 503
Validation rules	Per file: size <= 25 MB; content passes the AVIF signature and decoder-metadata check; pixels <= 50,000,000. Per request: files <= 20; total <= 150 MB; only the fields `files` and `tool`.
Failure mode	Size and count limit violations abort the whole request (multipart streaming limitation), and partial files are deleted. Content failures reject only the affected file.

Example request

```
POST /api/v1/batches HTTP/1.1
Content-Type: multipart/form-data; boundary=----pixora
Content-Length: 832801

------pixora
Content-Disposition: form-data; name="files"; filename="holiday photo.avif"
Content-Type: image/avif
<binary>
------pixora
Content-Disposition: form-data; name="files"; filename="notes.avif"
Content-Type: image/avif
<binary>
------pixora--
```

Example response

```
HTTP/1.1 201 Created
Location: /api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7
X-Request-Id: 9d3c6a0e-7e11-4b38-a3a0-5c2d9f1b6f52

{
  "batchId": "bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7",
  "tool": "avif-to-jpg",
  "status": "uploaded",
  "createdAt": "2026-11-01T10:30:00.000Z",
  "expiresAt": "2026-11-01T11:00:00.000Z",
  "pollIntervalMs": 1000,
  "options": null,
  "counts": { "total": 2, "accepted": 1, "rejected": 1, "queued": 0, "processing": 0, "completed":
0, "failed": 0 },
  "files": [
    {
      "fileId": "fil_a1b2c3d4e5f60718", "originalName": "holiday photo.avif", "status":
"uploaded",
      "original": { "sizeBytes": 812345, "width": 4000, "height": 3000, "hasAlpha": false,
"bitDepth": 8 },
      "output": null, "warnings": [], "error": null
    },
    {
      "fileId": "fil_9a8b7c6d5e4f3021",
      "originalName": "notes.avif",
      "status": "rejected",
      "original": null,
      "output": null,
      "warnings": [],
      "error": { "code": "UNSUPPORTED_FILE_TYPE", "message": "This file is not an AVIF image." }
    }
  ]
}
```

Error responses

Status	Error code	When
400	`NO_FILES_PROVIDED`	Request contains no `files` part
413	`FILE_TOO_LARGE`	A file exceeds 25 MB
413	`BATCH_TOO_LARGE`	Total exceeds 150 MB
422	`TOO_MANY_FILES`	More than 20 files
422	`NO_VALID_FILES`	Every file was rejected by content validation
415	`UNSUPPORTED_MEDIA_TYPE`	Not multipart/form-data

15.5.4 POST /api/v1/batches/{batchId}/conversions

Starts conversion of all accepted files in the batch with the given options. Returns immediately; processing is asynchronous.

Request parameters	Path `batchId` (pattern `^bch_[a-f0-9]{32}$`)
Request body	JSON `{ "options": { "quality": 1-100, "background": "#RRGGBB" } }`. `options` and both properties are optional; defaults come from config.
Headers	`Content-Type: application/json` (required if a body is sent)
Response	Batch object with status `processing`, files `queued` or `processing`, and `options` as applied
Status codes	202 accepted; 400 `VALIDATION_ERROR`; 404 `BATCH_NOT_FOUND`; 409 `BATCH_STATE_CONFLICT` (conversion already requested); 410 `BATCH_EXPIRED`; 415; 429; 503 `SERVICE_BUSY`
Validation rules	Quality is an integer 1 to 100. Background matches `^#[0-9A-Fa-f]{6}$`. Unknown properties rejected. Batch must be in status `uploaded`.

Example request

```
POST /api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7/conversions HTTP/1.1
Content-Type: application/json

{ "options": { "quality": 90, "background": "#FFFFFF" } }
```

Example response

```
HTTP/1.1 202 Accepted
Location: /api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7

{
  "batchId": "bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7",
  "tool": "avif-to-jpg",
  "status": "processing",
  "createdAt": "2026-11-01T10:30:00.000Z",
  "expiresAt": "2026-11-01T11:00:00.000Z",
  "pollIntervalMs": 1000,
  "options": { "quality": 90, "background": "#FFFFFF" },
  "counts": { "total": 1, "accepted": 1, "rejected": 0, "queued": 0, "processing": 1, "completed":
0, "failed": 0 },
  "files": [
    {
      "fileId": "fil_a1b2c3d4e5f60718", "originalName": "holiday photo.avif", "status":
"processing",
      "original": { "sizeBytes": 812345, "width": 4000, "height": 3000, "hasAlpha": false,
"bitDepth": 8 },
      "output": null, "warnings": [], "error": null
    }
  ]
}
```

15.5.5 GET /api/v1/batches/{batchId}

Returns the current state of the batch and every file. Clients poll this endpoint until the batch status is terminal.

Request parameters	Path `batchId`
Request body	None
Headers	None required
Response	Batch object. Completed files include `output` with `previewUrl` and `downloadUrl`. Failed and rejected files include `error`. A completed batch with an archive prepared includes `archive`.
Status codes	200; 400 invalid ID; 404; 410; 429
Validation	ID pattern

Example response

```
HTTP/1.1 200 OK

{
  "batchId": "bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7",
  "tool": "avif-to-jpg",
  "status": "completed",
  "createdAt": "2026-11-01T10:30:00.000Z",
  "expiresAt": "2026-11-01T11:00:00.000Z",
  "pollIntervalMs": 1000,
  "options": { "quality": 90, "background": "#FFFFFF" },
  "counts": { "total": 3, "accepted": 2, "rejected": 1, "queued": 0, "processing": 0, "completed":
2, "failed": 0 },
  "files": [
    {
      "fileId": "fil_a1b2c3d4e5f60718",
      "originalName": "holiday photo.avif",
      "status": "completed",
      "original": { "sizeBytes": 812345, "width": 4000, "height": 3000, "hasAlpha": false,
"bitDepth": 8 },
      "output": {
        "filename": "holiday photo.jpg", "mimeType": "image/jpeg", "sizeBytes": 1450221,
        "width": 4000, "height": 3000,
        "previewUrl":
"/api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7/files/fil_a1b2c3d4e5f60718/preview",
        "downloadUrl":
"/api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7/files/fil_a1b2c3d4e5f60718/download"
      },
      "warnings": [],
      "error": null
    },
    {
      "fileId": "fil_0f1e2d3c4b5a6978",
      "originalName": "logo.avif",
      "status": "completed",
      "original": { "sizeBytes": 20480, "width": 512, "height": 512, "hasAlpha": true, "bitDepth":
8 },
      "output": {
        "filename": "logo.jpg", "mimeType": "image/jpeg", "sizeBytes": 31870,
        "width": 512, "height": 512,
        "previewUrl":
"/api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7/files/fil_0f1e2d3c4b5a6978/preview",
        "downloadUrl":
"/api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7/files/fil_0f1e2d3c4b5a6978/download"
      },
      "warnings": [],
      "error": null
    },
    {
      "fileId": "fil_9a8b7c6d5e4f3021",
      "originalName": "notes.avif",
      "status": "rejected",
      "original": null,
      "output": null,
      "warnings": [],
      "error": { "code": "UNSUPPORTED_FILE_TYPE", "message": "This file is not an AVIF image." }
    }
  ]
}
```

15.5.6 GET /api/v1/batches/{batchId}/files/{fileId}/preview

Streams the converted JPEG for inline display. Never serves a file that is not `completed`.

Request parameters	Path `batchId`, `fileId` (pattern `^fil_[a-f0-9]{16}$`)
Request body	None
Headers	None required; `HEAD` supported
Response	Binary `image/jpeg` with `Content-Disposition: inline`, `Content-Length`, `Cache-Control: private, no-store`, `X-Content-Type-Options: nosniff`
Status codes	200; 400; 404 `BATCH_NOT_FOUND` / `FILE_NOT_FOUND`; 409 `FILE_NOT_READY`; 410; 429
Validation	ID patterns; file must belong to batch and be `completed`

Example response

```
HTTP/1.1 200 OK
Content-Type: image/jpeg
Content-Length: 1450221
Content-Disposition: inline
Cache-Control: private, no-store
<JPEG bytes>
```

15.5.7 GET /api/v1/batches/{batchId}/files/{fileId}/download

Downloads one converted JPEG as an attachment. The frontend sends `HEAD` first to verify availability (FR dld.notify).

Request parameters	Path `batchId`, `fileId`
Request body	None
Headers	None required; `HEAD` supported
Response	Binary `image/jpeg` with `Content-Disposition: attachment; filename="<ascii-fallback>"; filename*=UTF-8''<percent-encoded>`
Status codes	200; 400; 404; 409 `FILE_NOT_READY`; 410; 429
Validation	As endpoint 6

Example response

```
HTTP/1.1 200 OK
Content-Type: image/jpeg
Content-Length: 1450221
Content-Disposition: attachment; filename="holiday photo.jpg"; filename*=UTF-8''holiday
%20photo.jpg
```

15.5.8 POST /api/v1/batches/{batchId}/archive

Creates (or returns the already created) ZIP of all completed files. Generation uses store-only compression and writes to a temporary file that is renamed on success. The call is synchronous and typically takes under 10 s for 100 MB.

Request parameters	Path `batchId`
Request body	None
Headers	None required
Response	`archive` object: `filename`, `sizeBytes`, `fileCount`, `downloadUrl`, `createdAt`

Status codes	200; 400; 404; 409 `NO_COMPLETED_FILES` / `BATCH_STATE_CONFLICT` (still processing); 410; 429; 500 `ARCHIVE_FAILED`; 503 `STORAGE_UNAVAILABLE`
Validation	Batch status must be terminal (`completed`, `partially_completed`); at least one completed file

Example response

```
HTTP/1.1 200 OK
{ "archive": { "filename": "pixora-converted-20261101-103145.zip", "sizeBytes": 1482091,
"fileCount": 2,
              "downloadUrl": "/api/v1/batches/bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7/archive",
"createdAt": "2026-11-01T10:31:45.000Z" } }
```

15.5.9 GET /api/v1/batches/{batchId}/archive

Downloads the prepared ZIP. Served from the finished temporary file with `Content-Length`, so interrupted transfers are detectable by the browser.

Request parameters	Path `batchId`
Request body	None
Headers	None required; `HEAD` supported
Response	Binary `application/zip` with `Content-Disposition: attachment; filename="pixora-converted-YYYYMMDD-HHmmss.zip"`
Status codes	200; 400; 404; 409 `ARCHIVE_NOT_PREPARED`; 410; 429
Validation	Archive must exist

Example response

```
HTTP/1.1 200 OK
Content-Type: application/zip
Content-Length: 1482091
Content-Disposition: attachment; filename="pixora-converted-20261101-103145.zip"
```

15.5.10 DELETE /api/v1/batches/{batchId}

Deletes all files, the ZIP and metadata of the batch immediately. Idempotent: repeated calls return 204 while the tombstone exists.

Request parameters	Path `batchId`
Request body	None
Headers	None required
Response	Empty body
Status codes	204; 400; 404; 429
Validation	ID pattern

Example response

```
HTTP/1.1 204 No Content
```

15.6 Rate limit tiers

The tiers are applied per client IP and are configurable (Section 25.4). They are intentionally generous for normal use and restrictive for automated abuse.

Scope	Default limit	Environment variables
All `/api` requests	300 per minute	`RATE_LIMIT_GLOBAL_PER_MIN`
`POST /batches` (upload)	20 per 10 minutes	`RATE_LIMIT_UPLOAD_PER_10MIN`
`POST .../conversions` and `POST .../archive`	40 per 10 minutes	`RATE_LIMIT_ACTION_PER_10MIN`
`GET /batches/{id}` (status polling)	120 per minute	`RATE_LIMIT_STATUS_PER_MIN`
Preview and download endpoints	300 per 10 minutes	`RATE_LIMIT_FILES_PER_10MIN`

15.7 Error code catalogue

Code	HTTP	Used by	Meaning
`VALIDATION_ERROR`	400	All	Malformed ID, JSON, options or field
`NO_FILES_PROVIDED`	400	Upload	No `files` part
`UNSUPPORTED_TOOL`	400	Upload	Unknown `tool` value
`BATCH_NOT_FOUND`	404	Batch routes	Unknown batch (or expired longer than tombstone TTL)
`FILE_NOT_FOUND`	404	File routes	File not in batch
`ROUTE_NOT_FOUND`	404	Any	Unknown route
`BATCH_STATE_CONFLICT`	409	Conversions, archive	Operation not allowed in current batch status
`FILE_NOT_READY`	409	Preview, download	File is not `completed`
`NO_COMPLETED_FILES`	409	Archive	Nothing to archive
`ARCHIVE_NOT_PREPARED`	409	Archive GET	ZIP not created yet
`BATCH_EXPIRED`	410	Batch routes	Batch deleted by expiry or by the user
`FILE_TOO_LARGE`	413	Upload	File over per-file limit
`BATCH_TOO_LARGE`	413	Upload	Request over total limit
`UNSUPPORTED_MEDIA_TYPE`	415	Upload, JSON routes	Wrong Content-Type
`TOO_MANY_FILES`	422	Upload	More files than allowed
`NO_VALID_FILES`	422	Upload	All files rejected
`UNSUPPORTED_FILE_TYPE`	file-level	Upload result	Content is not AVIF
`CORRUPTED_FILE`	file-level	Upload / conversion result	AVIF damaged or undecodable
`IMAGE_DIMENSIONS_EXCEEDED`	file-level	Upload result	Image above 50 MP
`CONVERSION_FAILED`	file-level	Conversion result	Unexpected conversion error
`CONVERSION_TIMEOUT`	file-level	Conversion result	Exceeded 60 s
`ARCHIVE_FAILED`	500	Archive	ZIP creation failed
`RATE_LIMITED`	429	Any	Too many requests; see `Retry-After`
`SERVICE_BUSY`	503	Upload, conversions	Queue or storage bound reached; see `Retry-After`

Code	HTTP	Used by	Meaning
`STORAGE_UNAVAILABLE`	503	Upload, archive, health	Temporary storage not writable
`INTERNAL_ERROR`	500	Any	Unexpected server error

16. Data and File Handling Requirements

16.1 Data inventory and classification

Data	Class	Where	Lifetime
Original AVIF file	User content (potentially sensitive)	Temp disk `<batch>/<fileId>.avif`	Deleted immediately after its conversion attempt, at the latest at expiry
Converted JPEG	User content	Temp disk `<batch>/<fileId>.jpg`	Until expiry (default 30 min) or user deletion
ZIP archive	User content	Temp disk `<batch>/bundle.zip`	Deleted with the batch
Batch and file metadata (display name, sizes, dimensions, statuses)	Potentially personal (file names)	API process memory only	Until expiry; then removed
Expired batch ID tombstone	Technical	Memory	24 h; contains ID and expiry only
Request logs	Technical; IP truncated or hashed	Log pipeline	14 days
Theme preference, resume batch ID	Preference / technical	Browser storage	Until cleared; resume ID only for the session

16.2 Batch and file data model

The MVP has no database. The API keeps this model in memory (Section 21.2 explains the restart behaviour: a restart purges all data).

Entity	Fields
Batch	`id`, `tool`, `status`, `createdAt`, `expiresAt`, `options` {quality, background} or null, `files[]`, `archive` {path, sizeBytes, fileCount, createdAt} or null, `activeDownloads` (counter), `clientHash` (truncated IP hash for rate/abuse correlation)
File	`id`, `displayName` (sanitised), `outputName` (unique), `status`, `originalSizeBytes`, `originalWidth`, `originalHeight` (after orientation), `hasAlpha`, `bitDepth`, `outputSizeBytes`, `outputWidth`, `outputHeight`, `warnings[]`, `error` {code, message}, `startedAt`, `finishedAt`

16.3 File naming rules

Rule	Specification
Storage names	`<fileId>.avif`, `<fileId>.jpg`, `bundle.zip`, `*.part` (temporary). User text is never used in paths.
Display name	Original base name sanitised per SR display-names; extension removed.
Output name	`<display name>.jpg`; on case-insensitive collision append `-1`, `-2` in upload order.
ZIP name	`pixora-converted-YYYYMMDD-HHmmss.zip` using UTC at creation.
Download header	ASCII fallback `filename` plus RFC 6266 `filename*` UTF-8 form.

16.4 Image processing rules (conversion pipeline)

The converter applies these steps in order. They define behaviour, not code; the implementation uses Sharp.

Step	Operation	Detail and related requirement
1	Open with limits	Open the stored original with the pixel limit (50,000,000), error tolerance set to fail on errors but not on warnings, sequential read where possible (SR bomb).
2	Read metadata	Width, height, orientation, alpha, bit depth, ICC presence; compute oriented dimensions for reporting.
3	Apply orientation	Auto-orient using container transforms and EXIF (FR cnv.orient).
4	Flatten alpha	If alpha exists, composite on the background colour (FR cnv.alpha).
5	Colour conversion	Convert to sRGB using the embedded profile when present; reduce to 8-bit (FR cnv.color, cnv.hdr).
6	Encode JPEG	Quality from options; 4:4:4 at quality 90 or higher else 4:2:0; baseline (non-progressive) for maximum compatibility; no metadata copied (FR cnv.quality, cnv.meta).
7	Write atomically	Write to `<fileId>.jpg.part`, then rename to `<fileId>.jpg` (FR cnv.atomic).
8	Record result	Output size and dimensions; set status `completed`; delete the original (D-08).
9	On any error	Delete partial output, map to error code, set `failed`, delete the original (FR cnv.fail).

Spike required Behaviour for animated AVIF and for HDR/wide-gamut tone mapping depends on the installed libvips/libheif build. Phase 4 includes a time-boxed spike to confirm the behaviour and update tests INT-11 and INT-12 (risks R-08, R-09).

16.5 Default limits and configuration values (single source of truth)

All other sections refer to this table. The backend validates these variables at start-up and fails fast with a clear message when a value is missing, non-numeric or out of range.

Variable	Default	Meaning
`PORT`	3001	API listen port
`MAX_FILE_SIZE_MB`	25	Per-file upload limit
`MAX_FILES_PER_BATCH`	20	Files per batch
`MAX_BATCH_TOTAL_MB`	150	Total upload size per request
`MAX_INPUT_PIXELS`	50,000,000	Maximum decoded pixels (50 MP)
`DEFAULT_JPEG_QUALITY`	90	Default quality
`DEFAULT_BACKGROUND_COLOR`	#FFFFFF	Default flatten colour
`RETENTION_MINUTES`	30	Batch lifetime (allowed 5 to 1440)
`CLEANUP_INTERVAL_SECONDS`	60	Expiry sweeper interval
`ORPHAN_SWEEP_INTERVAL_MINUTES`	10	Orphan directory sweeper interval
`TOMBSTONE_TTL_HOURS`	24	Time that expired IDs return 410
`CONVERSION_CONCURRENCY`	2	Simultaneous conversions
`CONVERSION_TIMEOUT_SECONDS`	60	Per-file wall-clock timeout
`MAX_QUEUED_FILES`	200	Queue bound
`MAX_TOTAL_STORAGE_MB`	5120	Total temporary storage bound
`MIN_FREE_DISK_MB`	1024	Refuse uploads below this free space
`SHUTDOWN_GRACE_SECONDS`	10	Wait for in-flight work on SIGTERM

17. Non-Functional Requirements

17.1 Measurement basis

Reference environment: 2 vCPU, 4 GB RAM, SSD storage, Linux container, Node.js LTS, Sharp with bundled libvips; client for web metrics: Lighthouse mobile preset (simulated 4G, 4x CPU slowdown).

> **Targets versus guarantees** Every number in the "Proposed target" column is an engineering target to be validated by the stated test. It is not a promise to users and not a service-level agreement. If a target cannot be met on the reference environment the team raises a change request instead of silently relaxing it.

17.2 Requirements

ID	Category	Requirement	Proposed target	Verification
NFR-001	Performance	Largest Contentful Paint of the pre-rendered homepage	p75 <= 2.5 s (lab: Lighthouse mobile <= 2.5 s)	Lighthouse CI on every release; field data via Chrome UX Report once traffic exists
NFR-002	Performance	Cumulative Layout Shift and Interaction to Next Paint	CLS <= 0.1; INP p75 <= 200 ms	Lighthouse CI; Playwright traces
NFR-003	Performance	Initial JavaScript payload of the homepage (gzip)	<= 150 KB; lightbox and static pages code-split	Bundle-size budget enforced in CI (`vite build` + size check)
NFR-004	Performance	Lighthouse scores on the homepage (mobile)	Performance >= 90, Accessibility >= 95, Best Practices >= 95, SEO >= 95	Lighthouse CI gate
NFR-005	Performance	Server-side conversion time per image (excludes upload time)	2 MP image: median <= 0.8 s; 12 MP image: median <= 2.5 s and p95 <= 4 s	Benchmark script on fixtures in the reference environment (Phase 8)
NFR-006	Performance	End-to-end time to convert a batch of 20 images of about 12 MP each, excluding upload	<= 45 s with 2 concurrent conversions	Performance test PERF-02
NFR-007	Performance	API latency for lightweight endpoints (health, config, batch status) under 50 concurrent virtual users	p95 <= 150 ms	k6 load test
NFR-008	Performance	ZIP preparation time for 20 JPEG files totalling up to 100 MB	<= 10 s (store-only compression, level 0)	Performance test PERF-03
NFR-009	Scalability	Concurrent active users supported by a single reference instance with no degradation of the targets above	25 concurrent active batches (target; validated by load test, not guaranteed)	k6 scenario with realistic batch sizes
NFR-010	Scalability	Documented path to horizontal scaling without redesigning the API	Section 21.6 defines the migration (shared object storage with TTL, queue, shared state store)	Architecture review
NFR-011	Reliability	Conversion success rate for valid AVIF files in the conformance fixture set	>= 99% (100% for the mandatory fixtures listed in Section 23)	Automated integration suite
NFR-012	Reliability	Incomplete or partial files served to users	Zero: only atomically finalised files are served	Integration test INT-05; fault-injection test
NFR-013	Reliability	Graceful shutdown on SIGTERM	In-flight requests allowed up to 10 s, then all stored data purged	Integration test INT-09

ID	Category	Requirement	Proposed target	Verification
NFR-014	Availability	Monthly availability of the production service	Target 99.5% (about 3.6 h downtime per month). This is a planning target, not a contractual SLA	External uptime monitor on `/api/v1/health`
NFR-015	Availability	Automatic recovery from process crash	Restart within 30 s by process supervisor; startup purge removes orphaned data	Deployment checklist (Section 25)
NFR-016	Security	Conformance to the security requirements SR-001 onwards and to relevant OWASP ASVS Level 1 controls	0 open High or Critical findings at release	Security test suite (SEC-xx), ZAP baseline scan, `npm audit`
NFR-017	Privacy	Retention of user images on the server	Originals deleted immediately after processing; outputs deleted at expiry (default 30 min, configurable 5 to 1440 min); 100% of batch data gone within retention + 60 s sweep interval	Integration tests INT-06, INT-07
NFR-018	Privacy	Personal data and image data in logs	Zero image content, zero original file names, no full IP addresses retained beyond 14 days	Log review test SEC-09
NFR-019	Usability	First-time users completing "convert one AVIF and download it"	>= 9 of 10 test participants succeed unaided in <= 60 s; <= 3 deliberate actions (select, convert, download)	Moderated usability test in Phase 8
NFR-020	Accessibility	WCAG 2.2 Level AA conformance of all MVP pages	0 critical or serious axe-core violations; manual keyboard-only run passes; screen-reader pass with NVDA + Firefox and VoiceOver + Safari	A11Y-01 to A11Y-05
NFR-021	Maintainability	Static quality of the codebase	TypeScript `strict: true`; ESLint 0 errors; Prettier-clean; no circular dependencies between modules	CI lint and type-check
NFR-022	Maintainability	Automated test coverage	Backend services and utilities >= 85% line coverage; whole backend >= 80%; frontend >= 70%	Vitest coverage reports in CI
NFR-023	Maintainability	Developer documentation	README with setup in <= 5 commands, `.env.example` with every variable, OpenAPI file for the API	Documentation review in Phase 9
NFR-024	Browser compatibility	Supported browsers: the latest two major versions of Chrome, Edge, Firefox, Safari (macOS and iOS 16.4+), and Samsung Internet. The server-side design means the browser does NOT need AVIF decoding support to convert files	100% of E2E scenarios pass on Chromium, Firefox and WebKit	Playwright projects on three engines
NFR-025	Mobile responsiveness	Usability on phones and tablets	No horizontal scroll from 320 px; touch targets >= 44 x 44 CSS px; all flows pass on emulated iPhone and Pixel devices	Responsive tests RSP-01 to RSP-04
NFR-026	SEO	Technical SEO quality of public pages	Lighthouse SEO >= 95; pre-rendered HTML contains h1, body text and FAQ; valid structured data; sitemap and robots.txt reachable	SEO tests SEO-01 to SEO-04

ID	Category	Requirement	Proposed target	Verification
NFR-027	Code quality	API stability and discoverability	All endpoints versioned under `/api/v1`; OpenAPI 3.1 document kept in sync by a contract test	Contract test in CI

Categories covered: Performance, Scalability, Reliability, Availability, Security, Privacy, Usability, Accessibility, Maintainability, Browser compatibility, Mobile responsiveness, SEO, Code quality.

18. UI/UX Specifications

18.1 Design direction

Pixora should feel like a modern, premium SaaS product, not a student project. The interface is minimal and calm: a clean light background, generous white space, one confident brand colour (indigo) with a violet accent, subtle gradients, rounded cards, soft shadows and a very clear primary action. The upload component is the visual centrepiece.

- **Hierarchy:** one h1, one primary action per view; secondary actions are visually quieter.
- **Trust:** privacy cues (lock, auto-delete) sit directly under the upload area.
- **Consistency:** all spacing, radii, shadows and colours come from tokens; no ad-hoc values.
- **Craft:** consistent icon set (Lucide), crisp 1 px borders, restrained motion, no stock-photo clutter.
- **Accessibility is part of the design,** not an afterthought: all colours below were checked for WCAG 2.2 AA contrast.

18.2 Colour palette

Tokens are implemented as CSS custom properties on `:root` (light) and `:root[data-theme="dark"]`, and mapped in Tailwind configuration. Contrast ratios were computed with the WCAG relative-luminance formula and must be re-verified by test A11Y-04.

18.2.1 Light theme

Token	HEX	Usage	Contrast note
`--bg`	#FFFFFF	Page background	-
`--bg-subtle`	#F8FAFC	Alternating sections, footer	-
`--surface`	#FFFFFF	Cards, dialogs, inputs	-
`--surface-muted`	#F1F5F9	Skeletons, hover fills, code	-
`--border`	#E2E8F0	Decorative dividers and card borders	Decorative only
`--border-strong`	#64748B	Form-control and dropzone outlines	4.76:1 on white
`--text`	#0F172A	Headings, body text	17.85:1 on white
`--text-secondary`	#475569	Secondary text	7.58:1 on white; 6.92:1 on #F1F5F9
`--text-muted`	#64748B	Captions on white or `--bg-subtle` only	4.76:1 on white
`--primary`	#4F46E5	Primary buttons, links, focus ring	6.29:1 with white text
`--primary-hover`	#4338CA	Hover state	7.90:1 with white text
`--primary-soft`	#EEF2FF	Selected, drag-over, soft badges	Primary text 5.62:1
`--accent`	#7C3AED	Gradient end, highlights	5.70:1 with white text
`--success`	#15803D	Completed status	5.02:1 on white; 4.79:1 on #F0FDF4
`--warning`	#B45309	Expiry warning, animation notice	5.02:1 on white; 4.84:1 on #FFFBEB
`--error`	#B91C1C	Errors and destructive actions	6.47:1 on white; 5.91:1 on #FEF2F2
`--info`	#0369A1	Informational notices	5.93:1 on white

18.2.2 Dark theme

Token	HEX	Usage	Contrast note
`--bg`	#0B1020	Page background	-
`--bg-subtle`	#0F1630	Alternating sections, footer	-
`--surface`	#121933	Cards, dialogs, inputs	-
`--surface-muted`	#1A2347	Skeletons, hover fills	-
`--border`	#2A3563	Decorative dividers	Decorative only
`--border-strong`	#7C8DB5	Form-control and dropzone outlines	5.71:1 on bg; 4.61:1 on surface-muted
`--text`	#F1F5F9	Headings, body text	17.28:1 on bg
`--text-secondary`	#CBD5E1	Secondary text	12.75:1 on bg; 10.28:1 on surface-muted
`--text-muted`	#94A3B8	Captions	7.38:1 on bg; 6.75:1 on surface
`--primary` (text, links, ring)	#A5B4FC	Links, focus ring, outline accents	9.50:1 on bg; 8.69:1 on surface
Primary button fill	#4F46E5	Same as light theme; white label	6.29:1 with white text
`--primary-soft`	#1E2A5E	Selected, drag-over background	Primary text 7.0:1 or higher
`--success`	#4ADE80	Completed status	10.86:1 on bg
`--warning`	#FBBF24	Warnings	11.34:1 on bg
`--error`	#F87171	Errors	6.84:1 on bg
`--info`	#38BDF8	Informational notices	8.84:1 on bg

18.2.3 Gradients

Name	Definition	Used for
Brand	`linear-gradient(135deg, #4F46E5 0%, #7C3AED 100%)`	Logo mark, hero accent text, primary button (light and dark)
Hero (light)	`linear-gradient(180deg, #EEF2FF 0%, #FFFFFF 70%)` plus a soft radial highlight `rgba(124,58,237,0.10)` behind the upload card	Hero background
Hero (dark)	`radial-gradient(60% 50% at 50% 0%, rgba(99,102,241,0.25), transparent)` on `--bg`	Hero background

18.3 Typography

Headings: **Plus Jakarta Sans** (600, 700, 800). Body and UI: **Inter** (400, 500, 600). Both are self-hosted variable fonts (for example through Fontsource), Latin subset, `font-display: swap`, and the two critical files are preloaded. Fallback stack: `ui-sans-serif, system-ui, -apple-system, "Segoe UI", Roboto, Arial, sans-serif`. Self-hosting avoids third-party requests (privacy and performance).

Style	Desktop (>= 1024 px)	Mobile (< 640 px)	Weight	Use
Display	56 / 60 px, tracking -0.02em	36 / 42 px	800	Hero h1
Heading 2	36 / 44 px	28 / 36 px	700	Section titles
Heading 3	24 / 32 px	22 / 30 px	700	Card titles, sub-sections
Heading 4	20 / 28 px	18 / 26 px	600	Small headings

Style	Desktop (>= 1024 px)	Mobile (< 640 px)	Weight	Use
Body large	18 / 28 px	17 / 26 px	400	Hero sub-headline, intros
Body	16 / 24 px	16 / 24 px	400	Default text (never below 16 px on inputs)
Small	14 / 20 px	14 / 20 px	500	File metadata, helper text
Caption	12 / 16 px	12 / 16 px	500	Badges, legal text (minimum size)
Button	16 / 24 px (large 18)	16 / 24 px	600	Buttons; small buttons 14 / 20

18.4 Spacing, radius, elevation and motion

Token set	Values
Spacing (4 px base)	4, 8, 12, 16, 20, 24, 32, 40, 48, 64, 80, 96 px. Section padding 64 px (desktop) and 40 px (mobile). Card padding 24 px (16 px mobile). Gap between cards 16 px (mobile) and 24 px (desktop).
Layout	Container max width 1200 px; gutters 16 px (< 640 px) and 24 px (>= 640 px); readable text width 720 px.
Border radius	sm 8 px (badges, inputs), md 12 px (buttons, toasts), lg 16 px (cards), xl 24 px (upload card), full 9999 px (pills).
Elevation (light)	sm `0 1px 2px rgba(15,23,42,0.06)`; md `0 8px 24px -8px rgba(15,23,42,0.12)`; lg `0 24px 48px -16px rgba(79,70,229,0.25)`.
Elevation (dark)	Cards use a 1 px `--border` plus `0 8px 24px -8px rgba(0,0,0,0.5)`; no coloured glows.
Motion	Durations 120 ms (hover), 200 ms (state change), 300 ms (dialog, panel); easing `cubic-bezier(0.2, 0, 0, 1)`; animate `opacity` and `transform` only; hover lift `translateY(-1px)`; progress uses `transform: scaleX`. `prefers-reduced-motion` removes transforms and replaces spinners with a static indicator plus text.
Focus	2 px solid ring in `--primary`, 2 px offset, never removed; contrast against adjacent colours >= 3:1.
Icons	Lucide React, 20 px default (24 px in large buttons), `aria-hidden` when decorative; stroke width 1.75.

18.5 Components

18.5.1 Buttons

Variant	Appearance	States
Primary	Brand gradient fill, white label, radius 12; heights: small 40, default 48, large 56 px; horizontal padding 24 px	Hover: darker gradient (`--primary-hover` start) and shadow md; active: translateY(1px); focus ring; disabled: fill `#E2E8F0` with `#64748B` label (light) and cursor not-allowed; loading: spinner replaces icon, label stays, button not clickable
Secondary	Surface fill, 1 px `--border-strong`, text `--text`	Hover: `--surface-muted`; same focus and disabled rules
Ghost	Transparent, text `--primary`	Hover: `--primary-soft`
Destructive	Surface fill, 1 px `--error`, text `--error`	Hover: error tint background
Icon button	44 x 44 px hit area, radius 12, accessible name required	Same as ghost

18.5.2 Cards, upload component, badges and inputs

Component	Specification
Card	`--surface`, 1 px `--border`, radius 16, padding 24, shadow sm; interactive cards raise to shadow md on hover.
Upload card (dropzone)	Radius 24, 2 px dashed `--border-strong` on `--surface`, min height 280 px (desktop) / 220 px (mobile), centred icon in a 64 px gradient tile, headline 20 px, secondary line with limits (from config). Drag-over: solid `--primary` border, `--primary-soft` fill, scale 1.01. Error flash: `--error` border for 600 ms plus inline message. The whole card is a button (role and keyboard support) and also contains an explicit Browse files button.
File row	Height 64 px; thumbnail 40 px (or file icon), name (ellipsis), size (Small), status or validation badge, remove icon button (44 px hit area).
Status badge	Pill, caption text 600 with icon and text (never colour alone): Queued (clock, neutral), Processing (spinner, primary), Completed (check, success), Failed (alert, error), Rejected (x-circle, error).
Result card	Thumbnail with 4:3 crop-free `object-fit: contain` on `--surface-muted`, name, two metadata lines (original, converted), status badge, Preview and Download actions. Failed cards show the message in `--error` and omit thumbnail and download.
Progress bar	Height 8 px, radius full, track `--surface-muted`, fill brand gradient; `role="progressbar"` with `aria-valuenow`.
Inputs	Height 44 px, radius 12, 1 px `--border-strong`, label above, helper text below, error text with icon; slider thumb 24 px; colour swatches 44 px.
Toast	Radius 12, shadow lg, icon + message + close; bottom-right (desktop) and bottom-centre (mobile), max width 420 px; stacked, max three visible.
Dialog (lightbox)	Native `dialog`; backdrop `rgba(2,6,23,0.72)`; panel radius 20, max 92 vw by 92 vh; header with file name and close button; image `object-fit: contain`.
Skeleton	`--surface-muted` blocks with a subtle shimmer (disabled under reduced motion).

18.5.3 Navigation and footer

Element	Specification
Header	Sticky, 64 px high, `--bg` at 85% opacity with backdrop blur, 1 px bottom border. Left: logo mark (gradient tile with a stylised "P") and wordmark. Right (>= 768 px): How it works, Features, FAQ links and the theme toggle. Below 768 px: logo, theme toggle and a 44 px menu button that opens an accessible panel (focus trap, Escape closes).
Skip link	First focusable element: "Skip to main content", visible on focus.
Footer	`--bg-subtle`, 1 px top border. Desktop: four columns (brand and tagline; Product; Company: About, Contact; Legal: Privacy Policy, Terms). Mobile: stacked with 24 px gaps. Bottom line: copyright and "Files are deleted automatically" privacy note.

18.6 Responsive breakpoints and layout

Name	Min width	Layout rules
base	0 px (design from 320 px)	Single column; header collapsed; results one per row; feature cards one column; buttons full width
sm	640 px	Gutters 24 px; feature cards two columns; action buttons inline
md	768 px	Header links visible; results two columns; settings panel in two columns
lg	1024 px	Hero typography at desktop scale; features three columns; footer four columns

Name	Min width	Layout rules
xl	1280 px	Results three columns; container reaches max width 1200 px
2xl	1536 px	Container stays 1200 px; only outer margins grow

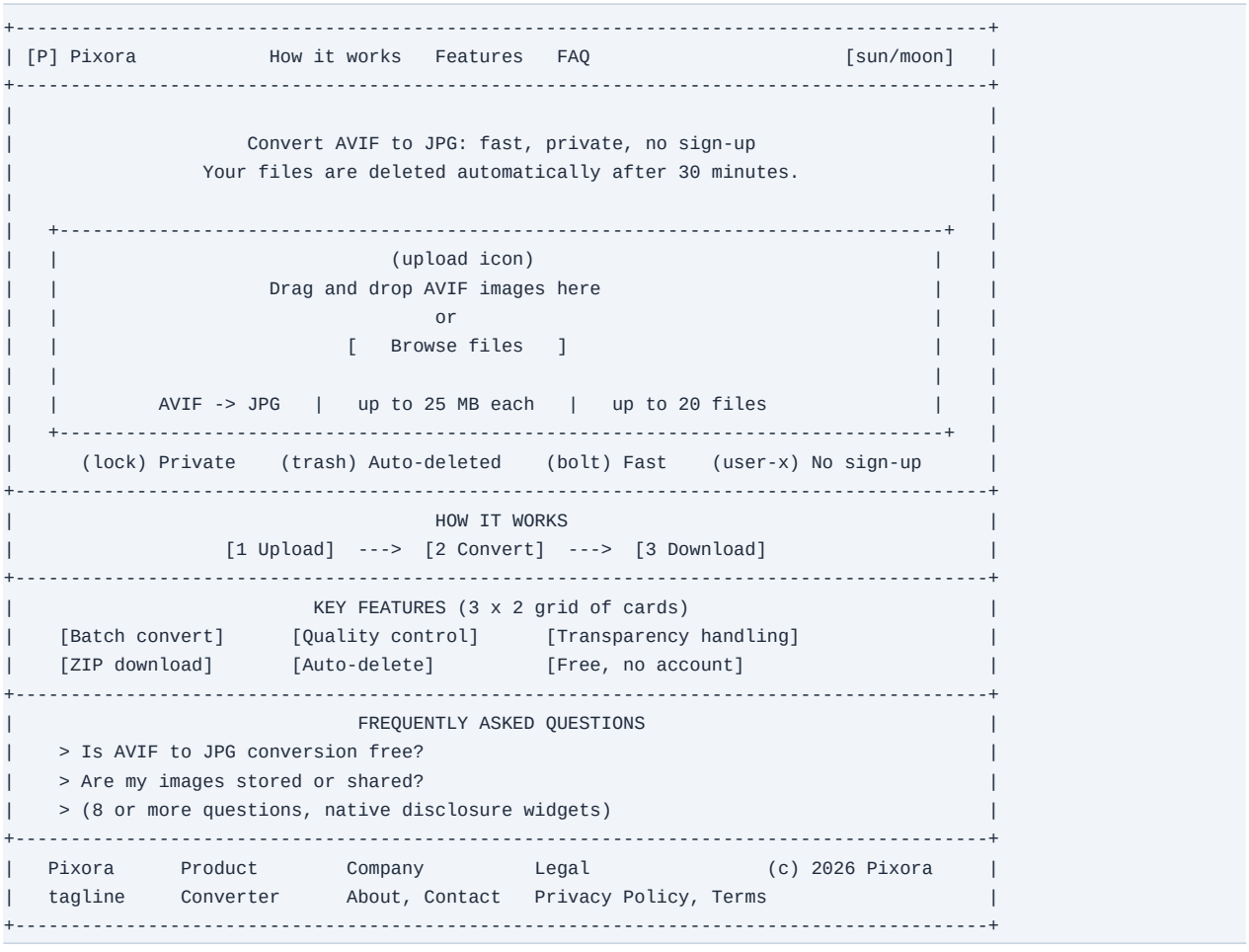
18.7 Light and dark mode

- Default follows `prefers-color-scheme`; a header toggle cycles Light, Dark and System; the choice is stored under `pixora-theme`.
- A tiny inline script in `index.html` sets `data-theme` before first paint (no flash); it is allowed in the CSP with a hash.
- Both themes are tested for contrast and screenshots (E2E-14, A11Y-04). Images and the logo have dark-mode variants where needed.
- Elevation in dark mode uses borders and darker surfaces instead of bright shadows.

18.8 Wireframe descriptions

The wireframes below are structural. Visual styling follows the design system above. Widths: desktop 1280 px, mobile 360 px.

18.8.1 Homepage (desktop)



Wireframe 1: Homepage. The upload card is in the first viewport; trust cues sit directly beneath it.

18.8.2 Upload section (files selected)

```
+-----------------------------------------------------------------------------------------+
|   Selected files (3)                                          [Clear all]               |
|   +-------------------------------------------------------------------------------+     |
|   | [img] holiday-photo.avif                    812.3 KB        Ready         [x] |     |
|   | [img] logo-transparent.avif                  20.0 KB        Ready         [x] |     |
|   | [!]   notes.png                               1.2 MB        Not an AVIF   [x] |     |
|   +-------------------------------------------------------------------------------+     |
|   (+) Add more files                                                                    |
|                                                                                         |
|   > Advanced settings                                                                   |
|   +-------------------------------------------------------------------------------+     |
|   |  Quality     ( High 90 )  ( Balanced 80 )  ( Small 70 )   [--o-------] 90     |     |
|   |  Background  ( White )  ( Black )  ( Custom #______ )   used for transparency |     |
|   +-------------------------------------------------------------------------------+     |
|                                                                                         |
|                                        [   Convert 2 images   ]                         |
|   Uploading... 64%  (8.1 of 12.7 MB)  [=========>        ]  [Cancel]                  |
+-----------------------------------------------------------------------------------------+
```

Wireframe 2: Upload section with file list, validation messages, advanced settings and upload progress. Convert shows the count of valid files.

18.8.3 Conversion processing state

```
+-----------------------------------------------------------------------------------------+
|   Converting your images                                          Time left: 28:40      |
|   Overall  [==============>                ]  2 of 5 done                               |
|   +-------------------------------------------------------------------------------+     |
|   | holiday-photo.avif    4000x3000    (check) Completed                        |     |
|   | logo.avif              512x512     (check) Completed                        |     |
|   | poster.avif           6000x4000    (spinner) Processing...                  |     |
|   | banner.avif           3840x1080    (clock) Queued                           |     |
|   | scan-07.avif          5200x7800    (clock) Queued                           |     |
|   +-------------------------------------------------------------------------------+     |
|   [ Download all (ZIP) ]  disabled until a conversion is finished     [Cancel]          |
|   Status changes are announced to screen readers (aria-live polite).                    |
+-----------------------------------------------------------------------------------------+
```

Wireframe 3: Processing. Each row has an icon and a text status. The overall bar uses `role="progressbar"`.

18.8.4 Conversion results section

```
+-----------------------------------------------------------------------------------------+
|   Done: 4 converted, 1 failed                 Files are deleted in 27:12                |
|   [ Download all (ZIP) ]    [ Convert more images ]    [ Delete my files now ]          |
|   +--------------------------+  +--------------------------+                           |
|   | +----------------------+ |  | +----------------------+ |                           |
|   | |   (JPEG thumbnail)   | |  | |   (JPEG thumbnail)   | |                           |
|   | +----------------------+ |  | +----------------------+ |                           |
|   | holiday-photo.jpg        |  | logo.jpg                 |                           |
|   | Original 812 KB 4000x3000|  | Original 20 KB 512x512   |                           |
|   | JPG 1.4 MB 4000x3000 +78%|  | JPG 31 KB 512x512 +55%   |                           |
|   | (check) Completed        |  | (check) Completed        |                           |
|   | [Preview]  [Download JPG]|  | [Preview]  [Download JPG]|                           |
|   +--------------------------+  +--------------------------+                           |
|   +-------------------------------------------------------------------------------+     |
|   | scan-07.avif             |   Failed card: no thumbnail, no download button.   |     |
|   | (alert) We could not     |   Shows message from the error code and the        |     |
|   | convert this file.       |   note "Other images were not affected."           |     |
|   +-------------------------------------------------------------------------------+     |
+-----------------------------------------------------------------------------------------+
```

Wireframe 4: Results grid (two columns at 768 px, three at 1280 px), batch actions, expiry countdown and a failed-card example.

18.8.5 Error state

```
+-------------------------------------------------------------------------------------+
|   (alert) Something went wrong                                          [x]         |
|   Pixora is temporarily unable to process files. Please try again in a              |
|   few minutes. Your selected files are still here.                [Try again]       |
+-------------------------------------------------------------------------------------+
|   (wifi-off) Reconnecting...  We lost connection. Retrying automatically (3/5).     |
+-------------------------------------------------------------------------------------+
|   +-------------------------------------------------------------------------+       |
|   | notes.png      1.2 MB     (x) Not an AVIF image.  Pixora converts .avif only. |  |
|   | scan.avif       31.2 MB     (x) Over the 25 MB limit per file.           |  |
|   +-------------------------------------------------------------------------+       |
+-------------------------------------------------------------------------------------+
|   (clock) These files were automatically deleted for your privacy.                  |
|   Please upload them again.                                          [Start over]   |
+-------------------------------------------------------------------------------------+
```

Wireframe 5: Error patterns at four levels: batch banner, reconnecting banner, per-file validation errors and the expired state.

18.8.6 Mobile homepage

```
+-------------------------------------+
| [P] Pixora                  [menu]  |
+-------------------------------------+
|                                     |
|  Convert AVIF to JPG                |
|  Fast, private, no sign-up.         |
|                                     |
|  +-------------------------------+  |
|  |        (upload icon)          |  |
|  |   Drag and drop or tap to     |  |
|  |   choose AVIF images          |  |
|  |      [  Browse files  ]       |  |
|  |  25 MB each | 20 files max    |  |
|  +-------------------------------+  |
|                                     |
|  (lock) Private  (trash) Auto-del.  |
+-------------------------------------+
|  HOW IT WORKS                       |
|  1 Upload  2 Convert  3 Download    |
+-------------------------------------+
|  KEY FEATURES (1 column)            |
|  [Batch convert]                    |
|  [Quality control]                  |
|  [ZIP download] ...                 |
+-------------------------------------+
|  FAQ  > Is it free?  > Private?      |
+-------------------------------------+
|  About  Contact  Privacy  Terms     |
+-------------------------------------+
```

Wireframe 6: Mobile homepage at 360 px. Single column; upload card within the first 640 px; menu collapses navigation.

18.9 Homepage content structure and microcopy

Section	Content	HTML structure
Hero	H1 "Convert AVIF to JPG: fast, private, no sign-up"; sub-headline "Your files are deleted automatically after 30 minutes." (value comes from config); upload card; trust row	`header`, `main > section#converter`, one `h1`

Section	Content	HTML structure
How it works	Three steps with icons and one-line descriptions: Upload your AVIF images / Convert to JPG in one click / Download single files or a ZIP	`section#how-it-works`, `h2`, ordered list
Key features	Six cards: Batch conversion; Quality control; Transparency handling; Private by design; ZIP download; No sign-up	`section#features`, `h2`, `ul` of `article`
About AVIF (SEO copy)	250 to 400 words explaining AVIF, why JPG is still needed (compatibility), and when to choose which quality setting	`section`, `h2`, paragraphs
FAQ	At least eight questions (list below)	`section#faq`, `details`/`summary`, JSON-LD
Footer	Links and copyright	`footer`, `nav`

Required FAQ questions: What is an AVIF file? Why convert AVIF to JPG? Is Pixora free? Do I need an account? Are my images stored or shared? How long are files kept? How many files can I convert at once? What is the maximum file size? Will image quality change? What happens to transparent images? Can I download all results together? Which browsers and devices are supported?

18.10 SEO and business-readiness requirements

18.10.1 URLs and indexing

Page	URL	Indexable	Title (<= 60 chars)	Meta description (<= 160 chars)
Home / converter	`/`	Yes	AVIF to JPG Converter: Free & Private | Pixora	Convert AVIF to JPG online in seconds. Batch convert up to 20 images, preview and download as ZIP. No sign-up; files are deleted automatically.
About	`/about`	Yes	About Pixora: Simple, Private Image Conversion	Learn what Pixora is, how it handles your images and why privacy and simplicity guide the product.
Contact	`/contact`	Yes	Contact Pixora	Questions, feedback or business enquiries? Contact the Pixora team by e-mail.
Privacy Policy	`/privacy-policy`	Yes	Privacy Policy | Pixora	How Pixora processes and deletes your images, what data is logged and your choices.
Terms	`/terms`	Yes	Terms of Service | Pixora	The terms that apply when you use Pixora.
404	any other	No (`noindex`)	Page not found | Pixora	The page you are looking for does not exist.

URLs are lowercase, hyphen-separated and extension-free. Trailing slashes are normalised to none. Each page declares a canonical URL. Future tools will use `/{source}-to-{target}` (for example `/png-to-jpg`), which is why the AVIF tool will move to `/avif-to-jpg` when the second tool launches (D-03).

18.10.2 Metadata, structured data and crawl files

- **Open Graph and Twitter:** `og:type=website`, `og:site_name`, `og:title`, `og:description`, `og:url`, `og:image` (1200 x 630 PNG, under 300 KB, with alt text), `twitter:card=summary_large_image`.
- **Structured data (JSON-LD, pre-rendered):** `WebApplication` (name, url, applicationCategory `MultimediaApplication`, operatingSystem `Any`, offers price 0), `Organization` (name, url, logo), `FAQPage` (mirrors the visible FAQ exactly), and `BreadcrumbList` for sub-pages. Marked-up content must match visible content.

- **`/sitemap.xml`:** the five public pages with absolute URLs and `lastmod`; **`/robots.txt`:** `User-agent: *`, `Allow: /`, a `Sitemap:` line; the API host serves `Disallow: /` and `X-Robots-Tag: noindex` on its responses.
- **Semantic HTML:** one `h1` per page, ordered headings, landmarks (`header`, `nav`, `main`, `footer`), descriptive link text, meaningful `alt` text.
- **Performance for SEO:** pre-rendered HTML (D-04), Core Web Vitals budgets (Section 17), image dimensions set to avoid layout shift, no render-blocking third-party scripts.
- **Rendering note:** the converter itself is client-rendered; all indexable copy exists in the pre-rendered HTML (test SEO-01).

19. Security and Privacy Requirements

Security is a core requirement. Pixora parses untrusted binary files with native image libraries, accepts anonymous traffic and handles potentially private photos. The design therefore combines strict validation, resource limits, short retention and minimal data.

19.1 Threat overview

Threat	Example	Primary mitigations
Malicious file (decoder exploit, polyglot)	Crafted AVIF or a file with appended payload	SR-001, SR-019, SR-022; output is re-encoded so no input bytes are passed through
Resource exhaustion (decompression bomb, volume)	Tiny file declaring huge dimensions; thousands of uploads	SR-004, SR-003, SR-008, SR-009
Path traversal and file-name abuse	`../../etc/passwd.avif`	SR-005, SR-006, SR-007
Unauthorised access to other users' files	Guessing or enumerating batch IDs	SR-017, SR-008
Cross-site scripting	File name containing HTML	SR-020, SR-011
Cross-origin misuse	Other sites calling the API	SR-010; no cookies, so CSRF has no session to ride on
Data leakage through logs or storage	File names or images in logs; stale files on disk	SR-016, SR-013, SR-014, SR-015
Supply-chain compromise	Malicious or vulnerable dependency	SR-019
Information disclosure	Stack traces, paths, versions	SR-018
Archive attacks	Zip-slip names inside the ZIP	SR-021

19.2 Security and privacy requirements

ID	Area	Requirement	Priority	Verified by
SR-001	File validation	The server SHALL determine file type from content, not from file name or declared MIME type: (1) read the first 32 bytes; (2) require an ISO-BMFF `ftyp` box at offset 4 whose major or compatible brands include `avif` or `avis`; (3) require Sharp metadata to report format `heif` with compression `av1`; (4) reject everything else (including HEIC, which reports `hevc`).	Must	UT-03, UT-04, SEC-01, SEC-02
SR-002	File validation	The declared part `Content-Type` SHALL be accepted only as `image/avif` or `application/octet-stream` (some systems send no precise type). It is advisory only; SR-001 is authoritative.	Must	SEC-01
SR-003	Resource limits	Multer and the request pipeline SHALL enforce: maximum file size, file count, total request size (checked from `Content-Length` before streaming and by a streaming counter), at most 1 non-file field of 64 bytes (`tool`), maximum header pairs and parts. Violations abort the request and delete partial uploads.	Must	API-08, API-09, SEC-06
SR-004	Resource limits	Decoders SHALL be run with `limitInputPixels` equal to MAX_INPUT_PIXELS (50,000,000); images above it are rejected with `IMAGE_DIMENSIONS_EXCEEDED` before decoding pixel data. Conversion has a wall-clock timeout and the process runs with a container memory limit.	Must	SEC-05

ID	Area	Requirement	Priority	Verified by
SR-005	File handling	Stored files SHALL use server-generated names `<fileId>.avif` and `<fileId>.jpg` where `fileId` comes from a cryptographically secure random generator. User-supplied names SHALL never be used in file-system paths.	Must	SEC-03
SR-006	File handling	Identifiers received in URLs SHALL be validated against strict patterns (`^bch_[a-f0-9]{32}$`, `^fil_[a-f0-9]{16}$`) before lookup, and every resolved path SHALL be verified to remain inside the storage root.	Must	UT-01, UT-06, API-16, SEC-03, SEC-04
SR-007	File handling	Original file names SHALL be sanitised for display and download: Unicode NFC normalisation; remove path separators, control characters and the characters `<>:"/|?*`; collapse whitespace; trim leading dots; limit to 100 characters; fall back to `image` when empty. Names are stored only in memory.	Must	UT-01, SEC-03, SEC-13
SR-008	Abuse prevention	Rate limiting SHALL be applied per client IP with the tiers in Section 19.3, returning `429 RATE_LIMITED` with a `Retry-After` header. The application SHALL be configured with the correct proxy trust setting so that client IPs are not spoofable through `X-Forwarded-For`.	Must	API-22, SEC-08, SEC-12
SR-009	Abuse prevention	The conversion queue SHALL be bounded (default 200 queued files) and total storage SHALL be bounded (default 5120 MB; minimum free disk 1024 MB). When a bound is reached new uploads SHALL be refused with `503 SERVICE_BUSY` and a `Retry-After` header.	Must	INT-15
SR-010	Transport	CORS SHALL use an explicit allow-list from `CORS_ORIGINS` (never `*` in production), allow methods GET, HEAD, POST, DELETE, OPTIONS, expose `Content-Disposition` and `X-Request-Id`, and not allow credentials.	Must	SEC-07
SR-011	Transport	Responses SHALL carry security headers (Helmet defaults plus: `Strict-Transport-Security`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`, `Permissions-Policy` denying unused features). API responses SHALL carry `Cache-Control: no-store`. The frontend host SHALL send a Content-Security-Policy restricting scripts, styles and images to the site origin (plus `blob:` and `data:` for images) and `frame-ancestors none`.	Must	SEC-10
SR-012	Input handling	All path parameters, query parameters and JSON bodies SHALL be validated with a schema library (Zod); unknown JSON properties SHALL be rejected; JSON bodies are limited to 10 KB; non-JSON bodies on JSON endpoints are rejected with 415.	Must	UT-05, API-10, API-13, SEC-04
SR-013	Storage	Temporary storage SHALL be a dedicated directory outside any web root, never served statically, created with mode 0700 (files 0600), owned by a non-root service user, excluded from backups and snapshots, and organised as one directory per batch.	Must	INT-08, INT-15, SEC-02
SR-014	Storage	The cleanup mechanisms in Section 19.4 (immediate original deletion, expiry sweeper, orphan sweeper, startup purge, graceful-shutdown purge, user-initiated delete, abort cleanup) SHALL all be implemented and tested.	Must	UT-10, API-21, INT-06, INT-08, INT-09, E2E-05
SR-015	Privacy	User images SHALL NOT be stored permanently, used to train models, analysed for any purpose other than conversion, or shared with third parties. The Privacy Policy SHALL state the retention period that is actually configured.	Must	INT-07, INT-13
SR-016	Privacy	Logs SHALL be structured JSON and SHALL include request ID, route, status, duration, batchId and fileId; they SHALL NOT include image bytes, original file names, full request bodies or full IP addresses (IPs truncated or hashed with a rotating salt).	Must	SEC-09

ID	Area	Requirement	Priority	Verified by
SR-017	Access control	Because the MVP has no accounts, batch access follows a capability-URL model: `batchId` SHALL be 128 bits from a CSPRNG, SHALL NOT appear in page URLs, analytics or logs of third parties, and no endpoint SHALL list or enumerate batches.	Must	UT-06, SEC-12
SR-018	Error handling	Unexpected errors SHALL return a generic `INTERNAL_ERROR` response with a request ID; internal details are logged server-side only. `unhandledRejection` and `uncaughtException` handlers SHALL log, attempt cleanup and exit so the supervisor restarts the process.	Must	UT-09, API-23, API-24, SEC-11
SR-019	Supply chain	Dependencies SHALL be locked with a lock file, scanned in CI (`npm audit`, container image scan), and updated through automated pull requests. Sharp and its native image libraries SHALL be updated promptly when security fixes are published because they parse untrusted binary input.	Must	SEC-15
SR-020	Frontend	The frontend SHALL render file names and server messages only as text (no `dangerouslySetInnerHTML`), SHALL build download links only from validated server-provided relative URLs, and SHALL not use inline event handlers.	Must	SEC-13
SR-021	File handling	ZIP entry names SHALL be generated server-side from sanitised names and SHALL contain no path components, so that extracting the archive cannot write outside the target folder.	Must	SEC-14
SR-022	Deployment	The backend container SHALL run as a non-root user with a read-only root file system (writable mount only for the storage directory), dropped Linux capabilities, and CPU and memory limits.	Should	SEC-16

19.3 Rate limiting tiers

The tiers in Section 15.6 apply. The API is deployed behind a known number of proxies and sets `trust proxy` accordingly; the client IP is taken from the trusted hop only. Counters live in memory (single instance, A-04). Rate-limit state contains only a hashed IP and counters.

19.4 Temporary storage lifecycle and cleanup

Files are temporary by design. The following independent mechanisms make deletion robust even after crashes. Each one has at least one automated test.

No.	Mechanism	Trigger	What is deleted	Test
M1	Immediate original deletion	After each file's conversion attempt (success or failure)	Original `.avif` file	INT-07
M2	Expiry sweeper	Every 60 s; batches with `expiresAt` in the past	Whole batch directory (JPEGs, ZIP, `.part` files) and in-memory entry; tombstone created	INT-06
M3	User-initiated delete	`DELETE /batches/{id}` (button "Delete my files now")	Whole batch immediately	API-21, E2E-08
M4	Abort cleanup	Client disconnect, Multer error or request abort during upload	Partial batch directory	E2E-05, API-08
M5	Orphan sweeper	Every 10 min	Any directory in the storage root that is not in the registry and is older than 5 minutes	INT-08
M6	Start-up purge	Process start	Everything in the storage root (the registry is lost on restart, so all data is orphaned)	INT-08

No.	Mechanism	Trigger	What is deleted	Test
M7	Shutdown purge	SIGTERM or SIGINT after the grace period	Everything in the storage root	INT-09

- **Active downloads:** the sweeper skips a batch that has an open download stream, up to 5 minutes past expiry, so users are not cut off mid-download; after that the batch is deleted regardless.
- **Failures:** a failed deletion is logged (`cleanup.error`) and retried at the next sweep. A directory that survives longer than twice the retention raises a `cleanup.stuck` warning for alerting.
- **Storage hygiene:** the storage root is never served statically, is excluded from backups and snapshots, and (optionally) can be placed on a RAM-backed or encrypted volume. Free-space and total-size bounds prevent disk exhaustion (SR queue).
- **Retention changes:** `RETENTION_MINUTES` is validated between 5 and 1440. The Privacy Policy text is generated from, or reviewed against, the configured value.

19.5 ZIP file handling and cleanup

1. `POST /archive` acquires a per-batch lock (a second concurrent request waits and receives the same result).
2. The service lists files with status `completed` in upload order and assigns unique, sanitised entry names (FR dld.names, SR zip).
3. It writes `bundle.zip.part` using store-only (level 0) compression, because JPEG data does not compress further, which keeps CPU use and latency low.
4. On successful finalisation the file is closed, its size recorded, and it is atomically renamed to `bundle.zip`. Only then does the response return the `downloadUrl`.
5. On any error the `.part` file is deleted, the lock released and `ARCHIVE_FAILED` returned. No partial archive is ever served.
6. `GET /archive` streams `bundle.zip` with `Content-Length`. The ZIP is deleted together with the batch (M1 to M7). Because JPEG outputs never change after completion, the ZIP never becomes stale.

19.6 Upload processing pipeline

1. Rate limit check and `Content-Length` check against `MAX_BATCH_TOTAL_MB`.
2. Content-Type must be `multipart/form-data`; otherwise 415.
3. Capacity check: queue bound, total storage bound and free disk. Otherwise 503 `SERVICE_BUSY`.
4. Generate a batch ID (CSPRNG), create `<STORAGE_DIR>/<batchId>/` with mode 0700.
5. Stream each file part to `<fileId>.avif.upload` with Multer limits (size, count, fields, parts).
6. Read the first 32 bytes and validate the ISO-BMFF `ftyp` signature and brand (SR content).
7. Read metadata with Sharp using the pixel limit; require `heif` with `av1` compression; record dimensions, alpha and bit depth.
8. On success rename to `<fileId>.avif`; on failure delete immediately and record the rejection.
9. If no file was accepted delete the batch directory and return 422 `NO_VALID_FILES`.
10. Register the batch in memory with `expiresAt`, return 201.

19.7 Privacy by design

The Privacy Policy SHALL state the following facts, each of which is backed by a requirement above and must remain true:

- Images are processed only to convert them; they are not viewed, analysed, shared, sold or used to train models.

- Original images are deleted immediately after conversion; converted images are deleted automatically after 30 minutes (or sooner if the user deletes them).
- Pixora does not require an account and does not set tracking cookies. Optional analytics, if enabled, is cookieless and aggregated.
- Server logs contain technical data only (time, route, status, duration, truncated or hashed IP) and are retained for 14 days.
- Metadata such as location data inside photos is removed from converted files.
- Contact details and the data-protection contact for requests.

Legal review This section describes engineering behaviour. The wording of the Privacy Policy and Terms must be reviewed by qualified counsel for the target jurisdictions before launch (assumption A-09).

19.8 Logging policy

Aspect	Rule
Format	Structured JSON (Pino) to stdout; the platform collects logs.
Mandatory fields	`time`, `level`, `requestId`, `route` (pattern, not raw URL), `status`, `durationMs`, `batchId`, `fileId` where relevant, `clientHash`.
Forbidden fields	Image bytes, original file names, request bodies, full IPs, query strings with identifiers outside the route pattern, full user-agent strings (only a browser family).
Key events	`upload.accepted`, `upload.rejected` (code only), `conversion.completed` (duration, sizes), `conversion.failed` (code), `archive.created`, `batch.deleted` (reason: expired, user, abort), `cleanup.error`, `cleanup.stuck`, `rate.limited`.
Levels	info for lifecycle events; warn for rejected input and cleanup errors; error for unexpected exceptions. Debug logging never includes forbidden fields.
Alerts	5xx rate above 2% over 5 minutes; health check failing for 2 minutes; free disk below `MIN_FREE_DISK_MB`; `cleanup.stuck` event; conversion p95 above target for 15 minutes.

20. System Architecture

20.1 Architectural overview

Pixora is a two-tier web application: a statically built React frontend and a stateless-looking Express API with a short-lived local file store. The API is internally modular (routes, controllers, services, middleware, configuration, utilities) so that each module can be understood, tested and replaced independently, which also helps AI coding agents.

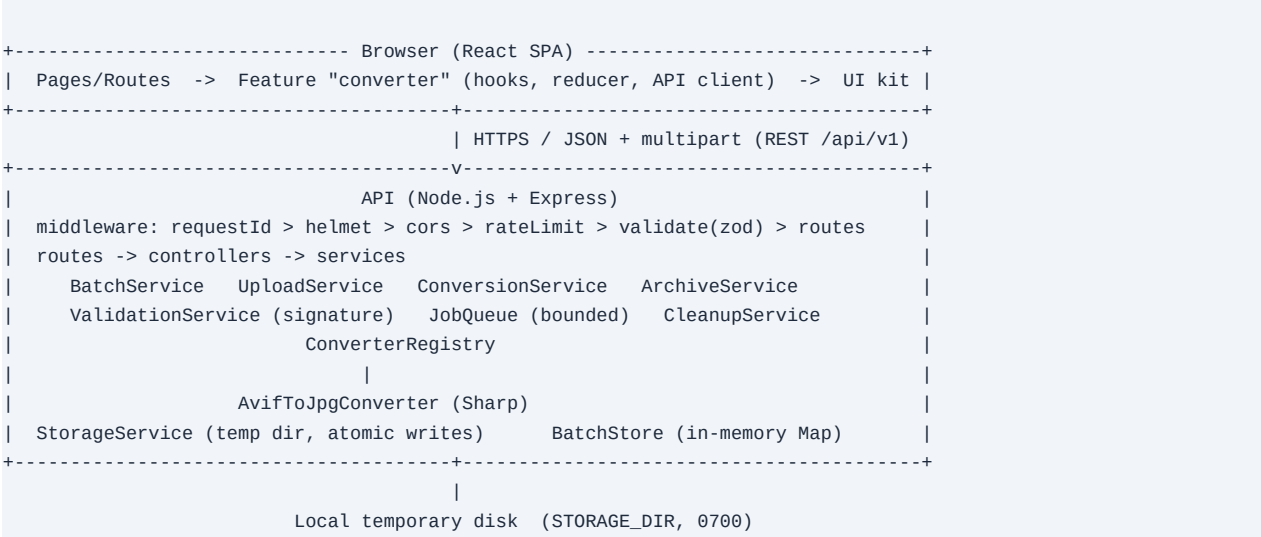


Figure 2: Logical architecture.

20.2 Technology stack and rationale

Layer	Technology	Why it is used
Frontend framework	React with TypeScript	Component model suits the stateful converter UI; static typing shared with the API contract; very large ecosystem and hiring pool.
Build tool	Vite	Fast development server and optimised production builds; simple configuration; supports code-splitting and pre-render scripting.
Styling	Tailwind CSS	Utility classes bound to design tokens give consistency and small CSS bundles; easy for AI agents to generate consistently.
Icons	Lucide React	Consistent, tree-shakable SVG icons; tiny runtime cost.
Routing	React Router	Five routes; works with build-time pre-rendering.
Frontend state	`useReducer` and context	A single converter state machine does not justify Redux or similar.
Backend runtime	Node.js (LTS) with Express	Mature HTTP stack, streams for uploads and downloads, same language as the frontend.
Backend language	TypeScript (`strict`)	Prevents a class of bugs; shared types with the frontend through `packages/shared`.
Image processing	Sharp (libvips)	High-performance, memory-efficient image pipeline with AVIF input and JPEG output, orientation, flatten, colour management, and pixel limits.
Uploads	Multer (current maintained major) with disk storage	Streams multipart bodies to disk instead of memory, supports size and count limits.
ZIP	Archiver (or equivalent streaming library)	Streaming ZIP writer with store-only mode.

Layer	Technology	Why it is used
Validation	Zod	Single schema language for params, bodies, environment, and shared types.
Security middleware	Helmet, cors, express-rate-limit	Well-reviewed building blocks for headers, CORS and throttling.
Logging	Pino	Fast structured logs with redaction support.
Testing	Vitest, Supertest, React Testing Library, Playwright (+axe-core), k6	Covers unit, API, component, E2E, accessibility and load testing with modern tooling.
Version control and CI	Git, GitHub, GitHub Actions	Required by the brief; pipeline defined in Section 25.9.
Packaging	Docker (multi-stage, non-root)	Reproducible builds and the same image locally and in production; also fixes the Sharp native-binary environment.

Not used on purpose No MongoDB or SQL database, no authentication library, no Redis, no message broker and no object storage in the MVP (D-01). Section 21.6 lists the triggers that would justify each.

20.3 Module responsibilities

Module	Responsibility	Must not
Routes and controllers	Map HTTP to service calls; shape responses	Contain business rules or file-system code
BatchService and BatchStore	Batch lifecycle, status aggregation, expiry, tombstones	Know about Sharp
UploadService and ValidationService	Receive files, enforce limits, verify content	Convert images
ConversionService and JobQueue	Queue files, enforce concurrency and timeouts, update statuses	Parse HTTP
ConverterRegistry and converters	Encapsulate each tool; expose input validator, options schema, `convert()`	Touch batch state directly (receive paths, return results)
ArchiveService	Build ZIPs safely	Serve HTTP
StorageService	All file-system access: paths, atomic writes, deletes, free space	Use user-supplied names
CleanupService	Sweepers, start-up and shutdown purge	Delete outside the storage root
Middleware	Cross-cutting concerns: request ID, security headers, CORS, rate limits, validation, errors	Hold state beyond counters

20.4 Conversion sequence

```
Browser                        API                          Disk / Queue
   | POST /batches (multipart)   |                                |
   |---------------------------->| validate size, sniff, metadata |
   |                             |------ write <fileId>.avif ---->|
   |<-- 201 batch(uploaded) -----|                                |
   | POST /batches/{id}/conversions {options}                     |
   |---------------------------->| enqueue files                  |
   |<-- 202 batch(processing) ---|                                |
   |                             |== worker (concurrency 2) =====>|
   |                             |  decode > orient > flatten >   |
   |                             |  sRGB > JPEG > <id>.jpg.part   |
   |                             |  rename to <id>.jpg; delete    |
   |                             |  original; status=completed    |
   | GET /batches/{id}  (every 1 s)                               |
   |---------------------------->|                                |
   |<-- 200 statuses ------------|                                |
   | GET .../preview | HEAD+GET .../download                      |
   | POST .../archive  -> GET .../archive  (ZIP)                  |
   | DELETE /batches/{id} (or expiry sweeper)  -> directory removed
```

Figure 3: Upload, conversion, status, preview, download and deletion.

20.5 Frontend architecture

- **State machine:** `empty` > `selected` > `uploading` > `processing` > `results`, with side states `error` (batch-level) and `expired`. Transitions are explicit reducer actions (`FILES_ADDED`, `FILE_REMOVED`, `UPLOAD_PROGRESS`, `BATCH_CREATED`, `BATCH_UPDATED`, `ERROR`, `EXPIRED`, `RESET`).
- **Polling:** `useConverter` polls `GET /batches/{id}` every `pollIntervalMs`; on network error it retries with exponential backoff (1, 2, 4, 8 s, then 8 s) for up to 60 s; it stops on terminal status, expiry, deletion or unmount.
- **Upload:** `XMLHttpRequest` provides upload progress and `abort()` for cancellation; other calls use `fetch` with `AbortController`.
- **Code splitting:** the lightbox and the static pages are lazy-loaded to meet the JavaScript budget.
- **Configuration:** limits, tool metadata and defaults are fetched from `/config` at start; the UI renders skeleton values until available and falls back to a safe error message if the call fails.
- **Pre-rendering:** a build script renders public routes to static HTML with the same React tree, injects metadata and JSON-LD, and the client hydrates. Interactive state starts empty, so hydration is predictable.

20.6 Project structure

The repository is an npm-workspaces monorepo (D-10). The tree below is the recommended structure; file names are normative where the document refers to them.

```
pixora/
  apps/
    web/                          # Frontend (React + TypeScript + Vite + Tailwind)
      index.html                  # Theme bootstrap script, base meta
      vite.config.ts, tailwind config, tsconfig.json, postcss config
      public/                     # robots.txt, sitemap.xml, favicon set, og-image.png, manifest
      scripts/prerender.ts        # Build-time pre-rendering of public routes (D-04)
      src/
        main.tsx, App.tsx         # Entry, router, providers
        routes/                   # HomePage, AboutPage, ContactPage, PrivacyPage, TermsPage, NotFoundPage
        features/converter/
          ConverterSection.tsx    # Orchestrates the converter views
          converterReducer.ts     # State machine (empty > selected > uploading > processing > results)
          useConverter.ts         # Hook: actions, polling, retries, cancellation
          api.ts                  # Typed API client (XHR for upload, fetch for the rest)
          components/             # Dropzone, FileList, FileRow, SettingsPanel, ProgressPanel,
                                  # ResultsGrid, ResultCard, PreviewDialog, BatchActions, ExpiryBanner
        components/
          layout/                 # Header, Footer, ThemeToggle, SkipLink
          sections/               # Hero, HowItWorks, Features, AboutAvif, Faq
          ui/                     # Button, Card, Badge, ProgressBar, Dialog, Toast, Skeleton, Icon wrappers
        content/                  # strings.ts, faq.ts, features.ts, seo.ts (titles, descriptions, JSON-LD)
        lib/                      # format.ts (sizes), validation.ts (client checks), theme.ts, config.ts
        styles/index.css          # Tailwind layers and design tokens (CSS variables)
        test/                     # Test setup, mocks (MSW handlers), fixtures
      tests/e2e/                  # Playwright specs
  packages/
    shared/                       # Shared TypeScript types, API contract, error codes, constants
      src/{api.ts, errors.ts, tools.ts}
  ...
```

```
apps/
  api/                          # Backend (Node.js + Express + TypeScript + Sharp)
    src/
      server.ts                 # Process entry: config, start, signals, graceful shutdown
      app.ts                    # Express app factory (testable without listening)
      config/
        env.ts                  # Zod-validated environment variables with defaults
        limits.ts               # Derived limits (bytes), public config projection
      routes/
        index.ts, health.routes.ts, config.routes.ts, batches.routes.ts
      controllers/
        batches.controller.ts   # HTTP <-> service mapping only; no business logic
      services/
        batch.service.ts        # Batch lifecycle, status aggregation, tombstones
        upload.service.ts       # Multer integration, per-file validation, rejection handling
        validation.service.ts   # ftyp signature check, Sharp metadata check
        conversion.service.ts   # Enqueue files, apply options, timeouts, state updates
        converters/
          converter.types.ts    # ImageConverter interface, ConversionOptions, ConversionResult
          converter.registry.ts# ConverterRegistry (tool id -> converter, schema, validator)
          avif-to-jpg.converter.ts  # Sharp pipeline (Section 16.4)
        archive.service.ts      # ZIP preparation with locking and atomic rename
        storage.service.ts      # Paths, atomic writes, deletion, free-space checks
        cleanup.service.ts      # Expiry sweeper, orphan sweeper, start-up and shutdown purge
        job-queue.ts            # Bounded FIFO queue with concurrency limit
        batch.store.ts          # In-memory Map and secondary indexes
      middleware/
        request-id.ts, security.ts (helmet), cors.ts, rate-limit.ts, validate.ts,
        upload.ts (Multer config), not-found.ts, error-handler.ts
      domain/
        models.ts               # Batch, FileRecord, status unions
        errors.ts               # AppError subclasses mapped to error codes
      schemas/                  # Zod schemas for params, options, responses
      utils/                    # ids.ts, filename.ts, bytes.ts, time.ts, logger.ts, hash.ts
    tests/
      unit/, integration/, api/, fixtures/avif/  # Real and malformed sample files
    openapi.yaml, Dockerfile, .env.example
tests/perf/                     # k6 scripts
.github/workflows/ci.yml        # Lint, type-check, unit, API, integration, e2e, audit, build
docker-compose.yml, package.json (workspaces), tsconfig.base.json, README.md, .nvmrc, .editorconfig
```

20.7 Purpose of key files and folders

Path	Purpose
`packages/shared/src/api.ts`	TypeScript types for every request and response in Section 15; imported by both apps so the contract cannot drift.
`packages/shared/src/errors.ts`	The error code union and default messages from Section 15.7 and Section 22.
`apps/api/src/config/env.ts`	The only place that reads `process.env`; validates and exposes typed configuration (Section 16.5).
`apps/api/src/app.ts`	Builds the Express app without listening, enabling Supertest to call it directly.
`apps/api/src/services/converters/`	The extensibility seam (FR ext.registry). Adding a tool = adding a converter file and registering it.
`apps/api/src/services/cleanup.service.ts`	Implements mechanisms M2, M5, M6 and M7 (Section 19.4).
`apps/api/tests/fixtures/avif/`	Real AVIF and malformed files listed in Section 23.4.
`apps/web/src/features/converter/`	All converter UI logic. Components inside are presentational; hooks hold behaviour.

Path	Purpose
`apps/web/src/content/`	Copy, FAQ, SEO text and strings in one place for review and later localisation.
`apps/web/scripts/prerender.ts`	Static HTML generation for SEO (D-04).
`apps/api/openapi.yaml`	Machine-readable API description kept in sync by a contract test.
`.github/workflows/ci.yml`	Pipeline defined in Section 25.9.

21. Database and Storage Strategy

21.1 No database in the MVP

The MVP intentionally has no database (D-01). The only durable-looking data are temporary files with a lifetime of minutes; batch metadata is derived at upload time and is useless after the files are gone. A database would add operational cost, a new privacy surface and failure modes without delivering a requirement.

Triggers that would justify adding one: user accounts or saved history; premium plans and billing; API keys and usage metering; multi-instance deployment needing shared batch state (a key-value store is then sufficient); analytics the business must own. Each is a Future item (Section 29).

21.2 In-memory state

- The `BatchStore` is a `Map` keyed by batch ID with the model from Section 16.2, plus a tombstone map and a download-guard counter.
- **Restart behaviour:** a restart or crash loses the registry. Because directories without a registry entry are orphans, the start-up purge deletes them (M6). Users with an open session receive 404 and see a friendly "session ended" message with a Start over button. This is acceptable for a tool whose sessions last minutes and is explicitly traded off against simplicity (A-04, R-04).
- **Concurrency:** Node.js is single-threaded for state changes; the only shared mutable structures are the store and the queue, accessed through service methods.

21.3 File storage layout

```
<STORAGE_DIR>/                          (mode 0700, owned by the service user)
  bch_3f9a1c7e5b2d48a6b0c1d2e3f4a5b6c7/    (one directory per batch, mode 0700)
    fil_a1b2c3d4e5f60718.avif.upload   (while streaming; removed on failure)
    fil_a1b2c3d4e5f60718.avif          (validated original; deleted right after conversion)
    fil_a1b2c3d4e5f60718.jpg.part      (while encoding)
    fil_a1b2c3d4e5f60718.jpg           (finished output)
    bundle.zip.part                    (while creating the ZIP)
    bundle.zip                         (prepared ZIP)
```

21.4 Retention policy summary

Item	Retention	Deleted by
Original file	Seconds to minutes (until its conversion attempt ends)	M1; fallbacks M2 to M7
Converted JPEG and ZIP	30 min after batch creation (configurable 5 to 1440)	M2, M3, M5 to M7
In-memory metadata	Same as outputs	M2, M3
Tombstone	24 h	Tombstone cleanup in the sweeper
Logs	14 days	Log platform policy

21.5 Capacity planning

Peak disk use is approximately concurrent active batches x files per batch x average output size, plus originals during upload. Example: 25 batches x 20 files x 6 MB = 3,000 MB, which fits within the 5120 MB bound. The worst case during simultaneous large uploads (25 x 150 MB originals) exceeds the bound, which

is exactly why uploads are refused with `SERVICE_BUSY` when `MAX_TOTAL_STORAGE_MB` or `MIN_FREE_DISK_MB` is reached. Provision at least 20 GB of fast disk for the reference environment.

21.6 Scale-out path (post-MVP)

Stage	Change	Trigger
0 (MVP)	Single instance, local disk, in-memory state, CDN for static assets	Launch
1	Vertical scaling (more vCPU and RAM), tuned concurrency, CDN caching for static assets, rate limits tuned	CPU above 70% sustained or conversion p95 above target
2	Multiple API instances: S3-compatible object storage with lifecycle (TTL) rules, Redis for batch state and rate limits, queue (for example BullMQ) with separate worker pool, optional SSE for status	One instance cannot meet load, or zero-downtime deploys required
3	Accounts, PostgreSQL, billing provider, API keys and metering, audit logging	Premium plans and business API launched

22. Error Handling

22.1 Principles

- **Say what happened and what to do next,** in plain language, without blame or jargon, and without exposing internals.
- **Be specific where safe:** name the file (as sanitised text) and the applicable limit; never reveal paths, stack traces or library messages.
- **Isolate failures:** a failing file never stops other files; a failing batch action never destroys the selection.
- **Recover automatically** when it is safe (polling backoff), otherwise offer a clear manual action (Retry, Start over, Reload).
- **Accessible:** errors are announced through live regions (assertive for blocking errors) and associated with the relevant control.

22.2 Scenario catalogue

Messages are the exact English strings for `content/strings.ts`; placeholders are in braces.

ID	Scenario	Code	User-facing message	Expected behaviour
ERR-01	Unsupported format	`UNSUPPORTED_FILE_TYPE`	“{name}” isn’t an AVIF image. Pixora currently converts .avif files only.	Client skips obviously wrong files; the server rejects by content. Other files proceed. Selection keeps the file marked with an error until removed.
ERR-02	Corrupted AVIF	`CORRUPTED_FILE`	“{name}” looks damaged and can’t be converted. Try exporting or downloading the original again.	File shown as Rejected or Failed; other files unaffected; no partial output kept.
ERR-03	Oversized file	`FILE_TOO_LARGE`	“{name}” is {size}, which is over the {limit} limit per file.	Client blocks before upload; if it reaches the server the request is rejected with 413 and partial data deleted; selection kept.
ERR-04	Too many files	`TOO_MANY_FILES`	You can convert up to {n} images at a time. Remove {k} to continue.	Client adds the first {n}, refuses the rest with one message; server enforces the same limit.
ERR-05	Empty upload	`NO_FILES_PROVIDED`	Choose at least one AVIF image to get started.	Convert is disabled when empty; the message appears if triggered otherwise; no request sent.
ERR-06	Network failure	(client)	We couldn’t reach Pixora. Check your connection and try again.	Upload: selection kept, Retry button. Polling: "Reconnecting" banner with backoff 1, 2, 4, 8 s up to 60 s, then manual retry. Server work continues.
ERR-07	Server failure	`INTERNAL_ERROR`	Something went wrong on our side. Please try again in a moment. (Reference: {requestId})	Generic message with request ID for support; error logged; selection kept; Retry offered.
ERR-08	Conversion failure	`CONVERSION_FAILED`	We couldn’t convert “{name}”. Your other images weren’t affected.	File marked Failed with no download; batch becomes `partially_completed`; ZIP skips it.
ERR-09	ZIP generation failure	`ARCHIVE_FAILED`	We couldn’t create your ZIP file. You can still download the images one by one.	Partial ZIP removed; per-file downloads stay enabled; Retry offered.
ERR-10	Expired download link	`BATCH_EXPIRED`	These files were automatically deleted for your privacy. Please upload them again.	Results replaced by the expired state with Start over; browser selection retained if still in memory.

ID	Scenario	Code	User-facing message	Expected behaviour
ERR-11	Temporary storage failure	`STORAGE_UNAVAILABLE`	Pixora is temporarily unable to process files. Please try again in a few minutes.	503 with Retry-After; health endpoint reports unavailable; alert raised; selection kept.
ERR-12	Too many requests	`RATE_LIMITED`	You're going a little fast. Please wait {n} seconds and try again.	Countdown from `Retry-After`; button re-enabled afterwards.
ERR-13	Service busy	`SERVICE_BUSY`	Pixora is busy right now. Please try again in a minute.	503 with Retry-After; same handling as ERR-12.
ERR-14	Conversion timeout	`CONVERSION_TIMEOUT`	"{name}" took too long to convert. Try a smaller image.	File Failed; slot released; late result discarded.
ERR-15	Image too large (pixels)	`IMAGE_DIMENSIONS_EXCEEDED`	"{name}" is larger than {mp} megapixels, which is the maximum Pixora can convert.	File Rejected before decoding; others unaffected.
ERR-16	Batch too large	`BATCH_TOO_LARGE`	These files add up to {size}. The limit per batch is {limit}. Remove some files and try again.	Client blocks first; server returns 413; selection kept.
ERR-17	Upload cancelled by user	(client)	Upload cancelled. Your files are still selected.	Request aborted; server removes partial data (M4).
ERR-18	Unexpected UI error	(client)	Something went wrong. Reload the page to continue.	Error boundary shows the full-page fallback with Reload button; no data is sent automatically.

22.3 Error levels in the user interface

Level	Pattern	Examples
File	Inline row or card message with icon and text; remove button remains	ERR-01, 02, 03, 08, 14, 15
Batch	Banner above the list with a primary recovery action; selection retained	ERR-04, 05, 07, 11, 12, 13, 16
Connection	Non-blocking "Reconnecting" banner with automatic retry	ERR-06
Session	Expired state replacing results with Start over	ERR-10
Application	Error boundary full-page fallback with Reload	ERR-18
Transient	Toast	ERR-09, 17 (together with inline state where needed)

22.4 Server-side behaviour

- A central error handler converts domain errors to the response structure of Section 15.2, assigns the request ID, logs at the correct level and never leaks internals (SR errors).
- Errors in background conversion are captured per file; they update state and are logged, but never crash the process.
- `unhandledRejection` and `uncaughtException` log, attempt cleanup and exit; the supervisor restarts the process (SR errors).
- Expected failures (validation, limits) are logged at warn without stack traces; unexpected failures at error with stack traces (server side only).

23. Testing Strategy

23.1 Objectives and principles

- Prove that every Must requirement works and that quality targets are met (Appendix B shows the traceability).
- Test with **real AVIF files** and deliberately broken files, not only mocks; Sharp behaviour must be verified against actual decoders.
- Keep the pyramid healthy: many fast unit tests, a solid layer of API and integration tests, and a small set of high-value end-to-end tests.
- Automate everything that can run in CI; reserve manual testing for screen readers, visual judgement and usability.
- Treat security and privacy tests (deletion, logging, validation) as first-class functional tests.

23.2 Test levels

Level	Scope	Tools	Runs	Targets
Unit (UT)	Pure functions and small modules: sanitisation, validators, ID generation, queue, option schemas, formatting, reducer	Vitest	Every commit	Backend services and utilities >= 85% lines; frontend >= 70%
Integration (INT)	Real Sharp and real file system: conversion pipeline, cleanup, retention, fault injection, concurrency	Vitest with real fixtures and temp directories	Every pull request	All P0 pass
API (API)	Every endpoint, status code and error structure through the Express app	Supertest	Every pull request	100% of endpoints and documented error codes
Frontend component (FE)	Components and hooks with mocked API	Vitest, React Testing Library, MSW	Every commit	All Must UI behaviours
End-to-end (E2E)	Full stack in a browser: user journeys, downloads, ZIP contents	Playwright (Chromium, Firefox, WebKit)	Every pull request (Chromium); nightly (all engines)	All P0 pass on three engines
Security (SEC)	Hostile inputs, headers, CORS, rate limits, logs, dependencies	Vitest, Playwright, ZAP baseline, `npm audit`, image scanner	Pull request (automated); release (ZAP)	0 High or Critical open
Performance (PERF)	Conversion benchmarks, load, soak, web vitals	Benchmark script, k6, Lighthouse CI	Nightly and before release	NFR targets (Section 17)
Responsive and accessibility (RSP, A11Y)	Breakpoints, devices, keyboard, screen reader, contrast, zoom, motion	Playwright projects, axe-core, manual with NVDA and VoiceOver	Automated per PR; manual per release	WCAG 2.2 AA
SEO (SEO)	Pre-rendered HTML, metadata, structured data, sitemap	curl scripts, validators, Lighthouse	Before release	NFR seo.score

23.3 Test environments and CI gating

Environment	Purpose	Notes
Local	Developer runs unit, API, integration and E2E	`docker compose up` provides the same Sharp binaries as production

Environment	Purpose	Notes
CI	Automated gate for pull requests	Containerised; fixtures committed; retention and sweeper intervals shortened by environment variables for tests
Staging	Release candidate verification, ZAP scan, load test	Same image and configuration as production with a staging domain
Production	Smoke tests and monitoring only	No load tests; synthetic single-image conversion every 15 minutes (uses a built-in sample that is not user data)

Merge gate: lint, type-check, unit, API, integration, frontend, E2E (Chromium), `npm audit`, bundle-size budget. **Release gate:** all engines, security scan, performance and Lighthouse budgets, manual accessibility pass, usability test result.

23.4 Test data (fixtures)

Fixtures are stored in `apps/api/tests/fixtures/avif/` and generated by a documented script (using an AVIF encoder and an EXIF tool) so they can be reproduced. No fixture contains personal data.

Fixture	Description	Used by
`photo-8bit.avif`	Opaque 8-bit sRGB photograph, 4000 x 3000 px (12 MP), about 3 to 6 MB	INT-01, E2E-01, PERF-01
`small-2mp.avif`	Opaque photograph, 1920 x 1080 px	PERF-01, API tests
`alpha-logo.avif`	512 x 512 logo with full and partial transparency	INT-02, E2E-04
`exif-orient-6.avif`, `irot-90.avif`, `orient-1.avif`	Orientation variants with a marker pattern in a corner	INT-03
`hdr-10bit.avif`	10-bit BT.2020 PQ image	INT-11
`p3-profile.avif`	Display-P3 image with embedded ICC profile	INT-16
`gps-exif.avif`	Image with EXIF GPS tags, XMP and thumbnail	INT-13
`sequence.avif`	AVIF image sequence (`avis`)	INT-12
`large-24mp.avif`	24 MP image, near the 50 MP limit when scaled in tests	E2E-10, PERF-02
`bomb-100mp.avif`	Small file that declares 100 MP (above the limit)	SEC-05
`truncated.avif`	Valid file cut at 60% of its length	INT-10, API-05
`fake-*.avif`	JPEG, PNG, WebP, HEIC, SVG, HTML and text content renamed to `.avif`	SEC-01, UT-04
`polyglot.avif`	Valid AVIF with appended payload	SEC-02
`empty.avif`	Zero-byte file	UT-04
`oversize-26mb.avif`	Valid AVIF padded above the per-file limit (generated at test time)	API-08, E2E-12
Name fixtures	Hostile and special file names (traversal, emoji, RTL, very long, duplicates)	UT-01, SEC-03, E2E-09

23.5 Defect severity and exit criteria

Severity	Definition	Release rule
S1 Critical	Data exposure, deletion failure, crash, security vulnerability, or conversion completely broken	Blocks release

Severity	Definition	Release rule
S2 High	Core flow fails for a class of valid inputs; accessibility blocker; wrong output (for example wrong orientation)	Blocks release
S3 Medium	Workaround exists; cosmetic issue on a primary screen	Fix or explicit sign-off
S4 Low	Minor cosmetic or wording issue	Backlog

Priorities in test cases: **P0** must pass for release; **P1** should pass (documented exception allowed); **P2** and **P3** are desirable.

24. Test Cases

This section defines 125 test cases. Each has an ID, scenario, preconditions, steps, expected result, priority and the requirements it covers. Positive and negative cases are both included (for example valid files versus spoofed, corrupted, oversized and bomb files). Real-file coverage: opaque, transparent, rotated, HDR, wide-gamut, large, corrupted and multiple uploads, plus ZIP downloads.

24.1 Unit tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
UT-01	Filename sanitisation	Utility module available	1. Pass names: `../../etc/passwd.avif`, `a<b>:c.avif`, 300-char name, `.hidden.avif`, empty string, name with emoji and RTL text	No path separators or control characters remain; length <= 100; leading dot removed; empty falls back to `image`; Unicode letters preserved (NFC)	P0	FR-043, SR-007, SR-006
UT-02	Output-name collision resolution	Utility module available	1. Provide `photo.avif`, `Photo.AVIF`, `photo.avif`, `photo-1.avif` in this order	Outputs are `photo.jpg`, `Photo-1.jpg`, `photo-2.jpg`, `photo-1-1.jpg` (case-insensitive uniqueness, upload order)	P0	FR-043
UT-03	AVIF signature accepted	Fixtures: `still.avif` (brand avif), `seq.avif` (brand avis)	1. Run signature validator on both files	Both pass; detected brand reported	P0	FR-015, SR-001
UT-04	Signature rejected for non-AVIF content	Fixtures: HEIC, JPEG, PNG, WebP, text file, 0-byte file, 20-byte truncated AVIF	1. Run signature validator on each file	Each is rejected with `UNSUPPORTED_FILE_TYPE` or `CORRUPTED_FILE` as appropriate; no exception escapes	P0	FR-015, FR-016, SR-001
UT-05	Conversion option validation	Zod schema available	1. Validate quality 0, 1, 90, 100, 101, 85.5, "90"; colours `#FFF`, `#ffffff`, `red`, `#GGGGGG`; extra property	Only integers 1-100 and six-digit HEX pass; extra properties rejected; defaults applied when omitted	P0	FR-023, FR-024, SR-012
UT-06	Identifier generation and validation	ID utility available	1. Generate 100 000 batch IDs; validate against patterns; test malformed IDs	All IDs match `^bch_[a-f0-9]{32}$`; no duplicates; malformed IDs rejected	P1	SR-017, SR-006
UT-07	Size formatting	Frontend utility available	1. Format 0, 1023, 1024, 1536, 1048576, 26214400 bytes	`0 B`, `1023 B`, `1.0 KB`, `1.5 KB`, `1.0 MB`, `25.0 MB`	P2	FR-013
UT-08	Client-side validation rules	Limits from mocked config (25 MB, 20 files, 150 MB)	1. Add files with wrong extension, uppercase `.AVIF`, oversize, 21st file, duplicate, total over limit	Each invalid file is skipped with a specific message; `.AVIF` accepted; duplicate flagged	P0	FR-014, FR-018
UT-09	Error catalogue mapping	Error module available	1. Throw each domain error through the error handler	HTTP status, `code` and message match the catalogue (Section 22); unknown errors map to `INTERNAL_ERROR` without internal text	P0	FR-031, SR-018

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
UT-10	Expiry calculation and tombstones	Fake timers	1. Create batch at T0; advance to T0+retention; run sweeper; request batch; advance 24 h; request again	Batch removed at expiry; first request returns 410; after tombstone TTL returns 404	P1	FR-045, SR-014
UT-11	Queue concurrency bound	Stub converter with 200 ms delay	1. Enqueue 10 jobs with concurrency 2; record maximum simultaneous jobs	Maximum simultaneous jobs = 2; order is FIFO; one rejected job does not stop others	P0	FR-028
UT-12	Environment parsing	Env module available	1. Provide missing, valid, out-of-range and non-numeric values for every variable	Defaults applied when missing; out-of-range values fail fast at start-up with a clear message; `RETENTION_MINUTES` limited to 5-1440	P0	FR-018, NFR-017
UT-13	Converter registry extensibility	Registry module available	1. Register a dummy `png-to-jpg` converter in a test; create a batch with that tool	Batch accepts the tool and runs the dummy converter; upload, queue, archive and cleanup modules are not modified	P1	FR-057, FR-058
UT-14	Strings are centralised	Lint rule or script available	1. Scan components for hard-coded user-facing text outside `content/strings.ts`	No violations in feature components; adding a second locale file does not require component changes	P2	FR-056

24.2 API tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
API-01	Health check	Server running with writable storage	1. `GET /api/v1/health`	200 with `status: "ok"`; with read-only storage returns 503 `status: "unavailable"`	P0	NFR-014
API-02	Public configuration	Env set to non-default limits	1. `GET /api/v1/config`	200; limits, defaults and retention equal the environment values; frontend shows the same numbers	P0	FR-004, FR-018
API-03	Upload one valid file	Fixture `photo-8bit.avif`	1. `POST /api/v1/batches` multipart with field `files`	201; `batchId` matches pattern; file status `uploaded`; original dimensions present; `expiresAt` = now + retention	P0	FR-011, FR-015
API-04	Upload several valid files	Three valid fixtures	1. POST three files in one request	201; `files` array order equals upload order; unique `fileId` per file	P0	FR-011, FR-028
API-05	Mixed valid and invalid upload	Two valid AVIF, one PNG renamed `.avif`, one truncated AVIF	1. POST the four files	201; two files `uploaded`, two `rejected` with `UNSUPPORTED_FILE_TYPE` and `CORRUPTED_FILE`; rejected files not stored on disk	P0	FR-016, FR-015

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
API-06	Upload with no files	None	1. POST multipart with no `files` part	400 `NO_FILES_PROVIDED`	P0	FR-020
API-07	Upload with only invalid files	Two non-AVIF files	1. POST both	422 `NO_VALID_FILES` with per-file details; no batch directory remains	P0	FR-016
API-08	Oversized file	File of 25 MB + 1 byte	1. POST the file	413 `FILE_TOO_LARGE`; partial upload deleted; message names the limit	P0	FR-018, SR-003
API-09	Too many files	21 small valid files	1. POST all files	422 `TOO_MANY_FILES`; nothing stored	P0	FR-018, SR-003
API-10	Wrong content type	None	1. POST JSON body to `/api/v1/batches`	415 `UNSUPPORTED_MEDIA_TYPE`	P1	SR-012
API-11	Unknown tool	None	1. POST with field `tool=png-to-gif`	400 `UNSUPPORTED_TOOL`	P1	FR-058
API-12	Start conversion	Batch in status `uploaded`	1. `POST /batches/{id}/conversions` with `{"options": {"quality":85,"background":"#000000"}}`	202; batch `processing`; `Location` header present; files move to `queued`/`processing`	P0	FR-021, FR-035, FR-023
API-13	Invalid conversion options	Batch in status `uploaded`	1. POST options with quality 0, 101, "high" and colour `blue`	400 `VALIDATION_ERROR` listing every invalid field; batch remains `uploaded`	P0	FR-023, FR-024, SR-012
API-14	Second conversion request	Batch already processing or completed	1. POST conversions again	409 `BATCH_STATE_CONFLICT`	P1	FR-029
API-15	Batch status polling	Conversion started on three files	1. Poll `GET /batches/{id}` every second until terminal	Counts update monotonically; terminal status is `completed`; completed files expose `output` with dimensions, size, URLs	P0	FR-029, FR-036
API-16	Unknown, malformed and expired batch	One expired batch (tombstone)	1. GET a never-issued valid ID, a malformed ID and the expired ID	404 `BATCH_NOT_FOUND`; 400 `VALIDATION_ERROR`; 410 `BATCH_EXPIRED`	P0	FR-045, SR-006
API-17	Preview endpoint	One completed and one queued file	1. GET `/files/{fileId}/preview` for each	Completed: 200 `image/jpeg`, `inline`, `no-store`. Queued: 409 `FILE_NOT_READY`	P0	FR-037, FR-030
API-18	Single download	Completed file with name `holiday photo.avif`	1. GET `/files/{fileId}/download`; also HEAD	200 `attachment; filename="holiday photo.jpg"` with RFC 6266 `filename*`; body is a valid JPEG; HEAD returns same headers without body	P0	FR-041, FR-044
API-19	ZIP preparation and download	Batch with 3 completed, 1 failed file	1. POST `/archive`; GET `/archive`	POST 200 with `fileCount: 3`; GET 200 `application/zip` with Content-Length; archive lists exactly 3 uniquely named entries	P0	FR-042, FR-043

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
API-20	ZIP rules	Batch without completed files, and batch still processing	1. POST `/archive` on each; GET `/archive` before preparing	409 `NO_COMPLETED_FILES`; 409 `BATCH_STATE_CONFLICT`; 409 `ARCHIVE_NOT_PREPARED`	P1	FR-042
API-21	Delete batch	Completed batch	1. `DELETE /batches/{id}`; then GET batch	204; directory removed from storage; subsequent GET returns 410 `BATCH_EXPIRED`	P0	FR-046, SR-014
API-22	Rate limiting	Upload limit set to 3 per window in test config	1. Send 4 upload requests from one client	4th returns 429 `RATE_LIMITED` with `Retry-After`; other endpoints still respond	P0	SR-008
API-23	Unknown route	None	1. GET `/api/v1/nope`	404 JSON `ROUTE_NOT_FOUND` using the standard error structure	P2	SR-018
API-24	Error structure and request ID	None	1. Trigger any 4xx; send `X-Request-Id` header on another request	Body has `error.code`, `error.message`, `error.requestId`; response header `X-Request-Id` equals body value	P1	FR-031, SR-018

24.3 Integration tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
INT-01	Opaque AVIF to JPEG	Fixture 8-bit sRGB 4000x3000	1. Upload and convert at quality 90	Output starts with `FFD8`, decodes as JPEG, dimensions 4000x3000, sRGB, completed within target	P0	FR-021, FR-022, FR-026, NFR-011
INT-02	Transparency flattened	Fixture AVIF with 50% and 100% transparent regions	1. Convert with default then with `#000000`	Fully transparent pixels become `#FFFFFF` (default) or `#000000`; no alpha channel in output	P0	FR-024
INT-03	Orientation handled	Fixtures: EXIF orientation 6, AVIF `irot` 90 degrees, orientation 1 control	1. Convert all three	Rotated outputs have swapped dimensions and upright content (marker pixel positions match reference); no orientation tag in output	P0	FR-025
INT-04	Quality control	Photographic fixture	1. Convert at quality 60, 90, 100	File size strictly increases with quality; chroma subsampling 4:2:0 at 60 and 4:4:4 at 90 and 100	P1	FR-023
INT-05	No incomplete output served	Fault injection: encoder throws after partial write	1. Convert; request preview and download; list batch directory	File `failed`; preview/download return 409 or 404-class error; no `.tmp` or partial `.jpg` remains	P0	FR-030, FR-031, NFR-012
INT-06	Retention expiry	Retention set to 1 minute, sweep interval 5 s	1. Convert a batch; wait 70 s; inspect storage and request all endpoints	Batch directory (including ZIP) removed; endpoints return 410	P0	FR-045, SR-014, NFR-017

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
INT-07	Originals deleted immediately	Batch of 3 files, one corrupt	1. Convert; inspect directory right after each file finishes	Original `.avif` is deleted after success and after failure; only outputs remain	P0	SR-015, NFR-017
INT-08	Orphan sweeper and start-up purge	Create stray directories older than 2x retention; restart server with stale data	1. Wait for orphan sweep; restart process	Strays removed by sweeper; all pre-existing data removed at start-up	P0	SR-014, SR-013
INT-09	Graceful shutdown	Active batch with processing files	1. Send SIGTERM	Process stops accepting uploads, waits up to 10 s, purges storage, exits with code 0	P1	SR-014, NFR-013
INT-10	Corrupted AVIF in a batch	Fixture truncated at 60% of its length plus two valid files	1. Upload and convert all three	Corrupt file `rejected` or `failed` with `CORRUPTED_FILE`; the two valid files `completed`; batch `partially_completed` if failure occurred at conversion stage	P0	FR-016, FR-031, FR-028
INT-11	High bit-depth AVIF	Fixture 10-bit BT.2020/PQ AVIF	1. Convert	Completes without error; output 8-bit sRGB; documented as fidelity-not-guaranteed	P1	FR-033
INT-12	Animated AVIF	Fixture AVIF image sequence (`avis`)	1. Convert	One JPEG of the first frame; file result contains warning `ANIMATION_FIRST_FRAME_ONLY`	P2	FR-032
INT-13	Metadata stripped	Fixture with EXIF GPS tags and XMP	1. Convert; inspect output with an EXIF tool	No GPS, XMP or thumbnail data in output	P0	FR-027, SR-015
INT-14	Conversion timeout	Test config timeout 2 s with slow stub converter	1. Convert one file	File `failed` with `CONVERSION_TIMEOUT`; late result discarded; queue slot released	P1	FR-034
INT-15	Storage unavailable	Storage directory made read-only; separately, free disk below minimum (mocked)	1. Attempt upload	503 `STORAGE_UNAVAILABLE` / `SERVICE_BUSY` with `Retry-After`; no crash; health endpoint reports unavailable	P0	SR-009, SR-013
INT-16	ICC profile applied	Fixture Display-P3 image with embedded profile	1. Convert; compare sampled saturated colours with reference sRGB rendering	Colour difference within agreed tolerance (delta E <= 3 at sampled points)	P1	FR-026
INT-17	Concurrency under real load	10 files of 12 MP, concurrency 2	1. Convert; sample process memory and simultaneous jobs	Never more than 2 concurrent conversions; peak RSS below container limit; all files complete	P1	FR-028, NFR-009

24.4 Frontend component tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
FE-01	File picker adds files	Home page rendered with mocked config	1. Choose two `.avif` files through the hidden input	Both listed with name and one-decimal size; Convert enabled	P0	FR-003, FR-011, FR-013
FE-02	Invalid selection feedback	Home page rendered	1. Add `notes.txt` and a file over the limit	Neither is listed; messages name each file and reason; focus and live region announce the problem	P0	FR-014
FE-03	Remove and clear	Three files listed	1. Remove the middle file; click Clear all	List updates immediately without network calls; empty state shown after clear	P0	FR-017, FR-052
FE-04	Convert disabled when empty	No files	1. Inspect Convert button; press Enter on it	Button has `aria-disabled`/`disabled`; no request is made	P1	FR-020
FE-05	Polling lifecycle	Mock API returning queued, processing, completed	1. Start conversion; advance timers	Statuses update each poll; polling stops at terminal status and on unmount	P0	FR-029, FR-051
FE-06	Result cards	Mock batch with one completed and one failed file	1. Render results	Completed card shows sizes, dimensions, delta, thumbnail and download; failed card shows message and no download	P0	FR-036, FR-037, FR-040
FE-07	Theme handling	System preference dark; localStorage empty	1. Load page; toggle to light; reload	Initial theme dark with no flash; toggle persists `pixora-theme=light`; reload keeps light	P1	FR-049
FE-08	Lightbox accessibility	Completed thumbnail visible	1. Open lightbox; Tab repeatedly; press Escape	Focus trapped inside; Escape closes; focus returns to thumbnail; 100% toggle works	P1	FR-038, FR-055
FE-09	Network failure during polling	Mock fetch rejecting	1. Start conversion; fail 5 consecutive polls; restore network	Backoff 1, 2, 4, 8 s; reconnecting banner shown; polling resumes and banner clears	P0	FR-053
FE-10	Expiry countdown	Fake timers; `expiresAt` in 6 minutes	1. Advance time to 5:00 remaining, then 0	Warning appears under 5 minutes; expired state replaces results at 0	P1	FR-045
FE-11	Toast announcements	Screen-reader test utilities	1. Trigger success and error toasts	Success in `polite` region, error in `assertive`; auto-dismiss at 5 s and 8 s	P2	FR-054
FE-12	Advanced settings	Home page rendered	1. Choose Balanced preset; set custom colour `#112233`; start conversion	Request body carries `quality: 80` and `background: "#112233"`; invalid HEX blocks submit with inline error	P0	FR-035
FE-13	Download verification	Mock HEAD returning 200, then 410	1. Click Download twice	200: download starts and toast shows; 410: expired message shown and no download	P0	FR-044

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
FE-14	Original thumbnails (optional)	Browser with and without AVIF decoding (stubbed)	1. Add a file; render list in both stubs	Supported: local thumbnail from object URL, revoked on removal; unsupported: neutral placeholder; no extra network request	P3	FR-039

24.5 End-to-end tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
E2E-01	Happy path, single file	Stack running; fixture available	1. Open `/`; drop `photo-8bit.avif`; click Convert; wait for completion; click Download JPG	Status badge Completed; file downloads as `photo-8bit.jpg`; it opens as a valid JPEG of the same dimensions	P0	FR-011, FR-021, FR-036, FR-041, FR-044
E2E-02	Batch with ZIP	Five valid fixtures	1. Select all; convert; click Download all (ZIP)	ZIP downloads; contains five `.jpg` files with unique names; each decodes correctly	P0	FR-011, FR-028, FR-042, FR-043
E2E-03	Mixed valid and invalid	Two valid AVIF, one PNG renamed, one corrupted	1. Select all; convert	Invalid files show per-file errors; valid files complete; ZIP contains only the valid files	P0	FR-016, FR-040, FR-053, FR-042
E2E-04	Transparency and background option	AVIF with transparency	1. Convert with White; repeat with Black via Advanced settings	Previews show white then black background; downloads match	P0	FR-024, FR-035
E2E-05	Cancel upload	Throttle network to slow 3G; 60 MB batch	1. Start conversion; click Cancel during upload	Request aborted; selection intact; no batch directory remains on server after cleanup	P1	FR-019, SR-014
E2E-06	Network loss and recovery	Batch of 10 files processing	1. Go offline for 15 s; go online	Banner "Reconnecting" appears; results continue and finish after reconnect; no duplicate batches created	P0	FR-053, FR-029
E2E-07	Expired files	Server retention set to 2 minutes	1. Convert; wait 125 s; click Download JPG	Expired state shown; message "These files were automatically deleted..."; Start over works	P0	FR-045, FR-044
E2E-08	Delete my files now	Completed batch	1. Click "Delete my files now"; confirm; try the old preview URL	Confirmation toast; empty upload state; old URLs return 410	P0	FR-046, FR-052
E2E-09	Name collisions	Files `a/photo.avif` and `b/photo.avif` (different folders), plus `Photo.AVIF`	1. Convert all; download ZIP; list entries	Entries `photo.jpg`, `photo-1.jpg`, `Photo-2.jpg`; individual downloads use the same names	P1	FR-043
E2E-10	Large image	12 MP AVIF (about 6 MB)	1. Convert on reference environment	Completes within the NFR target; preview opens; dimensions match	P1	NFR-005, FR-022

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
E2E-11	Resume after refresh	Feature enabled	1. Start a 10-file batch; reload page mid-processing	Results view resumes with current statuses; after expiry or deletion the stored ID is cleared	P3	FR-047
E2E-12	Client limits	Config limits default	1. Select 21 files; select a 26 MB file	Only 20 accepted; oversize file refused with message naming limit	P0	FR-014, FR-018
E2E-13	ZIP failure	Server fault injection on archive endpoint	1. Click Download all (ZIP)	Error toast "We couldn't create your ZIP file..."; individual downloads still work	P1	FR-042, FR-053
E2E-14	Theme switching and visuals	Chromium, Firefox, WebKit	1. Toggle theme in each state (empty, selected, processing, results, error)	No unreadable text; screenshots match approved baselines within threshold	P2	FR-049, FR-050, NFR-024
E2E-15	Homepage content	Stack running	1. Open `/`; inspect sections	Hero h1 with "AVIF to JPG", dropzone in first viewport, How it works (3 steps), >= 6 features, >= 8 FAQ items, footer links	P0	FR-001, FR-002, FR-004, FR-005, FR-006, FR-007, FR-008, FR-010
E2E-16	Static pages	Stack running	1. Visit `/about`, `/contact`, `/privacy-policy`, `/terms`, and a non-existent URL	Each static page renders with header/footer; contact shows mailto link and no form; unknown URL shows a designed 404	P1	FR-009
E2E-17	Processing state	Batch of 10 files	1. Observe UI during processing	Per-file spinners/statuses, overall progress bar, ZIP button disabled until at least one file completes	P1	FR-051, FR-029

24.6 Security tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
SEC-01	Extension and MIME spoofing	Fixtures: JPEG, PNG, WebP, HEIC, SVG, HTML, EXE renamed `.avif` and sent as `image/avif`	1. Upload each	All rejected by content validation; none reach the decoder beyond metadata read	P0	SR-001, SR-002, FR-015
SEC-02	Polyglot and trailing data	Valid AVIF with appended script and ZIP payload	1. Upload and convert; inspect output	Output is a clean JPEG with no appended data; nothing is executed or stored beyond temp files	P1	SR-001, SR-013

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
SEC-03	Malicious file names	Names: `../../etc/passwd.avif`, `..%2f..%2fx.avif`, `con.avif`, `x\0.avif`, very long name	1. Upload each; inspect storage and responses	Stored under random IDs; download names sanitised; no path appears in any response	P0	SR-005, SR-006, SR-007
SEC-04	Traversal through URL identifiers	None	1. Request `/batches/..%2f..%2fetc`, `/batches/bch_x/files/..%2f..`	400 `VALIDATION_ERROR`; no file-system access outside storage root (verified by strace/log)	P0	SR-006, SR-012
SEC-05	Decompression bomb	AVIF declaring 100 MP but small on disk	1. Upload and convert; monitor memory	Rejected with `IMAGE_DIMENSIONS_EXCEEDED` before pixel decode; memory remains stable	P0	SR-004, FR-018
SEC-06	Request-shape abuse	None	1. Send 1000 parts, 10 000 header pairs, an unterminated multipart body, slow-drip upload	Requests rejected or timed out without exhausting memory; server stays responsive	P1	SR-003
SEC-07	CORS policy	Allow-list contains one origin	1. Send preflight and actual requests from allowed and disallowed origins	Allowed origin receives CORS headers; disallowed gets none; no wildcard; no credentials	P0	SR-010
SEC-08	Rate-limit evasion	Production-like proxy settings	1. Send requests with spoofed `X-Forwarded-For` values	Spoofed headers do not reset or bypass the limit	P1	SR-008
SEC-09	Log privacy review	Run a full session at debug level	1. Inspect all logs for file names, bytes, IP addresses	No original file names, image content or full IPs appear	P0	SR-016, NFR-018
SEC-10	Security headers	Staging deployment	1. Scan frontend and API responses	All headers in SR-011 present; CSP blocks inline script injection; API has `no-store`	P0	SR-011
SEC-11	Information disclosure	Fault injection for exceptions	1. Trigger 500 errors, malformed JSON, invalid IDs	No stack traces, paths, library names or versions in bodies or headers (`X-Powered-By` absent)	P0	SR-018, FR-031
SEC-12	Batch enumeration	Several active batches	1. Attempt listing endpoints; brute-force 10 000 random IDs	No listing endpoint exists; random IDs return uniform 404; rate limiting engaged	P1	SR-017, SR-008
SEC-13	Stored-XSS style file names	File named `<img src=x onerror=alert(1)>.avif`	1. Upload, convert, preview, download	Name rendered as text everywhere; no script execution; ZIP name sanitised	P0	SR-020, SR-007
SEC-14	ZIP safety	Batch with hostile names	1. Download and extract ZIP with a strict extractor	All entries are flat file names; extraction writes only inside target folder	P1	SR-021
SEC-15	Dependency and image scans	CI pipeline	1. Run `npm audit --omit=dev` and container scan	0 High/Critical findings or documented, time-boxed exceptions	P0	SR-019, NFR-016

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
SEC-16	Dynamic scan	Staging deployment	1. Run OWASP ZAP baseline against frontend and API	No High findings; Medium findings triaged	P1	NFR-016, SR-022

24.7 Performance tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
PERF-01	Per-image conversion time	Fixtures 2 MP and 12 MP; reference environment	1. Convert each 50 times sequentially	2 MP median <= 0.8 s; 12 MP median <= 2.5 s, p95 <= 4 s	P1	NFR-005
PERF-02	Batch time	20 x 12 MP fixtures	1. Convert as one batch; measure from conversion start to terminal status	<= 45 s with concurrency 2	P1	NFR-006
PERF-03	ZIP preparation time	20 JPEG outputs, 100 MB total	1. POST `/archive`; measure	<= 10 s; Content-Length matches bytes received	P1	NFR-008
PERF-04	API load	k6 with 50 virtual users	1. Run health, config, status polling mix for 5 min	p95 <= 150 ms; error rate < 0.1%	P1	NFR-007
PERF-05	Soak test	25 concurrent batches of 10 files for 30 min	1. Run k6 scenario; sample memory and disk every 10 s	No memory growth trend; storage returns to baseline within retention + sweep interval; no failed valid conversions	P1	NFR-009, NFR-011
PERF-06	Web performance budgets	Production build served	1. Run Lighthouse CI 5 times (mobile preset)	LCP <= 2.5 s, CLS <= 0.1, JS <= 150 KB gzip, scores per NFR	P1	NFR-001, NFR-002, NFR-003, NFR-004

24.8 Responsive tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
RSP-01	Minimum width 320 px	Chromium viewport 320 x 640	1. Walk through empty, selected, processing, results, error states	No horizontal scroll; no clipped text or controls	P0	FR-048, NFR-025
RSP-02	Phone flow	Emulated iPhone 14 and Pixel 7	1. Complete the happy path including download	Dropzone within first 640 px; touch targets >= 44 px; flow completes	P0	FR-002, FR-048, NFR-025
RSP-03	Tablet layout	Viewport 768 x 1024 and 1024 x 768 landscape	1. Open home and results	Two-column results grid; navigation expanded at 768 px or higher; no overlaps	P1	FR-048

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
RSP-04	Large desktop	Viewport 1920 x 1080	1. Open home and results	Content constrained to 1200 px container; spacing remains balanced	P2	FR-048

24.9 Accessibility tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
A11Y-01	Automated accessibility scan	axe-core in Playwright	1. Scan every page and every converter state, both themes	0 critical or serious violations	P0	FR-055, NFR-020, FR-007
A11Y-02	Keyboard-only operation	No mouse	1. Complete convert and download using Tab, Shift+Tab, Enter, Space, Escape	Every control reachable; visible focus; logical order; no traps (except intentional lightbox trap)	P0	FR-055, FR-012
A11Y-03	Screen-reader pass	NVDA + Firefox; VoiceOver + Safari	1. Complete the happy path and an error path	Selection, progress, per-file status, errors and download confirmation are announced	P0	FR-055, FR-054
A11Y-04	Contrast, zoom and reflow	Both themes	1. Check tokens; zoom to 200% and 400%; enable forced-colors	Text >= 4.5:1, UI components >= 3:1; content reflows without horizontal scroll; controls visible in forced-colors	P1	FR-055, FR-049
A11Y-05	Reduced motion	OS reduced-motion enabled	1. Run through all transitions	No non-essential motion; spinners replaced by static or minimal indicators	P2	FR-050

24.10 SEO tests

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
SEO-01	Pre-rendered HTML	Production build	1. `curl` each public URL with JavaScript disabled	HTML contains h1, main copy, how-it-works, features and FAQ text	P0	NFR-026, FR-001, FR-007
SEO-02	Metadata	Production build	1. Inspect head of each page	Unique title (<= 60 chars) and description (<= 160), canonical, Open Graph and Twitter tags, `lang`, viewport	P0	NFR-026
SEO-03	Structured data	Production build	1. Validate JSON-LD with Rich Results Test and Schema validator	`WebApplication`, `Organization` and `FAQPage` valid with no errors	P1	NFR-026, FR-007

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
SEO-04	Sitemap and robots	Production build	1. Fetch `/sitemap.xml` and `/robots.txt`	Sitemap lists all public pages with absolute URLs; robots references sitemap and disallows the API host paths	P1	NFR-026

24.11 Usability test

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
UX-01	Moderated usability test	10 first-time participants, mixed devices	1. Task: convert one AVIF image and download it, without help; record time and errors	>= 9 of 10 succeed unaided in <= 60 s using <= 3 deliberate actions; findings logged	P1	NFR-019

24.12 Reviews

Test ID	Scenario	Preconditions	Test steps	Expected result	Prio	Covers
REV-01	Architecture extensibility review	Code complete	1. Review checklist: registry used, no AVIF literals outside the converter and config, tool-driven UI labels	Checklist passes; reviewer signs off	P2	FR-059, FR-057
REV-02	Maintainability gates	CI pipeline	1. Check strict TypeScript, ESLint, Prettier, coverage thresholds, README setup, `.env.example`, OpenAPI contract test	All gates green; setup in 5 commands or fewer; OpenAPI matches implementation	P1	NFR-021, NFR-022, NFR-023, NFR-027
REV-03	Scale path and recovery review	Staging deployment	1. Kill the API process during processing; review Section 21.6 against the code seams	Process restarts within 30 s; start-up purge runs; no seam requires API changes for the Stage 2 migration	P2	NFR-015, NFR-010

25. Deployment Requirements

25.1 Deployment topologies

Option	Description	Strengths	Trade-offs
A. Static CDN host + container host	Frontend on a static hosting platform with CDN (for example Cloudflare Pages, Netlify or Vercel in static mode); API as a container on a managed container platform (for example Render, Railway or Fly.io) at `api.<domain>`	Global CDN for the site, simple frontend deploys, managed TLS and restarts, easy scaling of the API	Cross-origin (CORS must be configured); container platform disk is ephemeral (acceptable); mid-range monthly cost for 2 vCPU / 4 GB; vendor request-body and timeout limits must be checked
B. Single VPS with Docker Compose	One virtual server (for example Hetzner, DigitalOcean or Lightsail) running Caddy (automatic HTTPS) which serves the static build and reverse-proxies `/api` to the Node container	Lowest cost; same-origin (no CORS in practice); full control of body limits and timeouts; simple mental model	You operate the server (patching, backups of config, monitoring); no global CDN unless added; single point of failure
C. AWS or similar cloud	S3 + CloudFront for the site; ECS Fargate or App Runner for the API; later S3 for temporary files, ElastiCache for state	Clear scale-out path, rich monitoring, compliance options	Highest complexity and cost for an MVP; over-engineered at launch
D. Serverless functions for the API	Functions on a serverless platform	No servers	Not suitable for the MVP: request-body size limits, no shared local disk, in-memory state is lost between invocations, cold starts with native libraries

Recommendation Use Option B for the lowest cost and simplest behaviour if the team can operate a Linux server; use Option A if the team prefers managed services. Both satisfy the requirements. Option C is the Stage 2 target (Section 21.6). The final choice is an open decision (OD-02). Prices and platform limits change; verify current limits (request body size, timeouts, disk, memory) before deciding.

25.2 Frontend hosting requirements

- Serve the Vite build output with the pre-rendered public pages; unknown paths return the 404 page with status 404 (not a soft 404).
- Cache policy: hashed assets `Cache-Control: public, max-age=31536000, immutable`; HTML `no-cache` (or short max-age with revalidation).
- Security headers on HTML responses per SR headers (CSP with hash for the theme script, HSTS, Referrer-Policy, Permissions-Policy, `frame-ancestors none`).
- Compression (Brotli and gzip), HTTP/2 or HTTP/3, 301 redirect from `www` to the apex (or the reverse) and from HTTP to HTTPS.

25.3 Backend hosting requirements

- Container image built from a Debian-based Node LTS slim image with Sharp installed from prebuilt binaries; runs as a non-root user (SR container).
- Minimum reference size: 2 vCPU, 4 GB RAM, 20 GB disk; set `UV_THREADPOOL_SIZE` to at least 4 and keep `CONVERSION_CONCURRENCY` at 2 until benchmarks justify more.
- Writable volume only at `STORAGE_DIR`; root file system read-only; `--cap-drop ALL`; memory limit (for example 3 GB) so a runaway decode cannot take down the host.
- Restart policy `unless-stopped` (or platform equivalent) and health check on `/api/v1/health`.
- Graceful shutdown: stop signal timeout of at least `SHUTDOWN_GRACE_SECONDS` plus 5 s.

25.4 Environment configuration

All variables in Section 16.5 apply. Additional variables:

Variable	Default	Meaning
`NODE_ENV`	`production`	Runtime mode
`HOST`	`0.0.0.0`	Bind address inside the container
`STORAGE_DIR`	`/data/pixora`	Temporary storage root (writable volume)
`CORS_ORIGINS`	(required)	Comma-separated list of exact allowed origins, for example `https://pixora.example`
`TRUST_PROXY`	`1`	Number of trusted proxy hops for client IP detection
`LOG_LEVEL`	`info`	Pino level
`RATE_LIMIT_GLOBAL_PER_MIN`	300	Section 15.6
`RATE_LIMIT_UPLOAD_PER_10MIN`	20	Section 15.6
`RATE_LIMIT_ACTION_PER_10MIN`	40	Section 15.6
`RATE_LIMIT_STATUS_PER_MIN`	120	Section 15.6
`RATE_LIMIT_FILES_PER_10MIN`	300	Section 15.6
`APP_VERSION`	from build	Reported by the health endpoint
`VITE_API_BASE_URL` (frontend build)	`/api/v1`	API base; absolute URL in Option A
`VITE_SITE_URL` (frontend build)	(required)	Canonical origin for metadata and sitemap
`VITE_CONTACT_EMAIL` (frontend build)	(required)	Shown on the Contact page
`VITE_ANALYTICS_DOMAIN` (frontend build)	(empty)	Enables privacy-friendly analytics when set (A-08)

Secrets: the MVP has none. Variables are supplied by the platform or an untracked `.env` file; `.env.example` documents every variable and is committed; `.env` is git-ignored.

25.5 Production build

1. `npm ci` (lock file respected).
2. Build `packages/shared`, then `apps/web` (Vite build followed by the pre-render script that emits static HTML, sitemap and metadata) and `apps/api` (TypeScript compile).
3. Build the API Docker image (multi-stage: dependencies, build, slim runtime with only production dependencies).
4. Tag with the Git SHA and semantic version; publish the image to a registry; publish the web build as a deploy artifact.
5. Source maps are generated but not publicly served.

25.6 Reverse proxy and platform settings

Setting	Requirement
Maximum request body	At least 150 MB plus multipart overhead (for example 160 MB). Check every layer: reverse proxy, platform and any CDN in front of the API; some CDN plans cap request bodies (for example 100 MB), in which case lower `MAX_BATCH_TOTAL_MB` or route API traffic without that CDN (R-15).
Request buffering	Disable buffering of upload bodies where possible so the API streams to disk and memory stays flat.

Setting	Requirement
Timeouts	Client body timeout at least 120 s for slow connections; response timeout at least 60 s for ZIP preparation; idle timeout longer than the polling interval.
Compression	Do not compress `image/jpeg` or `application/zip` responses.
Forwarded headers	Proxy sets `X-Forwarded-For` and `X-Forwarded-Proto`; `TRUST_PROXY` matches the number of hops.
HTTP versions	HTTP/1.1 to the API is sufficient.

25.7 CORS, HTTPS and domain configuration

Topic	Requirement
DNS	Apex and `www` records for the site; `api` subdomain for Option A; short TTL during launch, then raise; optional CAA records.
HTTPS	Valid certificates (automatic renewal) for every hostname; TLS 1.2 or higher; redirect HTTP to HTTPS; enable HSTS after verifying all subdomains work (start with a short max-age).
CORS	Option A: `CORS_ORIGINS` contains exactly the production site origin (and staging origin in staging). Option B: same origin, CORS list still set. Never `*` in production (SR cors).
Canonical host	One canonical host (apex or `www`); the other redirects with 301; canonical tags and sitemap use the canonical host.
E-mail	Mailbox for the Contact page and the data-protection contact exists and is monitored before launch.

25.8 File storage

- Local disk or attached volume at `STORAGE_DIR`; not backed up; not shared between instances (A-04).
- Monitor free space; alert below `MIN_FREE_DISK_MB`.
- Optional hardening: mount as tmpfs (RAM) for the shortest data lifetime if memory allows, or an encrypted volume.

25.9 CI/CD pipeline

Stage	Actions	Blocks merge or release
1 Static checks	Install, lint, Prettier check, TypeScript type-check	Yes
2 Unit and component	Vitest with coverage thresholds	Yes
3 API and integration	Supertest and integration suites against real fixtures in the container image	Yes
4 Build	Production build, bundle-size budget, pre-render check	Yes
5 E2E	Playwright against the built stack (Chromium per PR; all engines nightly)	Yes
6 Security	`npm audit --omit=dev`, container image scan, secret scan	Yes (High and Critical)
7 Web quality	Lighthouse CI budgets (performance, accessibility, SEO)	Release
8 Deploy staging	Deploy image and web build; run smoke tests and ZAP baseline	Release
9 Approval and production	Manual approval; deploy; post-deploy smoke test (one conversion); automatic rollback on failure	-

Branching: short-lived feature branches with pull requests to `main`; conventional commit messages; protected `main` requiring the merge gate.

25.10 Logging, monitoring and alerting

- Logs: JSON to stdout, collected by the platform or a log service, retained 14 days, searchable by `requestId` (SR logging).
- Uptime: external monitor on `/api/v1/health` every minute from at least two regions, plus a synthetic conversion every 15 minutes.
- Metrics (from logs or a lightweight exporter): request rate, status codes, conversion duration, queue length, storage used, free disk, active batches, rate-limited requests, cleanup events.
- Alerts: thresholds in Section 19.8; notification to e-mail or chat; documented runbook for each alert (restart, free disk, investigate abuse, rotate configuration).

25.11 Go-live checklist

Item	Evidence
All P0 tests pass; no open S1 or S2 defects	CI report
Security scan and dependency audit clean	SEC-15, SEC-16 reports
Retention and deletion verified on staging with real clock	INT-06, E2E-07 results
Privacy Policy and Terms reviewed by counsel and match configuration	Sign-off record (A-09)
Domain, DNS, HTTPS, redirects, HSTS verified	Checklist run
CORS list and rate limits set to production values	Configuration review
Proxy and platform body-size and timeout limits verified with a maximum-size batch	Manual test
Monitoring, alerts and runbook active; on-call contact defined	Alert test
Sitemap submitted to search consoles; robots.txt live	Search console status
Rollback procedure rehearsed	Runbook

26. Development Roadmap

The nine phases below follow the project brief. Effort is an indicative estimate in person-days for one experienced full-stack developer (about 31 days, 6 to 7 weeks); AI-assisted development can shorten implementation but not review, testing or deployment. Phases build on each other; Phase 8 may start in parallel with Phase 7 for suites that are already stable.

Phase	Days	Deliverables	Acceptance criteria	Requirements delivered
Phase 1 Project setup and architecture	2	Monorepo with workspaces, TypeScript base config, ESLint and Prettier, shared package, env schema with validation, Express app factory with `health` and `config`, Vite app shell with Tailwind tokens, test runners, CI skeleton, Dockerfile skeleton, fixture-generation script, Sharp AVIF spike inside the production base image	`npm ci && npm run dev` starts web and API; CI green; `GET /api/v1/health` returns 200; spike converts a sample AVIF in the container and records findings on orientation, alpha, HDR and sequences; versions pinned (A-10)	see NFR and SR
Phase 2 Frontend design and homepage	4	Design tokens, light and dark themes, header, footer, static pages, hero, upload card (visual and interactions without network), How it works, Features, FAQ, responsive layout, strings module, 404	Pages match Section 18 on 320, 768, 1280 and 1920 px; axe 0 critical or serious; RSP-01, RSP-03, RSP-04 pass; Lighthouse accessibility >= 95 on the home page	FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-048, FR-049, FR-050, FR-052, FR-055, FR-056
Phase 3 File upload and validation	3	Client selection and validation, file list and removal, config-driven limits, upload progress and cancel; API `POST /batches` with Multer, signature and metadata validation, storage service, per-file rejection, rate limiter skeleton	API-03 to API-11, UT-01, UT-03 to UT-06, UT-08, UT-12, FE-01 to FE-04, SEC-01, SEC-03 to SEC-05 pass; rejected files never remain on disk	FR-011, FR-012, FR-013, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019, FR-020, FR-058
Phase 4 Backend image conversion	4	ConverterRegistry and AVIF-to-JPG converter, job queue, conversions endpoint, batch status endpoint, options validation, atomic outputs, timeouts, metadata stripping, orientation and colour handling; spike results turned into tests (HDR, sequences)	API-12 to API-15, INT-01 to INT-04, INT-10 to INT-14, INT-16, INT-17, UT-02, UT-05, UT-11, UT-13 pass; 12 MP median time within target on the reference environment	FR-021, FR-022, FR-023, FR-024, FR-025, FR-026, FR-027, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035, FR-057
Phase 5 Preview and download	3	Polling hook and state machine, progress UI, result cards, thumbnails, lightbox, preview and download endpoints with correct headers, HEAD verification, expiry countdown and expired state, delete action, optional resume	API-16 to API-18, API-21, FE-05 to FE-08, FE-10, FE-12, FE-13, E2E-01, E2E-07, E2E-08, E2E-15 to E2E-17 pass	FR-036, FR-037, FR-038, FR-039, FR-040, FR-041, FR-043, FR-044, FR-045, FR-046, FR-047, FR-051
Phase 6 ZIP download and batch processing	3	Archive preparation and download endpoints with locking and atomic rename, ZIP UI, collision-free naming, partial-success handling, batch UX polish	API-19, API-20, E2E-02, E2E-03, E2E-09, E2E-13, SEC-14, PERF-03 pass; ZIP never contains failed files	FR-042
Phase 7 Security and error handling	4	Rate limiting tiers, CORS, Helmet and CSP, log redaction, all cleanup mechanisms M1 to M7, error catalogue and messages in the UI, reconnect logic, toasts, error boundary, container hardening	SEC-06 to SEC-13, INT-05 to INT-09, INT-15, API-22 to API-24, FE-09, FE-11, E2E-05, E2E-06 pass; every ERR scenario demonstrated	FR-053, FR-054
Phase 8 Testing and optimisation	5	Complete all suites, coverage gates, performance benchmarks and soak test, bundle-size optimisation, manual accessibility passes, usability test with 10 participants, extensibility review, fix backlog	All P0 and at least 95% of P1 tests pass; NFR targets met or change requests approved; no open S1 or S2 defects; REV-01 signed	FR-059

Phase	Days	Deliverables	Acceptance criteria	Requirements delivered
Phase 9 SEO and deployment	3	Pre-rendering, metadata, Open Graph image, JSON-LD, sitemap, robots, production Docker image, hosting and DNS, HTTPS, monitoring and alerts, runbook, final Privacy Policy and Terms text, go-live	SEO-01 to SEO-04 pass; Lighthouse budgets met; staging smoke tests and ZAP baseline clean; go-live checklist (Section 25.11) complete	see NFR and SR

26.1 Definition of done (all phases)

- Code merged through a pull request with passing CI and at least one review.
- Tests written for the requirements in the phase; coverage thresholds not reduced.
- No new lint or type errors; documentation and `.env.example` updated.
- Accessibility checked for new UI (keyboard and axe).
- The traceability matrix (Appendix B) updated when requirements or tests change.

27. Acceptance Criteria

The MVP is accepted for release when every criterion below is met and evidenced.

No.	Area	Criterion	Evidence
AC-01	Functionality	All 52 Must functional requirements are implemented and their linked tests pass; Should requirements are implemented or have an approved deferral	Appendix B matrix, CI report
AC-02	Core journey	A first-time user can select an AVIF file, convert it and download a valid JPG in at most 3 deliberate actions, and 9 of 10 usability participants succeed unaided	E2E-01, usability report (NFR use.task)
AC-03	Batch	A batch of 20 files converts, each file shows its own status, failures are isolated, and the ZIP contains exactly the completed files with unique names	E2E-02, E2E-03, API-19
AC-04	Quality of output	Output dimensions equal oriented input dimensions; orientation, transparency and colour fixtures pass; metadata is stripped	INT-01 to INT-04, INT-13, INT-16
AC-05	Limits and validation	Limits are configurable, shown in the UI from config, and enforced by server; spoofed, corrupted and bomb files are rejected	API-08, API-09, SEC-01, SEC-05
AC-06	Privacy	Originals are deleted right after conversion; all batch data is gone within retention plus sweep interval; logs contain no file names or image data	INT-06, INT-07, INT-08, SEC-09
AC-07	Security	All SR requirements implemented; 0 open High or Critical findings; headers, CORS and rate limits verified	SEC-01 to SEC-16
AC-08	Performance	Performance targets in Section 17 are met on the reference environment, or a change request records the deviation	PERF-01 to PERF-06
AC-09	Accessibility	WCAG 2.2 AA: 0 critical or serious axe violations; keyboard-only and screen-reader runs pass	A11Y-01 to A11Y-05
AC-10	Responsive	No horizontal scroll from 320 px; touch targets at least 44 px; flows pass on emulated phones	RSP-01 to RSP-04
AC-11	Browsers	E2E suite passes on Chromium, Firefox and WebKit	Nightly E2E results
AC-12	SEO	Pre-rendered content, metadata, structured data, sitemap and robots verified; Lighthouse SEO >= 95	SEO-01 to SEO-04
AC-13	Errors	Every scenario ERR-01 to ERR-18 shows the specified message and behaviour	E2E and FE tests
AC-14	Operations	Deployed on HTTPS with monitoring, alerts and a runbook; rollback rehearsed	Go-live checklist
AC-15	Documentation	README, `.env.example`, OpenAPI file and this SRS (kept up to date) are delivered	Review record
AC-16	Scope discipline	No Future feature, database, authentication, payments or advertising exists in the MVP code base	REV-01, code review

28. Risks and Mitigation

Likelihood (L) and impact (I): Low, Medium, High.

ID	Risk	L	I	Mitigation	Owner
R-01	A crafted AVIF exploits a decoder vulnerability in libheif or libaom	Low	High	Content validation before decode; update Sharp promptly; non-root read-only container with memory limits; later option: run conversions in an isolated child process or sandbox	Tech lead
R-02	Resource exhaustion by large images or many uploads	Med	High	Pixel, size, count and storage bounds; bounded queue; timeouts; rate limits; container limits; `SERVICE_BUSY` back-pressure	Tech lead
R-03	A client-rendered SPA is indexed poorly	Med	High	Build-time pre-rendering (D-04) with a Phase 9 spike and test SEO-01; fallback to a static HTML shell with full copy	Frontend lead
R-04	Single instance: restart loses sessions; outage affects everyone	Med	Med	Fast restart, health monitoring, friendly "session ended" state, documented scale-out path (Section 21.6)	Ops
R-05	Bandwidth or hosting cost grows through abuse or popularity	Med	Med	Rate limits, limits per batch, cost alerts, CDN for static assets, monitor egress	Product owner
R-06	Strong SEO competition among converter sites	High	Med	Fast, clean, trustworthy UX; high-quality content; structured data; plan additional tool pages after MVP	Product owner
R-07	Sharp or libheif AVIF support differs by OS or CPU architecture	Low	High	Phase 1 spike in the production image; pin base image; CI runs on the same image	Tech lead
R-08	HDR or wide-gamut AVIF looks wrong after conversion to 8-bit sRGB	Med	Med	Document as a known limitation; test fixtures INT-11 and INT-16; evaluate tone-mapping options after the MVP	Tech lead
R-09	Animated AVIF handling differs from the specification	Med	Low	Phase 4 spike; show a warning; fallback to rejecting sequences with a clear message if the first frame cannot be extracted	Tech lead
R-10	Browsers do not report whether a file download finished	High	Low	HEAD pre-check, Content-Length on all downloads, honest wording ("Download started")	Frontend lead
R-11	Privacy or legal statements do not match behaviour or local law	Med	High	Counsel review (A-09); statements generated from configuration; tests for deletion and logging	Product owner
R-12	Scope creep delays the MVP	Med	Med	Strict MVP versus Future labelling; AC-16; change control via document versioning	Product owner
R-13	Sharp operations cannot be cancelled after a timeout and keep using CPU	Med	Med	Pixel and size limits, low concurrency, container CPU limit; later move to worker processes that can be killed	Tech lead
R-14	A batch URL is leaked and someone downloads the files within the retention window	Low	Med	128-bit IDs, IDs not in page URLs, short retention, delete button, no-referrer policy (SR ids)	Tech lead
R-15	Proxy or CDN limits on request size or time are lower than configured limits	Med	Med	Verify each layer (Section 25.6); lower limits or bypass the CDN for the API; test with a maximum-size batch before launch	Ops
R-16	Vulnerability in a middleware dependency such as Multer	Low	High	Use the maintained major; automated audits and update pull requests; security test suite	Tech lead

29. Future Enhancements

> **Not in the MVP** Everything in this section is a roadmap. None of it is to be implemented in the MVP; the MVP only keeps the architectural seams listed in 29.4.

29.1 Additional conversion tools

Tool	Route (future)	Notes	Indicative stage
AVIF to PNG	`/avif-to-png`	Lossless output; keep alpha; larger files	Stage 1
WebP to JPG	`/webp-to-jpg`	High search demand; same pipeline with different input validator	Stage 1
PNG to JPG	`/png-to-jpg`	Background colour option already exists	Stage 1
JPG to PNG, JPG to WebP, PNG to WebP	`/jpg-to-png`, `/jpg-to-webp`, `/png-to-webp`	WebP encoding options (quality, lossless)	Stage 2

29.2 Additional image tools

Tool	Description	Architecture note
Image compression	Reduce file size with target size or quality	New converter type with options schema
Image resizing	Resize by pixels or percent, keep aspect ratio	Extends options; frontend needs a controls panel
Image cropping	Interactive crop	Needs client-side UI with coordinates sent to the API
Format comparison	Show size and quality across formats side by side	Multiple outputs per input; result model must support variants

29.3 Monetisation roadmap

Monetisation starts only after the MVP is validated (A-01). Payments and advertising are not implemented in the MVP.

Stage	Offering	Prerequisites	Architecture and UX impact
M1	Display advertising	Traffic and consent management; ad network selection; privacy policy update	Reserved fixed-size slots to avoid layout shift; no ads inside the dropzone, processing or download controls; evaluate effect on the "private" brand and on Core Web Vitals; cookie consent where required
M2	Higher upload limits and priority processing	Accounts, subscription billing, a database (Stage 3)	Limits become per-plan; queue gains priority lanes; the config endpoint becomes plan-aware
M3	Premium batch conversion	As M2	Larger batches, longer retention options, saved settings
M4	Additional formats as premium or free-with-ads	Converter registry (already in MVP)	New converters and landing pages
M5	API access for businesses	API keys, usage metering, quotas, documentation portal, SLA	Authenticated variant of the existing API with versioning, per-key rate limits and billing integration
M6	Business partnerships (white-label, integrations)	Commercial agreements, branding options	Embeddable widget, partner configuration

29.4 Architectural seams reserved in the MVP

- Converter registry and `tool` field in the API (FR ext.registry, ext.tool).
- Limits and defaults served by `/config` rather than hard-coded.
- Versioned API (`/api/v1`) and shared contract package.
- Tokens-based design system and layout containers able to host reserved slots.
- Stateless request handling aside from the documented in-memory store (replaceable by a shared store).

30. Glossary

Term	Definition
AVIF	A modern image format based on the AV1 video codec, stored in an ISO base media file container; efficient but not supported by all software.
Alpha channel / transparency	Per-pixel opacity data. JPEG cannot store it, so Pixora flattens it onto a background colour.
Atomic write	Writing to a temporary name and renaming on success so readers never see a partial file.
Backoff	Retrying with increasing waits (1, 2, 4, 8 s) to avoid overloading a recovering service.
Batch	A group of files uploaded together, converted together and expiring together; identified by `batchId`.
Capability URL	An unguessable identifier that grants access to a resource without a login.
Chroma subsampling	Storing colour at lower resolution than brightness; 4:4:4 keeps full colour detail, 4:2:0 saves space.
Content validation	Checking what a file really is from its bytes rather than trusting its name or declared type.
Converter	A backend component that turns one input format into one output format; registered in the `ConverterRegistry`.
Core Web Vitals	Google user-experience metrics: LCP, INP and CLS.
Decompression bomb	A small file that expands into an enormous image when decoded, used to exhaust memory.
EXIF orientation	A metadata value telling software to rotate or mirror an image for display.
Flatten	Compositing a transparent image over a solid colour.
`ftyp` box	The file-type box near the start of ISO base media files that declares brands such as `avif` or `avis`.
Hydration	React attaching behaviour to server-generated or pre-rendered HTML.
ICC profile	Data describing the colour space of an image.
Orphan	A stored directory with no matching in-memory batch, for example after a crash.
Polling	The client repeatedly asking the server for status.
Pre-rendering	Generating static HTML for pages at build time so crawlers and users see content immediately.
Quality (JPEG)	Encoder setting from 1 to 100 trading file size against visual fidelity.
Rate limiting	Restricting how many requests a client may make in a time window.
Retention period	How long converted files are kept before automatic deletion.
sRGB	The standard colour space for web images.
Store-only ZIP	A ZIP written without compression, used for already compressed JPEG data.
Sweeper	A scheduled job that deletes expired or orphaned data.
Tombstone	A short-lived record of an expired batch ID so the API can answer 410 instead of 404.
Tool	A named conversion capability such as `avif-to-jpg`.

31. References

No.	Reference
1	ISO/IEC/IEEE 29148:2018, Systems and software engineering: Life cycle processes: Requirements engineering (successor to IEEE 830-1998).
2	AOMedia, AV1 Image File Format (AVIF) specification; ISO/IEC 23008-12 (HEIF) and ISO/IEC 14496-12 (ISO base media file format).
3	ITU-T T.81 / ISO/IEC 10918-1, JPEG; CIPA DC-008, Exif; IEC 61966-2-1, sRGB.
4	W3C, Web Content Accessibility Guidelines (WCAG) 2.2, https://www.w3.org/TR/WCAG22/
5	IETF RFC 9110 (HTTP Semantics), RFC 6266 (Content-Disposition), RFC 6585 (429 status), RFC 9457 (Problem Details; informs the error design).
6	OWASP, Application Security Verification Standard; File Upload Cheat Sheet; Top 10, https://owasp.org/
7	Sharp documentation, https://sharp.pixelplumbing.com/
8	Express, Multer, Zod, Pino, Helmet, express-rate-limit documentation (npm package pages and project repositories).
9	React, Vite, Tailwind CSS, Lucide and Playwright documentation.
10	Google Search Central, SEO starter guide and structured data documentation, https://developers.google.com/search
11	web.dev, Core Web Vitals, https://web.dev/
12	Schema.org vocabulary, https://schema.org/; Open Graph protocol, https://ogp.me/

Library versions and platform limits referenced in this SRS must be confirmed against the current documentation at the start of Phase 1 (assumption A-10).

Appendix A: Consistency Review and Open Decisions

A.1 Automated consistency checks (run when this document was generated)

Check	Result
All requirement and test IDs are unique (233 IDs)	Pass
Every Must FR traces to at least one test (52 Must of 59 FR)	Pass
Every FR (all priorities) has a test or review case	Pass
Every SR (22) is covered by at least one test	Pass
NFRs with a named test case (27 of 27); the rest state another verification method	All
All test IDs mentioned anywhere in the text exist in Section 24 (125 distinct references)	Pass
All FR IDs mentioned in the text exist	Pass
Every FR is assigned to exactly one development phase	Pass
Risk IDs referenced (R-01 to R-16) all defined	Pass
Decision IDs referenced (D-01 to D-10) all defined	Pass
Default limits are generated from one configuration table (Section 16.5), so API examples, UI copy, requirements and tests use identical values	Pass (by construction)

A.2 Issues found and resolved during the review

Issue	Resolution
Per-file limit times file count (25 MB x 20) exceeds the batch limit of 150 MB	The batch limit governs; both limits are shown in the UI and enforced by the server (SR limits).
Multer aborts the whole request when a size limit is hit, so a per-file 413 cannot coexist with other files in the same request	Documented: size and count violations fail the request; content failures reject individual files. The client pre-validates to avoid surprises (FR upl.client).
Re-converting with another quality conflicts with deleting originals immediately	D-08: originals are deleted; the browser still holds the files and re-uploads as a new batch.
Secondary text on a muted surface fell below 4.5:1 (4.34:1 for #64748B on #F1F5F9)	`--text-muted` is restricted to white and subtle backgrounds; `--text-secondary` is used on muted surfaces; dedicated `--border-strong` token meets 3:1 for controls.
Some CDN plans cap request bodies below the 150 MB batch limit	Added R-15, the reverse-proxy settings table and a go-live test with a maximum-size batch.
Download success cannot be confirmed by the browser	FR dld.notify uses a HEAD pre-check and honest wording; R-10 recorded.
ZIP streamed on the fly could be truncated without warning	D-07: ZIP prepared as a finished file with Content-Length; failure is reported before download starts.
Restart loses in-memory state while files remain on disk	Start-up purge (M6) and friendly session-ended message; A-04 and R-04 recorded.

A.3 Open decisions and unresolved assumptions

ID	Decision or assumption to resolve	Default used in this SRS	Needed by
OD-01	Final product name availability, domain and trademark check	`Pixora`, `https://pixora.example` placeholder (A-02)	Phase 9 (earlier for branding)
OD-02	Hosting option and provider (Section 25.1)	Option A or B; no provider fixed	Phase 1 (staging) and Phase 9
OD-03	Pre-rendering tool for Vite (library or custom script)	Custom build script (D-04)	Phase 2 spike, Phase 9
OD-04	Animated AVIF behaviour confirmed by spike	First frame plus warning	Phase 1 and 4
OD-05	HDR and wide-gamut tone-mapping approach	Convert to 8-bit sRGB; fidelity not guaranteed	Phase 4
OD-06	Use of web analytics and which provider	Disabled unless configured	Phase 9
OD-07	Contact mailbox and data-protection contact	Placeholder address	Phase 2 and 9
OD-08	Legal text of Privacy Policy and Terms; jurisdiction	Draft from Section 19.7; counsel review required	Before launch
OD-09	Confirm 30-minute retention and 25 MB / 20 files / 150 MB / 50 MP limits after benchmarks	Values in Section 16.5	Phase 8
OD-10	Final versions of Node.js, React, Vite, Tailwind, Sharp, Multer and others	Latest stable at Phase 1 (A-10)	Phase 1
OD-11	Whether to ship the optional resume-after-refresh feature (FR dld.resume)	Could; implement if time permits	Phase 5
OD-12	Whether to offer an option to keep metadata in a later release	Metadata always stripped (D-09)	Post-MVP
OD-13	Process isolation (child process or sandbox) for conversions	In-process Sharp with container limits	Review after Phase 8 benchmarks
OD-14	Advertising network and consent platform	Not in MVP	Post-MVP

A.4 Recommended first actions

1. Approve this SRS and resolve OD-01, OD-02 and OD-07.
2. Start Phase 1 with the Sharp AVIF spike in the production base image; record results against R-07, R-08 and R-09.
3. Generate the fixtures listed in Section 23.4 early; they unblock Phases 3 and 4.
4. Commission legal review of the Privacy Policy and Terms in parallel with Phases 2 to 8.

Appendix B: Requirement-to-Test Traceability Matrix

B.1 Functional requirements

FR	Requirement (abridged)	Priority	Phase	Test cases
FR-001	The homepage SHALL show a hero section with one `h1` containing the phrase "AVIF to JPG" and a sub-headline of at...	Must	2	E2E-15, SEO-01
FR-002	The drag-and-drop upload area SHALL be visible without scrolling on a 1280 x 720 desktop viewport, and its top edge...	Must	2	E2E-15, RSP-02
FR-003	The upload area SHALL contain a "Browse files" button that opens the operating-system file picker with `multiple`...	Must	2	FE-01
FR-004	The homepage SHALL display supported-format information: input AVIF (.avif), output JPG, maximum size per file,...	Must	2	API-02, E2E-15
FR-005	The homepage SHALL include a "How it works" section with exactly three steps: Upload, Convert, Download.	Must	2	E2E-15
FR-006	The homepage SHALL include a "Key features" section of at least six feature cards (for example: batch conversion, no...	Must	2	E2E-15
FR-007	The homepage SHALL include an FAQ of at least eight questions using accessible disclosure widgets (native...	Must	2	E2E-15, A11Y-01, SEO-01, SEO-03
FR-008	Every page SHALL include a footer with links to About, Contact, Privacy Policy and Terms, plus a copyright line.	Must	2	E2E-15
FR-009	The MVP SHALL provide static content pages at `/about`, `/contact`, `/privacy-policy` and `/terms`. The Contact page...	Must	2	E2E-16
FR-010	Every page SHALL include a header with the Pixora logo (link to `/`), anchor links to How it works, Features and FAQ...	Should	2	E2E-15
FR-011	The user SHALL be able to add AVIF files by drag-and-drop onto the upload area or through the file picker; multiple...	Must	3	API-03, API-04, FE-01, E2E-01, E2E-02
FR-012	The upload area SHALL provide a distinct drag-over visual state, SHALL be operable by keyboard (Enter and Space open...	Must	3	A11Y-02
FR-013	Each selected file SHALL be listed with its file name (truncated with an ellipsis when too long, full name available...	Must	3	UT-07, FE-01
FR-014	The frontend SHALL pre-validate selections for UX purposes only: extension `.avif` (case-insensitive) or MIME...	Must	3	UT-08, FE-02, E2E-12
FR-015	The backend SHALL validate the actual content of every uploaded file (ISO-BMFF `ftyp` signature with an AVIF brand...	Must	3	UT-03, UT-04, API-03, API-05, SEC-01
FR-016	The backend SHALL reject unsupported or corrupted files individually with a machine-readable error code and a...	Must	3	UT-04, API-05, API-07, INT-10, E2E-03
FR-017	The user SHALL be able to remove any single file, or all files ("Clear all"), from the selection before conversion...	Must	3	FE-03
FR-018	Upload limits SHALL be configurable through environment variables with these defaults: 25 MB per file, 20 files per...	Must	3	UT-08, UT-12, API-02, API-08, API-09, E2E-12, SEC-05
FR-019	While files are uploading, the UI SHALL show overall upload progress (percentage and bytes sent) and SHALL allow the...	Should	3	E2E-05

FR	Requirement (abridged)	Priority	Phase	Test cases
FR-020	The "Convert" button SHALL be disabled while the selection is empty, and an empty submission attempted by any other...	Must	3	API-06, FE-04
FR-021	The system SHALL convert each accepted AVIF image to a JPEG file (`image/jpeg`, extension `.jpg`) using server-side...	Must	4	API-12, INT-01, E2E-01
FR-022	By default the output pixel dimensions SHALL equal the oriented input dimensions (no resampling). No resize option...	Must	4	INT-01, E2E-10
FR-023	The JPEG quality SHALL be an integer from 1 to 100, defaulting to 90 (environment variable `DEFAULT_JPEG_QUALITY`)....	Must	4	UT-05, API-12, API-13, INT-04
FR-024	Images that contain an alpha channel SHALL be flattened onto a solid background colour supplied as a six-digit HEX...	Must	4	UT-05, API-13, INT-02, E2E-04
FR-025	Image orientation SHALL be applied to the pixel data, honouring both AVIF container transforms (`irot`, `imir`) and...	Must	4	INT-03
FR-026	Output SHALL be 8-bit sRGB. Embedded ICC profiles in the input SHALL be applied when converting to sRGB.	Must	4	INT-01, INT-16
FR-027	By default, EXIF, XMP and embedded thumbnails (including GPS location data) SHALL be removed from the output. There...	Must	4	INT-13
FR-028	Batches SHALL be processed with bounded concurrency (default 2 simultaneous conversions, environment variable...	Must	4	UT-11, API-04, INT-10, INT-17, E2E-02
FR-029	Each file SHALL expose one of the statuses `uploaded`, `queued`, `processing`, `completed`, `failed` or `rejected`,...	Must	4	API-14, API-15, FE-05, E2E-06, E2E-17
FR-030	The output file SHALL be written to a temporary name and atomically renamed only after the encoder finishes...	Must	4	API-17, INT-05
FR-031	Every conversion failure SHALL be caught, mapped to an error code from the error catalogue, recorded in the file...	Must	4	UT-09, API-24, INT-05, INT-10, SEC-11
FR-032	For animated or multi-image AVIF (image sequences) the system SHALL output a single JPEG of the primary (first)...	Should	4	INT-12
FR-033	AVIF files with 10-bit or 12-bit depth, including wide-gamut or HDR signalling, SHALL convert without error to 8-bit...	Must	4	INT-11
FR-034	A conversion that exceeds 60 seconds (environment variable `CONVERSION_TIMEOUT_SECONDS`) SHALL be marked `failed`...	Must	4	INT-14
FR-035	The UI SHALL provide an "Advanced settings" panel with a quality control (presets High 90, Balanced 80, Small 70,...	Must	4	API-12, FE-12, E2E-04
FR-036	For each file the results view SHALL show: file name, original size, original dimensions, output size, output...	Must	5	API-15, FE-06, E2E-01
FR-037	For each completed file the results view SHALL show a thumbnail of the converted JPEG loaded from the preview...	Must	5	API-17, FE-06
FR-038	Activating a thumbnail SHALL open an accessible modal preview (native `dialog` element) showing the converted image...	Must	5	FE-08
FR-039	When the user's browser can decode AVIF, the file list MAY show a small client-side thumbnail of the original;...	Could	5	FE-14
FR-040	For a failed or rejected file the card SHALL show the user-facing error message for its error code and SHALL NOT...	Must	5	FE-06, E2E-03

FR	Requirement (abridged)	Priority	Phase	Test cases
FR-041	Each completed file SHALL have a "Download JPG" action that downloads that single file with `Content-Disposition:...	Must	5	API-18, E2E-01
FR-042	When at least one file is completed, a "Download all (ZIP)" action SHALL be available. It SHALL first request ZIP...	Must	6	API-19, API-20, E2E-02, E2E-03, E2E-13
FR-043	Output file names SHALL be `<sanitised original base name>.jpg`. Name collisions within a batch (case-insensitive)...	Must	5	UT-01, UT-02, API-19, E2E-02, E2E-09
FR-044	Before starting any download the frontend SHALL verify availability with a `HEAD` request. On success it SHALL start...	Must	5	API-18, FE-13, E2E-01, E2E-07
FR-045	The results view SHALL show a countdown to the batch expiry (computed from `expiresAt`), SHALL warn when fewer than...	Must	5	UT-10, API-16, INT-06, FE-10, E2E-07
FR-046	The results view SHALL provide a "Delete my files now" action that calls `DELETE /api/v1/batches/{batchId}` and,...	Must	5	API-21, E2E-08
FR-047	The frontend MAY store the opaque `batchId` in `sessionStorage` so that a page refresh during processing can resume...	Could	5	E2E-11
FR-048	The interface SHALL be fully usable from 320 px to 1920 px viewport width with no horizontal page scrolling, using...	Must	2	RSP-01, RSP-02, RSP-03, RSP-04
FR-049	The interface SHALL support light and dark themes, default to the operating-system preference, allow manual override...	Must	2	FE-07, E2E-14, A11Y-04
FR-050	Animations SHALL last no longer than 300 ms, SHALL animate only `opacity` and `transform` where possible, and SHALL...	Must	2	E2E-14, A11Y-05
FR-051	The UI SHALL show loading indicators for: uploading (progress bar), processing (per-file spinner and an overall...	Must	5	FE-05, E2E-17
FR-052	The UI SHALL provide a designed empty state for the upload area (no files selected) and for the results area (all...	Must	2	FE-03, E2E-08
FR-053	The UI SHALL provide distinct error states at three levels: per file (inline), per batch (banner with retry action),...	Must	7	FE-09, E2E-03, E2E-06, E2E-13
FR-054	Transient feedback SHALL be shown as toasts: success and info toasts auto-dismiss after 5 s, error toasts after 8 s...	Must	7	FE-11, A11Y-03
FR-055	The UI SHALL conform to WCAG 2.2 Level AA: every control keyboard-operable, visible focus indicators, programmatic...	Must	2	FE-08, A11Y-01, A11Y-02, A11Y-03, A11Y-04
FR-056	All user-facing strings SHALL be defined in a central module (`content/strings.ts`) so that localisation can be...	Should	2	UT-14
FR-057	The backend SHALL implement conversions behind an `ImageConverter` interface and a `ConverterRegistry`, so that a...	Must	4	UT-13, REV-01
FR-058	Batches SHALL carry a `tool` identifier (MVP value: `avif-to-jpg`). The API SHALL reject unknown tool values with...	Must	3	UT-13, API-11
FR-059	The architecture SHALL NOT preclude the future features listed in Section 29 (additional formats, compression,...	Should	8	REV-01

B.2 Security and privacy requirements

SR	Key	Test cases
SR-001	content	UT-03, UT-04, SEC-01, SEC-02
SR-002	mime	SEC-01
SR-003	limits	API-08, API-09, SEC-06
SR-004	bomb	SEC-05
SR-005	names	SEC-03
SR-006	traversal	UT-01, UT-06, API-16, SEC-03, SEC-04
SR-007	display-names	UT-01, SEC-03, SEC-13
SR-008	rate	API-22, SEC-08, SEC-12
SR-009	queue	INT-15
SR-010	cors	SEC-07
SR-011	headers	SEC-10
SR-012	validate	UT-05, API-10, API-13, SEC-04
SR-013	storage	INT-08, INT-15, SEC-02
SR-014	cleanup	UT-10, API-21, INT-06, INT-08, INT-09, E2E-05
SR-015	privacy	INT-07, INT-13
SR-016	logging	SEC-09
SR-017	ids	UT-06, SEC-12
SR-018	errors	UT-09, API-23, API-24, SEC-11
SR-019	supply	SEC-15
SR-020	xss	SEC-13
SR-021	zip	SEC-14
SR-022	container	SEC-16

B.3 Non-functional requirements

NFR	Requirement (abridged)	Test cases or verification
NFR-001	Performance: Largest Contentful Paint of the pre-rendered homepage	PERF-06
NFR-002	Performance: Cumulative Layout Shift and Interaction to Next Paint	PERF-06
NFR-003	Performance: Initial JavaScript payload of the homepage (gzip)	PERF-06
NFR-004	Performance: Lighthouse scores on the homepage (mobile)	PERF-06
NFR-005	Performance: Server-side conversion time per image (excludes upload time)	E2E-10, PERF-01
NFR-006	Performance: End-to-end time to convert a batch of 20 images of about 12 MP each, e	PERF-02
NFR-007	Performance: API latency for lightweight endpoints (health, config, batch status) u	PERF-04

NFR	Requirement (abridged)	Test cases or verification
NFR-008	Performance: ZIP preparation time for 20 JPEG files totalling up to 100 MB	PERF-03
NFR-009	Scalability: Concurrent active users supported by a single reference instance with	INT-17, PERF-05
NFR-010	Scalability: Documented path to horizontal scaling without redesigning the API	REV-03
NFR-011	Reliability: Conversion success rate for valid AVIF files in the conformance fixtur	INT-01, PERF-05
NFR-012	Reliability: Incomplete or partial files served to users	INT-05
NFR-013	Reliability: Graceful shutdown on SIGTERM	INT-09
NFR-014	Availability: Monthly availability of the production service	API-01
NFR-015	Availability: Automatic recovery from process crash	REV-03
NFR-016	Security: Conformance to the security requirements SR-001 onwards and to relevan	SEC-15, SEC-16
NFR-017	Privacy: Retention of user images on the server	UT-12, INT-06, INT-07
NFR-018	Privacy: Personal data and image data in logs	SEC-09
NFR-019	Usability: First-time users completing "convert one AVIF and download it"	UX-01
NFR-020	Accessibility: WCAG 2.2 Level AA conformance of all MVP pages	A11Y-01
NFR-021	Maintainability: Static quality of the codebase	REV-02
NFR-022	Maintainability: Automated test coverage	REV-02
NFR-023	Maintainability: Developer documentation	REV-02
NFR-024	Browser compatibility: Supported browsers: the latest two major versions of Chrome, Edge, Fir	E2E-14
NFR-025	Mobile responsiveness: Usability on phones and tablets	RSP-01, RSP-02
NFR-026	SEO: Technical SEO quality of public pages	SEO-01, SEO-02, SEO-03, SEO-04
NFR-027	Code quality: API stability and discoverability	REV-02